

Earthquake Catalogues of Aotearoa New Zealand, 1960–2026

A Registry-style Inventory of National, Regional, Temporary-Deployment, and Derived Catalogues*

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Abstract

New Zealand’s earthquake-catalogue record is curated nationally by GeoNet (Earth Sciences New Zealand / GNS Science) but is complemented by a large, fragmented body of regional, temporary-deployment, volcano-seismic, slow-earthquake, and derived catalogues that are individually valuable yet difficult to discover, compare, and cite. This report inventories **107 distinct earthquake catalogues** that were developed for, or that cover, Aotearoa New Zealand and its offshore margins with coverage overlapping the period 1960–2026, with particular emphasis on **temporary seismic deployments** (passive-seismic experiments, ocean-bottom deployments, and aftershock rapid-response arrays). Each catalogue is profiled against the minimum-attributes *Catalogue Submission Profile* proposed for a national “catalogue of catalogues”, recording what it contains and how to obtain it. Of the 107 catalogues, 90 carry a resolvable DOI or stable dataset identifier. Temporary deployments form the single largest group, underscoring both the richness of New Zealand’s regional seismology and the discoverability gap that a federated registry is designed to close. The inventory is offered as a working population of that registry and as a reproducible, citable reference for researchers, hazard modellers, and engineers.

Keywords: earthquake catalogue; seismicity; temporary deployment; ocean-bottom seismometer; slow slip; FAIR data; Aotearoa New Zealand; GeoNet; metadata.

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*Each entry is a discovery record summarising what a catalogue contains and where to obtain it; confirm details against the cited source before quantitative use.

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1 Introduction and scope

An earthquake catalogue is the evidence base for seismicity research, seismic-hazard estimation, and engineering design. In Aotearoa New Zealand that evidence base is distributed across one authoritative national operational catalogue and many high-value but loosely coupled regional, temporary, and derived catalogues. Because these use different formats, identifiers, magnitude scales, and metadata conventions, combining them is manual and error-prone—a cost made explicit by national products such as the 2022 National Seismic Hazard Model, which required dedicated effort to standardise magnitudes across sources.

This report addresses the first practical step toward a federated capability: *knowing which catalogues exist*. It inventories the earthquake catalogues developed for, or covering, New Zealand and its offshore margins (including the Hikurangi and Kermadec subduction systems and the subantarctic) whose coverage overlaps the period 1960–2026. Catalogues that begin earlier and extend into the period, and historical or derived catalogues used for New Zealand even where compiled recently, are included. Special attention is given to **temporary deployments**, which are scientifically rich yet routinely invisible to the national system.

Each catalogue is described using the *Catalogue Submission Profile* (Appendix A): a concise, controlled set of attributes covering identification and contact, coverage, content, classification, and availability. The profile records *what* a catalogue contains and *how* to obtain it, so that a reader can assess relevance before consulting the full documentation. Consistent with that scope, each entry here is a discovery record rather than a quality certification; deeper detail is carried by the catalogue’s own dataset, DOI, or defining publication and is cited accordingly.

2 Scope, sources, and compilation

This inventory was compiled from peer-reviewed publications, dataset repositories (for example Zenodo, IRIS/EarthScope, and PANGAEA), FDSN network records, and the catalogues and reports of GeoNet / GNS Science and partner institutions. Each catalogue was profiled against the submission schema (Appendix A) and cross-checked against its defining publication or dataset; persistent identifiers and citations were checked against the original source, and fields that could not be established from the sources are left blank rather than inferred.

The inventory is a *discovery* resource: each entry summarises what a catalogue contains and how to obtain it, and is not a certification of scientific adequacy—confirm details against the cited source before quantitative use. It is also not a census: earthquake-catalogue production in Aotearoa New Zealand is decentralised and continuous, so a residual long tail of specialised and thesis-derived catalogues remains (Section 6). Of the 107 catalogues listed, 90 carry a resolvable DOI or stable dataset identifier.

3 The shape of New Zealand’s catalogue ecosystem

New Zealand’s earthquake-catalogue landscape has three tiers.

(1) One authoritative national operational catalogue. The GeoNet catalogue, produced by GNS Science / Earth Sciences New Zealand, provides continuous instrumental coverage (locations from the 1930s; descriptive events from ~1460) and is underpinned by the permanent National Seismograph Network (FDSN code NZ). It is the parent dataset for nearly every derived New Zealand catalogue.

(2) A large, scattered body of regional, temporary-deployment, volcano/geothermal, and special-purpose catalogues, produced mostly by universities (Victoria University of Wellington, Otago, Auckland, Canterbury) and international partners. These deliver far higher *local* precision—dense arrays, ocean-bottom and borehole sensors, and matched-filter / template-matching detection—but are typically published as one-off datasets discoverable only through the originating paper.

(3) Derived products: relocated and reprocessed catalogues, magnitude-homogenised and declustered hazard-model catalogues (NSHM 2010 and 2022), and physics-based synthetic (RSQSim) catalogues.

Temporary deployments are the **single largest group** in this inventory (20 onshore + 5 offshore, plus a large share of the volcano-seismic and aftershock-response catalogues), yet they are precisely the catalogues *least* discoverable in the national system. Scattered across journal supplements, Zenodo, IRIS/EarthScope, PANGAEA, and FDSN temporary-network DOIs, and using heterogeneous magnitude and depth conventions, they are exactly the material a federated registry is designed to surface.

4 Historical development, 1960–2026

1960s–1970s (analogue era). The national catalogue is maintained by the DSIR Seismological Observatory; the *Computer File of New Zealand Earthquakes* runs to 1974, while Eiby’s *Descriptive Catalogue* and *Annotated List (1460–1965)* bridge the instrumental and historical records. Globally relocated coverage exists via ISC-GEM (from 1904) and later ISC-EHB (from 1964).

1980s (digitisation). Continuous digital waveforms are archived from September 1986. The 1987 Edgecumbe sequence is an early instrumented aftershock response.

1990s (first large passive experiments). SAPSE (Southern Alps Passive Seismic Experiment, 1995–96) is a landmark temporary deployment; Wellington-region relocation work begins; Downes’ *Atlas of Iseoseismal Maps* (1995) systematises historical macroseismic intensities.

2000s (targeted arrays and double-difference relocation). A wave of temporary arrays (Marlborough XB, Tongariro, Taranaki, CNIPSE/NIGHT, SAHKE, ALFA08/09, and the SAMBA borehole array from 2008); double-difference relocation becomes standard; the 2003 Fiordland response.

2010s (GeoNet modernises; sequences drive dense catalogues). GeoNet moves to SeisComP with 3-D NonLinLoc (2012). The Canterbury sequence (Darfield 2010, Christchurch 2011) and Kaikōura 2016 drive dense aftershock and ocean-bottom catalogues (HOBITSS, 2014–15). Matched-filter / template-matching catalogues emerge, and slow-slip and tremor / low-frequency catalogues mature.

2020s (machine-scale relocation, synthetics, multi-year geodetic catalogues). High-precision matched-filter and nested-tomography catalogues (Taupō Volcanic Zone; Hikurangi); long RSQSim synthetic catalogues for NSHM 2022; multi-year GNSS slow-slip catalogues; dense onshore arrays (DWARFS, SALSA, SOSA); volcanic-unrest catalogues at Taupō (2019; 2022–23); and geothermal induced-seismicity monitoring (Rotokawa, Ngatamariki).

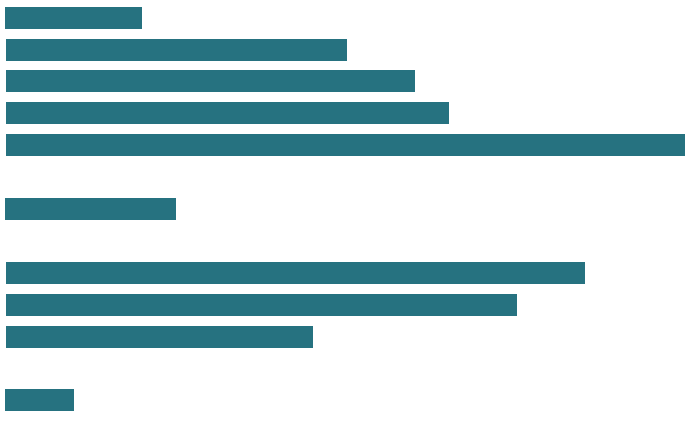
5 Summary statistics

Table 1 gives the distribution of the 107 catalogues across categories; Table 2 is a master index, numbered as in the detailed inventory (Section 7).

Table 1: Catalogues by category.

Category	Count
National operational and network catalogues	4
Relocated and reprocessed catalogues	10
Historical and macroseismic (felt-report) catalogues	12
Aftershock-sequence rapid-response catalogues	13
Temporary onshore passive-seismic deployments	20
Temporary offshore / ocean-bottom (OBS) deployments	5
Volcano-seismic, geothermal and induced catalogues	17
Slow-slip, tremor, LFE and moment-tensor catalogues	15
Derived, homogenised and synthetic / hazard-model catalogues	9
Strong-motion and ground-motion databases	2
Total	107

Distribution by category

National operational and network		4
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Slow-slip, tremor, LFE and moment-tensor		15
Derived, homogenised and synthetic / hazard-model		9
Strong-motion and ground-motion databases		2

Availability. 90 of the 107 catalogues carry a resolvable DOI or stable dataset identifier.

Table 2: Master index of catalogues (number matches the detailed entry in Section 7). The final column gives availability, the primary reference, and the DOI / dataset link where one exists.

#	Catalogue	Period	Type	Availability, ref & DOI
1	GeoNet Aotearoa New Zealand Earthquake Catalogue	1460–ongoing	background	open (DOI), [52], 10.21420/OS8P-TZ38
2	New Zealand National Seismograph Network (NZNSN) seismic waveform dataset	1916–ongoing	background	open (DOI), [59], 10.21420/G19Y-9D40
3	National Earthquake Information Database (NEID)	?–ongoing	compilation	open (DOI), [1]
4	Computer File of New Zealand Earthquakes (DSIR Seismological Observatory)	1460–1974	compilation	not avail., [124]
5	Eberhart-Phillips & Reyners 2001-2011 NZ-Wide 3-D Relocated Catalogue	2001–2011	background	open (DOI), [4], 10.1080/00288306.2022.2089171
6	Hikurangi Subduction Zone Nested-Tomography Precise Relocation Catalogue (Aziz Zanjani et al. 2021)	2012–2016	background	on request, [8], 10.1093/gji/ggab294
7	ISC-EHB Bulletin (New Zealand coverage)	1964–2021	compilation	open (DOI), [154], 10.31905/PY08W6S3
8	ISC-GEM Global Instrumental Earthquake Catalogue (New Zealand coverage)	1904–2014	large-events-only	open (DOI), [30], 10.31905/D808B825
9	Taupo Volcanic Zone Consistent High-Precision Earthquake Catalogue (2007-2023)	2007–2023	background	open (DOI), [69], 10.26443/seismica.v4i1.1490
10	Wellington Region Double-Difference Relocated Catalogue (Du et al. 2004)	1990–2001	background	not avail., [36], 10.1111/j.1365-246X.2004.02366.x
11	Central Alpine Fault Relocated Microseismicity Catalogue (Michailos 2008-2017 / SAMBA)	2008–2017	background	open (DOI), [93], 10.5281/zenodo.3529755
12	Central Southern Alps Matched-Filter Microseismicity Catalogue (Michailos 2009-2020)	2009–2020	background	open (DOI), [94], 10.5281/zenodo.15002912
13	ISC Bulletin / Reviewed ISC Bulletin (New Zealand component)	1900–ongoing	compilation	open (DOI), [74], https://doi.org/10.31905/D808B830

Table 2 continued

#	Catalogue	Period	Type	Availability, ref & DOI
14	USGS NEIC Preliminary Determination of Epicenters (PDE) / ANSS Comprehensive Catalog (ComCat) — New Zealand component	1973–ongoing	compilation	open (DOI), [138], https://doi.org/10.5066/F74T6GJC
15	Atlas of Iseoseimal Maps of New Zealand Earthquakes (1st edition, Downes 1995)	1848–1990	compilation	open (DOI), [32], https://www.naturalhazards.govt.nz/assets/Publications-Resources/7-Atlas-of-iseoseimal-maps-of-New-Zealand-earthquakes-compressed-v2.pdf
16	Atlas of Iseoseimal Maps of New Zealand Earthquakes 1843-2003 (2nd edition, Downes & Dowrick 2014)	1843–2003	compilation	on request, [33]
17	A Descriptive Catalogue of New Zealand Earthquakes (Eiby 1968/1973)	pre-1845 (Part I); 1846 (Part II)–1854	historical	open (DOI), [43], 10.1080/00288306.1968.10423671
18	An Annotated List of New Zealand Earthquakes, 1460-1965 (Eiby 1968)	1460–1965	compilation	open (DOI), [44], 10.1080/00288306.1968.10420275
19	GeoNet Felt Classic / Community Modified Mercalli Intensity (CMMI) Dataset	2004–2016	compilation	on request, [61], 10.1785/0220170163
20	GeoNet Aotearoa New Zealand Felt Rapid Dataset	2015–ongoing	compilation	open (DOI), [56], 10.21420/RS7F-VE53
21	GeoNet Felt Detailed Macroseismic Intensity Dataset	2016–ongoing	compilation	on request, [60], 10.5459/bnzsee.50.2.352-362
22	GeoNet Aotearoa New Zealand Shaking Layers Dataset	?–ongoing	compilation	open (DOI), [2], 10.21420/J856-2J84
23	Historical Incidence of Modified Mercalli Intensity in New Zealand (1840-1997)	1840–1997	compilation	open (DOI), [34], 10.5459/bnzsee.36.1.1-24
24	Patterns of Earthquake-Related Mortality in New Zealand, 1840-2017	1840–2017	historical	open (DOI), [5], 10.1177/8755293019878190
25	Magnitudes of New Zealand Earthquakes, 1901-1993 (Dowrick & Rhoades 1998)	1901–1993	large-events-only	open (DOI), [35], 10.5459/bnzsee.31.4.260-280
26	Pre-2010 Historical Seismicity near Christchurch (1869 Christchurch & 1870 Lake Ellesmere)	1869–1870	historical	open (DOI), [31], 10.1080/00288306.2012.690767
27	1987 Edgecumbe Earthquake Sequence (ML 6.3, Bay of Plenty)	1987-02–1987-03	aftershock	on request, [123], 10.1080/00288306.1989.10421386
28	2003 Mw 7.2 Fiordland (Secretary Island) Aftershock Relocations	2003-08–unknown — not documented	aftershock	on request, [86], 10.1111/j.1365-246X.2007.03336.x
29	2010 Darfield (Canterbury) Mw 7.1 High-Resolution Aftershock Relocations	2010-09–2011	aftershock	on request, [130], 10.1002/jgrb.50301
30	2011 Christchurch (Mw 6.2) Fine-Scale Aftershock Relocations	2011-02–unknown — not documented	aftershock	on request, [11], 10.1785/gssr1.82.6.839
31	Pegasus Bay Aftershock Sequence (2011-2012, Canterbury)	2011-12–2012	aftershock	on request, [108], 10.1093/gji/ggt222
32	2015 ML 6 Wanaka Aftershock Matched-Filter Catalogue	2015-05–2015-05	aftershock	on request, [148], 10.1785/0220170016

Table 2 continued

#	Catalogue	Period	Type	Availability, ref & DOI
33	2016 Kaikoura (Mw 7.8) Aftershock Relocations (Lanza et al. 2019, STREWN + GeoNet)	2016-11 – 2017-05	aftershock	on request, [82], 10.1029/2019GL082780
34	2016 Kaikoura 10-Year Matched-Filter Seismicity Catalogue (Chamberlain et al. 2021)	2009 – 2020	background	on request, [25], 10.1029/2021JB022304
35	Southern Cook Strait / Seddon-Lake Grassmere 2013 Earthquake Response Deployment	2013-09 – 2014-01	aftershock	on request, [114]
36	Kermadec 2021 Refined Earthquake Catalogue and Slip Models (Lythgoe et al.)	2021-03 – 2021	aftershock	open (DOI), [85], 10.7488/ds/7534
37	2016 Mw 7.1 Te Araroa (East Cape) Matched-Filter Foreshock & Aftershock Catalogue (Warren-Smith et al. 2018)	2016-08 – 2016-11	aftershock	on request, [151]
38	1994 Mw 6.7 Arthur’s Pass Earthquake Finely Relocated Aftershock Catalogue (Bannister et al. 2006)	1994-06 – unknown — not documented	aftershock	on request, [10]
39	2009 Mw 7.8 Dusky Sound (Fiordland/Puysegur) Aftershock Deployment Relocations	2009-07 – unknown — not documented (aftershock deployment ran from 17 July 2009; broader host catalogue spans 2001-2011)	aftershock	on request, [40]
40	SAPSE — Southern Alps Passive Seismic Experiment (1995-96)	1995-11 – 1996-04	background	open (DOI), [83], 10.7914/SN/XC_1995
41	SAMBA — Southern Alps Microearthquake Borehole Array (network deployment)	2008 – ongoing	background	open (DOI), [143], 10.7914/SN/9F_2008
42	DFDP-2 Real-Time / Matched-Filter Earthquake Catalogue, Whataroa (2014-2015)	2014-08 – 2015-01	background	on request, [24]
43	DFDP10 / Whataroa-Wanganui Passive Seismology Experiment 2010 (FDSN XO)	2010 – 2010	background	open (DOI), [145], 10.7914/SN/XO_2010
44	ALFA08 / ALFA09 Alpine Fault Arrays (FDSN YR)	2008 – 2010	background	open (DOI), [54], 10.7914/SN/YR_2008
45	WIZARD Array — Wisconsin, New Zealand And Rensselaer Deployment (FDSN ZT / 6F)	2012 – 2020	background	open (DOI), [136], 10.7914/SN/ZT_2012
46	DWARFS — Dense Westland Arrays Researching Fault Segmentation (FDSN 7S/8N)	2019 – 2021	background	open (DOI), [58], 10.7914/SN/8N_2019
47	SALSA — Southern Alps Long Skinny Array (FDSN ZX)	2021 – 2023	background	open (DOI), [146], 10.7914/SN/ZX_2021
48	SOSA — Southland Otago Seismic Array Catalogue (2022-2023)	2022-10 – 2023-10	background	open (DOI), [157], 10.7914/jr68-qq17

Table 2 continued

#	Catalogue	Period	Type	Availability, ref & DOI
49	Otago Temporary Broadband Network (FDSN 6K, 2014–2015)	2014-03 – 2015-04	background	open (DOI), [55], 10.7914/SN/6K_2014
50	Marlborough Fault Zone Anisotropy/Converted-Phase Array (FDSN XB, 2000–2002)	2000-12 – 2002-05	background	open (DOI), [76], 10.7914/SN/XB_2000
51	SAHKE — Seismic Array HiKurangi Experiment (Wellington Geophysical Transect, passive)	2009-11 – 2010-04	background	open (DOI), [62], 10.7914/SN/X2_2009
52	CNIPSE / NIGHT Passive — Central North Island Passive Seismic Experiment (FDSN XQ)	2001-01 – 2001-06	background	open (DOI), [103], 10.7914/SN/XQ_2001
53	HADES — Deep Geothermal HADES Seismic Array (Taupo, FDSN Z8)	2009 – 2011	background	open (DOI), [9], 10.7914/SN/Z8_2009
54	Taranaki Dense Temporary Network Crustal Seismicity Catalogue (Sherburn & White 2005)	2001-12 – 2002-09	background	on request, [121]
55	Tongariro Volcanic Centre Local Earthquake Tomography Catalogue (Rowlands et al. 2005)	2001-01 – 2001-06	background	on request, [113]
56	Local Earthquake and Focal Mechanism Catalogue for the Northern Hikurangi Margin, December 2017–October 2018 (NZ3D onshore network)	2017-12 – 2018-10	background	open (DOI), [159], 10.5281/zenodo.17063327
57	Southern Lakes / Northern Fiordland High-Resolution Microseismicity Catalogue (Central Otago Seismic Array, COSA)	2012-06 – 2013-03	background	on request, [150]
58	Dunedin Region Short-Term Broadband Array Microseismicity Catalogue (Akatore Fault)	? – unknown — not documented	background	on request, [132]
59	Hawke’s Bay Region Microearthquake Catalogue (Bannister 1988; subducting Pacific plate fine structure)	? – unknown — not documented (two-month deployment)	background	not avail., [14]
60	HOBITSS Microearthquake Catalogue (2014–2015, northern Hikurangi, manual)	2014-06 – 2015-05	background	open (DOI), [161], 10.5281/zenodo.2022405
61	HOBITSS Automatic-Detection Earthquake Catalogue (2014 SSE, REST detector)	2014-09 – 2014-10	background	open (DOI), [163], 10.5281/zenodo.7274399
62	Brothers Volcano Ocean-Bottom Hydrophone Seismicity Catalogue (south Kermadec arc)	2004 – 2005	background	not avail., [37]
63	Monowai Submarine Volcano Long-Range Hydroacoustic (T-phase) Activity Catalogue	2003-07 – 2017-01	induced	open (DOI), [89], 10.1029/2018JB015888
64	Dataset of Earthquakes in the Northern Hikurangi Subduction Zone During the 2019 Slow Slip Event (2018–2019 OBS deployment)	2018 – 2019	background	open (DOI), [78], 10.5281/zenodo.15713529

Table 2 continued

#	Catalogue	Period	Type	Availability, ref & DOI
65	Okataina Volcanic Centre Double-Difference Relocated Catalogue (2010-2021)	2010–2021	background	on request, [13], 10.1016/j.jvolgeores.2022.107653
66	Taupo Volcano Magmatic-Seismicity Catalogue (2019-2022 ECLIPSE temporary deployment)	2019–2022	background	on request, [87], 10.1029/2025JB031644
67	Taupo 2019 Volcanic Unrest Earthquake Catalogue (Illsley-Kemp et al. 2021)	2019–2019	background	on request, [70], 10.1029/2021GC009803
68	Taupo 2022-2023 Volcanic Unrest Seismicity Catalogue (Lamb et al. 2024)	2022-05–2023-05	background	open (DOI), [81], 10.26443/seismica.v3i2.1125
69	Cascading Earthquake Swarms, Northern Taupo Volcanic Zone (2019, Aber et al. 2025)	2019-03–2019-09	background	on request, [6], 10.1029/2024GC012079
70	Tarawera 2019 Dyke-Intrusion Earthquake Catalogue (Puhipuhi, Okataina; Benson et al. 2021)	2019-03–2019-03	background	open (DOI), [18], 10.5281/zenodo.4035171
71	Waimangu-Rotomahana-Mt Tarawera Geothermal Field Earthquake Swarm Catalogue (2004-2015)	2004–2015	background	not avail., [12], 10.1016/j.jvolgeores.2015.07.024
72	Mt Ruapehu Earthquake Swarm Catalogue (1990-2023)	1990–2023	background	open (DOI), [96], 10.3389/feart.2024.1343874
73	Whakaari/White Island Very-Long-Period (VLP) Earthquake Catalogue (2007-2019)	2007–2019	background	on request, [99], 10.1186/s40623-020-01224-z
74	Ngatamariki Geothermal Injection-Induced Microseismicity Catalogue (2012-2015)	2012–2015	induced	on request, [63], 10.1029/2019GC008243
75	Rotokawa Geothermal Field Induced Microseismicity Catalogue (2012-2015)	2012–2015	induced	on request, [64], 10.1016/j.geothermics.2019.101750
76	Rotokawa & Ngatamariki Combined Matched-Filter Catalogue (2012-2015, anisotropy)	2012–2015	induced	on request, [97], 10.1093/gji/ggz400
77	Rotorua and Kawerau Geothermal Systems Relocated Seismicity Catalogue (1984-2004; Clarke et al. 2009)	1984–2004	background	on request, [28], 10.1016/j.jvolgeores.2008.11.004
78	Rotokawa Geothermal Field Microseismicity Catalogue 2008-2012 (Sherburn et al. 2015)	2008–2012	induced	open (DOI), [122], 10.1016/j.geothermics.2014.11.001
79	Wairakei Geothermal Field Microseismicity Catalogue (Wairakei Seismic Network; Sepulveda et al. 2015)	2009-03–ongoing	induced	on request, [116]
80	Mt Ruapehu 2022 Volcanic Unrest Drumbeat / Tremor Seismicity Catalogue (Bramwell et al. 2025)	2022-03–2022-07	background	open (DOI), [22], 10.5281/zenodo.13910455
81	Auckland Volcanic Field Matched-Filter Earthquake Catalogue (van Wijk et al. 2021, COVID-19 lockdown study)	2019-11–2020-06	background	on request, [142], 10.5194/se-12-363-2021

Table 2 continued

#	Catalogue	Period	Type	Availability, ref & DOI
82	GeoNet New Zealand Earthquake Moment Tensor Solutions (MwNZ)	2003-08 – ongoing	background	open (DOI), [53], 10.21420/MMJ9-CZ67
83	Hikurangi Slow Slip Event Inventory (Wallace & Beavan 2010)	2002 – 2010	compilation	on request, [147], 10.1029/2010JB007717
84	14-Year GNSS Slow Slip Event Catalogue for the Hikurangi Subduction Zone (Michel et al. 2025)	2009 – 2023	compilation	on request, [95], 10.1029/2024JB030865
85	Shallow Hikurangi Slow Slip Event Catalogue (Perez-Silva et al. 2025)	2006 – 2024	compilation	open (DOI), [100], 10.5281/zenodo.14848128
86	Northern Hikurangi Tectonic Tremor Catalogue 2010-2015 (Todd & Schwartz 2016)	2010 – 2015	compilation	on request, [131], 10.1002/2016JB013480
87	Ambient Tectonic Tremor Catalogue: Manawatu, Cape Turnagain, Marlborough & Puysegur (Romanet & Ide 2019)	2005 – 2016	compilation	on request, [112], 10.1186/s40623-019-1039-1
88	Alpine Fault Low-Frequency Earthquake Catalogue (Chamberlain et al. 2014)	2009 – 2012	background	on request, [23], 10.1002/2014GC005436
89	Alpine Fault Low-Frequency Earthquake Catalogue with Focal Mechanisms (Baratin et al. 2018)	2009 – 2017	background	on request, [15], 10.1016/j.epsl.2017.12.021
90	Kaimanawa (Deep Hikurangi) Low-Frequency Earthquake Catalogue (Aden-Antoniow et al. 2024)	2010 – 2022	background	on request, [7], 10.1029/2023JB027971
91	Raukumara Peninsula Repeating Earthquake Catalogue 2003-2020 (Hughes et al. 2021)	2003 – 2020	background	on request, [65], 10.1029/2021GC009670
92	New Zealand-Wide Earthquake Focal Mechanism Catalogue for Tectonic Stress (Townend et al. 2012)	2004-01 – 2011-02	compilation	on request, [133], 10.1016/j.epsl.2012.08.003
93	Southeastern South Island / Otago Earthquake Focal Mechanism Catalogue (Williams et al. 2026)	? – unknown — not documented	compilation	open (DOI), [156], 10.5281/zenodo.17228270
94	HOBITSS Ocean-Bottom Focal-Mechanism / Stress-Tensor Cycling Catalogue, Northern Hikurangi (Warren-Smith et al. 2019)	2014-05 – 2015-06	background	on request, [152], 10.1038/s41561-019-0367-x
95	Manawatu Deep Short-Term Slow-Slip and Tremor Catalogue (Fasola et al. 2023)	2005 – 2016	background	on request, [48], 10.1029/2023GL105428
96	Global Centroid Moment Tensor (GCMT) catalogue (New Zealand component)	1976 – ongoing	compilation	open (DOI), [38], https://doi.org/10.1016/j.pepi.2012.04.002
97	NSHM 2010 Earthquake Catalogue (Stirling et al. 2012)	1840 – 2009	background	on request, [125]
98	Mw-Homogenised NZ Earthquake Catalogue for NSHM 2022 (Christophersen et al. 2022)	1931 – 2020	compilation	open (DOI), [26], 10.21420/A2SN-XM76

Table 2 continued

#	Catalogue	Period	Type	Availability, ref & DOI
99	Integrated / Augmented Earthquake Catalogue for Aotearoa New Zealand (NSHM 2022; Rollins et al.)	1840–2020	compilation	open (DOI), [109], 10.21420/XT4Y-WY45
100	NSHM 2022 Seismicity Rate Model Distributed-Seismicity / Declustered Catalogue (Gerstenberger et al. 2024)	1840–2020	background	open (DOI), [51], 10.1785/0120230165
101	RSQSim Synthetic Earthquake Catalogue for New Zealand (NSHM2012 fault system, 276 kyr)	synthetic (276,000-year simulated span; published 2021)–synthetic (276,000-year simulated span)	background	open (DOI), [117], 10.5281/zenodo.5534462
102	RSQSim Synthetic Earthquake Catalogue + Ground Motions for New Zealand (NZNSHM22 fault system)	synthetic (RSQSim simulated span; published 2025)–synthetic (RSQSim simulated span)	background	open (DOI), [118], 10.5281/zenodo.14532399
103	Hikurangi Trench Earthquake-Swarm Catalogue (ETAS-based, Nishikawa et al. 2021)	1997–2015	compilation	[98], 10.1029/2020JB020618
104	EEPAS Lag-Compensated Earthquake Forecast for Central New Zealand (Rhoades et al. 2022)	2019–2030	compilation	on request, [105], 10.3390/geosciences12090349
105	New Zealand Earthquake Forecast Testing Centre (CSEP-NZ) Authorized Test Catalogue	2008–ongoing	compilation	on request, [50]
106	New Zealand Strong Motion Database (NZSMDB)	1968–ongoing	compilation	open (DOI), [140], https://doi.org/10.5459/bnzsee.50.1.1-20
107	New Zealand Ground-Motion Database (NZGMDB)	2000–2022-12	compilation	open (DOI), [67], 10.1785/0120230184 (2023versionpaper,BSSA);datasethostedonOSFat https://osf.io/q9yrg/ .Predecessor:GNSScience Report2021/56,DOI10.21420/Z20E-5507

6 Coverage gaps and FAIR weaknesses

- **Not a census.** Because earthquake-catalogue production is decentralised and continuous, a residual long tail remains: likely-missing material includes further PhD-thesis catalogues, additional dedicated catalogues for New Zealand $M_w \geq 6$ aftershock responses since 1960, more geothermal fields (Ohaaki, Mokai, Ngāwhā), and the SISIE/MOANA/SHIRE offshore experiments not yet retained as distinct event catalogues.
- **Discoverability is the core problem.** Temporary-deployment and regional catalogues live across journal supplements, Zenodo, IRIS/EarthScope, and FDSN network DOIs—the fragmentation a registry exists to fix.

- **Heterogeneity.** Magnitude scales vary (M_L ; M_w via the GeoNet moment-tensor catalogue from 2003; the homogenised $M_{LNZ20} \rightarrow M_w$ used for NSHM 2022); the magnitude of completeness M_c is rarely stated and is spatially variable; depths span 0–~600 km for subduction seismicity, while many regional catalogues are shallow-crustal only.
- **A few entries are not conventional event lists** (e.g. the National Seismograph Network *waveform* dataset; the `nz3drx` 3-D *location method* carried within the GeoNet catalogue DOI). They are included for completeness and flagged in their records.
- **A few entries are provisional** (e.g. the 2013 Seddon / Lake Grassmere and 2009 Dusky Sound response deployments, and two forecast / test-centre catalogues), included pending a clearly attributable standalone dataset.

7 The inventory

Each entry follows the submission-profile schema (Appendix A). The italic header line gives *catalogue type* · *focus* · *detection method* · *review status*. References are cited to the bibliography at the end of the report.

7.1 National operational and network catalogues

The authoritative national record and the permanent-network data products that underpin it.

1. GeoNet Aotearoa New Zealand Earthquake Catalogue

background · *geographical* · *automatic* · *partially reviewed*

- **Also known as:** GeoNet earthquake catalogue; NZ Earthquake Catalogue; QuakeSearch; FDSN network NZ catalogue product; `nz3drx` 3-D NonLinLoc locations; NEID catalogue component
- **Coverage:** 1460 — ongoing
- **Region:** Aotearoa New Zealand (national), incl. Hikurangi margin, Kermadec, Southern Alps, offshore regions
- **Bounding box:** New Zealand and offshore margins, approx 34S–48S, 165E–180E/–175W (incl. Kermadec and Hikurangi margins and offshore regions)
- **Depth range:** 0–~600 km (subduction-zone seismicity to deep events)
- **Magnitude:** Roughly M0 to M8+ instrumental; magnitude of completeness varies spatially and over time (network-dependent), M_c not stated on landing page
- **Events:** ~700,000+ (approx; ~20,000 new events added per year)
- **Content:** hypocentral locations; phase arrival-time picks; amplitude picks; quality information; historical observations
- **Producer / contact:** GNS Science (Earth Sciences New Zealand) / GeoNet; Geohazards Information Manager (info@geonet.org.nz)
- **Data source:** Continuous waveforms from the New Zealand National Seismograph Network (FDSN code NZ) plus historical felt/observational records; routinely processed by GeoNet (SeisComP since 2012). Successor to the DSIR Seismological Observatory / IGNS national catalogue lineage.
- **Availability:** available (DOI) · DOI [10.21420/OS8P-TZ38](https://doi.org/10.21420/OS8P-TZ38)

The authoritative national operational earthquake catalogue for Aotearoa New Zealand, produced by GNS Science/GeoNet to provide a continuous public record of NZ seismicity for monitoring, research, and hazard assessment. It spans historical (descriptive) events from ~1460 and instrumental locations from the 1930s to the present, and is the parent dataset for nearly every derived/relocated/hazard NZ catalogue. Location methods evolved over time (LOCAL+`nz1d` pre-1987; CUSP/GROPE+`nz1dr` ~1987-2012; SeisComP from 2012 with NonLinLoc '`nz3drx`' 3-D and LOCSAT/`iasp91`, LOCSAT routine since the New Generation Magnitude Calibration Dec 2018). The 3-D NonLinLoc '`nz3drx`' is a location method within this single DOI, not a separate catalogue.

Key references: [52, 153].

2. New Zealand National Seismograph Network (NZNSN) seismic waveform dataset

background · geographical · automatic · non-reviewed

- **Also known as:** NZ; NZNSN; FDSN network NZ; Aotearoa/New Zealand GeoNet Seismic Digital Waveform Dataset
- **Coverage:** 1916 — ongoing
- **Region:** Aotearoa New Zealand (national)
- **Bounding box:** Mainland New Zealand plus Raoul Island and Chatham Islands
- **Magnitude:** Not applicable (waveform dataset, not an event list)
- **Content:** other
- **Producer / contact:** GNS Science (Earth Sciences New Zealand) / GeoNet
- **Data source:** Permanent national network of broadband, short-period and strong-motion sensors (FDSN code NZ), covering mainland NZ plus Raoul and Chatham Islands; continuous miniSEED waveform recordings.
- **Availability:** available (DOI) · DOI [10.21420/G19Y-9D40](https://doi.org/10.21420/G19Y-9D40)

The permanent national seismograph network (FDSN code NZ) whose continuous waveforms are the instrumental basis for the GeoNet earthquake catalogue and moment tensors. This is a waveform dataset (network record) rather than an event list, but it is the foundational instrumental data product and is registered with its own network/dataset DOI distinct from the catalogue DOI. The permanent network dates from 1916 (Wellington), with continuous digital miniSEED archived from September 1986.

Key references: [59].

3. National Earthquake Information Database (NEID)

compilation · geographical · other · partially reviewed

- **Also known as:** NEID
- **Coverage:** unknown — not documented — ongoing
- **Region:** Aotearoa New Zealand (national)
- **Bounding box:** Aotearoa New Zealand (national)
- **Magnitude:** Not applicable (umbrella database, not a single event list)
- **Content:** hypocentral locations; phase arrival-time picks; focal mechanisms; fault-geometry; historical observations; quality information; other
- **Producer / contact:** Earth Sciences New Zealand (formerly GNS Science) / GeoNet; Paul Viskovic (Geomodeller)
- **Data source:** Umbrella database aggregating GeoNet/GNS raw and derived earthquake data products across seven interlinked datasets.
- **Availability:** available (DOI)

A nationally significant umbrella database, managed by Earth Sciences New Zealand (formerly GNS Science), that provides raw and derived earthquake data for New Zealand to support geohazards research and response. It is the institutional custodian/compilation rather than a single event list: it comprises ~seven interlinked components (earthquake catalogue, moment tensors, waveform archive, station metadata, felt reports, strong-motion data products, and fault-rupture models). It is kept distinct from the GeoNet catalogue, which is one of its components.

Key references: [1].

4. Computer File of New Zealand Earthquakes (DSIR Seismological Observatory)

compilation · geographical · manual · reviewed

- **Also known as:** Smith (1976) computer file; DSIR Seismological Observatory master file
- **Coverage:** 1460 — 1974
- **Region:** Aotearoa New Zealand (national)
- **Bounding box:** New Zealand region

- **Events:** >13,000 origins
- **Content:** hypocentral locations; historical observations
- **Producer / contact:** W. D. Smith, DSIR Seismological Observatory, Wellington, New Zealand
- **Data source:** Master file assembled mid-1970s by the DSIR Seismological Observatory from Observatory bulletins/origin determinations and Eiby descriptive catalogues for pre-instrumental events; stored on magnetic tape for the Burroughs B6700 computer at Victoria University of Wellington.
- **Availability:** not available

The foundational computerised national earthquake catalogue for New Zealand, assembled in the mid-1970s by W. D. Smith at the DSIR Seismological Observatory to satisfy recurrent requests for earthquake lists selected by geography and other criteria. It contains over 13,000 earthquake origins and is the direct institutional ancestor of the modern GeoNet/IGNS national catalogue. It draws on Eiby’s descriptive catalogues for pre-instrumental (historical) events. It is a legacy magnetic-tape dataset and is not publicly available as such.

Key references: [124].

7.2 Relocated and reprocessed catalogues

Catalogues that re-pick, relocate, or re-time existing events with improved velocity models or double-difference / matched-filter methods, including the New Zealand components of global relocation efforts.

5. Eberhart-Phillips & Reyners 2001–2011 NZ-Wide 3-D Relocated Catalogue

background · geographical · manual · reviewed

- **Also known as:** NZ-Wide 3-D relocated catalogue 2001–2011
- **Coverage:** 2001 — 2011
- **Region:** New Zealand (nationwide); southern South Island comparison
- **Bounding box:** New Zealand (nationwide), approx. 34S–48S, 166E–179E, plus southern South Island comparison region
- **Content:** hypocentral locations; relative relocations; quality information
- **Producer / contact:** Donna Eberhart-Phillips & Martin Reyners, GNS Science (Te Pū Ao)
- **Data source:** GeoNet / New Zealand National Seismograph Network catalogue earthquakes (2001–2011), relocated using the NZ-Wide 3-D seismic velocity model (Eberhart-Phillips et al. NZ-Wide v2.2/v2.3); includes comparison with 2019–2020 auto-detected earthquakes in the southern South Island.
- **Availability:** available (DOI) · DOI [10.1080/00288306.2022.2089171](https://doi.org/10.1080/00288306.2022.2089171)

Produced to improve hypocentral accuracy over routine 1-D GeoNet locations by relocating the 2001–2011 national earthquake catalogue within the NZ-Wide 3-D velocity model, giving locations consistent with the velocity structure for use in seismic-hazard and tectonic studies. A secondary aim was to assess detection/location performance in the sparsely instrumented southern South Island by comparing relocated events with 2019–2020 auto-detected seismicity. A derived child of the GeoNet catalogue.

Key references: [4, 41, 42].

6. Hikurangi Subduction Zone Nested-Tomography Precise Relocation Catalogue (Aziz Zanjani et al. 2021)

background · geographical · manual · reviewed

- **Coverage:** 2012 — 2016
- **Region:** Hikurangi margin / Hikurangi subduction zone
- **Bounding box:** Hikurangi subduction zone, North Island and northern South Island, New Zealand
- **Depth range:** 0–250
- **Magnitude:** Approx. M2–5 (local/regional events used in relocation)
- **Events:** 57233 relocated (of 61654 catalogued); 43025 (75%) further refined with cross-correlation
- **Content:** hypocentral locations; relative relocations; phase arrival-time picks; other

- **Producer / contact:** Farzaneh Aziz Zanjani (University of Miami) with Guoqing Lin (Univ. Miami) & Clifford H. Thurber (Univ. Wisconsin-Madison)
- **Data source:** GeoNet local/regional P-wave arrival times (2012-2016) combined with ISC-EHB teleseismic P-wave data (1964-2016); relocated with teletomoDD nested regional-global double-difference tomography plus waveform cross-correlation differential times.
- **Availability:** available on request · DOI [10.1093/gji/ggab294](https://doi.org/10.1093/gji/ggab294)

Produced to obtain precise hypocentres and a nested regional-global velocity model along the Hikurangi subduction zone, sharpening imaging of the subducting slab and the double seismic zone. The teletomoDD inversion relocated the 2012-2016 GeoNet seismicity, with a large subset further refined using waveform cross-correlation to reduce relative location errors. A derived child of the GeoNet catalogue augmented by ISC-EHB teleseismic data.

Key references: [8].

7. ISC-EHB Bulletin (New Zealand coverage)

compilation · geographical · manual · reviewed

- **Also known as:** EHB; Engdahl-van der Hilst-Buland Bulletin; ISC-EHB
- **Coverage:** 1964 — 2021
- **Region:** New Zealand / Kermadec / Hikurangi subset of global coverage
- **Bounding box:** Global; NZ/Kermadec/Hikurangi subset approx. 25S-55S, 160E-170W
- **Magnitude:** Teleseismically well-constrained events (typically ~M4.5+); magnitudes from ISC scheme
- **Events:** ~208,480 events globally (NZ/Kermadec/Hikurangi subset not separately reported)
- **Content:** hypocentral locations; phase arrival-time picks; quality information; other
- **Producer / contact:** International Seismological Centre (ISC), Thatcham, UK (E.R. Engdahl; D. Di Giacomo; D.A. Storchak)
- **Data source:** Groomed subset of the ISC Bulletin: teleseismically well-recorded events relocated with the EHB algorithm using primary and depth-phase arrival times; NZ/Kermadec/Hikurangi events are the qualifying subset.
- **Availability:** available (DOI) · DOI [10.31905/PY08W6S3](https://doi.org/10.31905/PY08W6S3)

A global, internally consistent 'groomed' dataset of teleseismically well-constrained earthquakes (1964-2021; the 1964-2008 portion rebuilt for this release) relocated with the EHB procedure to minimise location and especially depth bias. The qualifying New Zealand / Kermadec / Hikurangi subset provides homogeneous high-quality teleseismic hypocentres and depths suitable for slab and structure studies. It is a global product, not a standalone NZ catalogue; the NZ events are a regional subset.

Key references: [47, 73, 154].

8. ISC-GEM Global Instrumental Earthquake Catalogue (New Zealand coverage)

large-events-only · geographical · manual · reviewed

- **Also known as:** ISC-GEM
- **Coverage:** 1904 — 2014
- **Region:** New Zealand / Kermadec / Hikurangi subset of global coverage
- **Bounding box:** Global; NZ/Kermadec/Hikurangi subset approx. 25S-55S, 160E-170W
- **Magnitude:** Large instrumental events: $M_w \geq 7.5$ (pre-1918), ≥ 6.25 (1918-1959), ≥ 5.5 (1960 onward); recomputed M_w
- **Events:** NZ subset not separately reported (global catalogue, large-events-only)
- **Content:** hypocentral locations; quality information; other
- **Producer / contact:** International Seismological Centre (ISC) / Global Earthquake Model (GEM) Foundation (D. Di Giacomo; E.R. Engdahl; D.A. Storchak)
- **Data source:** Reprocessed historical/instrumental station parametric data (1904 onward); hypocentres relocated and M_w recomputed under consistent methodology; NZ/Kermadec/Hikurangi large events are the qualifying subset.

- **Availability:** available (DOI) · DOI [10.31905/D808B825](https://doi.org/10.31905/D808B825)

A homogeneous global catalogue of large instrumental earthquakes (1904–2014, extended toward 2018+) with relocated hypocentres and recomputed moment magnitudes (M_w), selected by time-dependent magnitude thresholds. The qualifying large New Zealand / Kermadec / Hikurangi events provide a long-period, large-event source/benchmark used in NZ seismic-hazard work (e.g. NSHM 2022). It is a global product, not NZ-specific; NZ events are a subset.

Key references: [30, 72, 128].

9. Taupo Volcanic Zone Consistent High-Precision Earthquake Catalogue (2007–2023)

background · geographical · machine-learning · reviewed

- **Also known as:** TVZ catalogue; Illsley-Kemp & Mestel 2025
- **Coverage:** 2007 — 2023
- **Region:** Taupō Volcanic Zone (TVZ)
- **Bounding box:** Taupō Volcanic Zone, central North Island, New Zealand (~300 km long zone)
- **Events:** 86579 events (~65% high-precision relocated)
- **Content:** hypocentral locations; relative relocations; phase arrival-time picks; quality information
- **Producer / contact:** Finnigan Illsley-Kemp & Eleanor Mestel, Te Herenga Waka - Victoria University of Wellington
- **Data source:** GeoNet permanent network waveforms integrated with temporary TVZ deployments (incl. HADES array, Bannister 2009; back-arc rifting arrays); EQTransformer ML phase picking, 3-D velocity model location, GrowClust3D relative relocation.
- **Availability:** available (DOI) · DOI [10.26443/seismica.v4i1.1490](https://doi.org/10.26443/seismica.v4i1.1490)

Produced to create a single internally consistent, high-precision earthquake catalogue spanning the whole ~300 km Taupō Volcanic Zone over 2007–2023, overcoming the heterogeneity of routine locations across many separate deployments. Uses EQTransformer machine-learning phase picking, location in a 3-D velocity model, and GrowClust3D relative relocation. A flagship ML + 3-D reprocessing effort merging permanent GeoNet data with temporary TVZ arrays.

Key references: [69].

10. Wellington Region Double-Difference Relocated Catalogue (Du et al. 2004)

background · geographical · manual · reviewed

- **Also known as:** Du et al. 2004 Wellington hypoDD catalogue
- **Coverage:** 1990 — 2001
- **Region:** Wellington region / SW Hikurangi margin
- **Bounding box:** Wellington region, southern North Island, New Zealand
- **Magnitude:** Approx. M1.8–6.5
- **Events:** 6592 relocated of 6825 (96.5%)
- **Content:** hypocentral locations; relative relocations; phase arrival-time picks
- **Producer / contact:** Wen-xuan Du & Clifford H. Thurber (Univ. Wisconsin-Madison); with M. Reyners & D. Eberhart-Phillips (GNS Science) and H. Zhang
- **Data source:** Wellington-region seismicity 1990–2001 from the NZ National Seismograph Network plus some temporary stations (47 stations within 200 km; 22 primary control stations); catalogue and waveform-based differential times relocated with hypoDD double-difference.
- **Availability:** not available · DOI [10.1111/j.1365-246X.2004.02366.x](https://doi.org/10.1111/j.1365-246X.2004.02366.x)

An early double-difference (hypoDD) relocation of Wellington-region earthquakes (1990–2001) to obtain precise relative hypocentres and resolve fine seismicity structure. The relocations sharpened both planes of the double seismic zone to <10 km thickness (~20 km apart) and revealed intraslab faulting at the SW end of the Hikurangi margin. A derived child of the national catalogue using permanent NZNSN plus some temporary stations.

Key references: [36].

11. Central Alpine Fault Relocated Microseismicity Catalogue (Michailos 2008-2017 / SAMBA)

background · geographical · manual · reviewed

- **Also known as:** SAMBA microseismicity catalogue; Michailos et al. 2019
- **Coverage:** 2008 — 2017
- **Region:** Central Southern Alps / Alpine Fault (Ka Tiritiri o te Moana)
- **Bounding box:** Central Southern Alps / Alpine Fault, South Island, New Zealand
- **Magnitude:** Microseismicity; local magnitudes provided per event
- **Events:** 8137 earthquakes; 7719 high-precision relocated hypocentres
- **Content:** hypocentral locations; relative relocations; phase arrival-time picks; quality information
- **Producer / contact:** Konstantinos Michailos with Euan G.C. Smith, Calum J. Chamberlain, Martha K. Savage & John Townend, Victoria University of Wellington
- **Data source:** GeoNet permanent stations plus temporary Southern Alps arrays (SAMBA, ALFA08/09, WIZARD, DFDP); local-earthquake locations relocated with hypoDD double-difference; QuakeML catalogue deposited on Zenodo.
- **Availability:** available (DOI) · DOI [10.5281/zenodo.3529755](https://doi.org/10.5281/zenodo.3529755)

A decade-long (2008-2017) double-difference (hypoDD) relocated microseismicity catalogue of the central Southern Alps / Alpine Fault, produced to map along-strike variations in seismogenic thickness and the fault's microseismic behaviour. Built by combining permanent GeoNet data with multiple temporary Southern Alps deployments. Predecessor to the 2025 matched-filter catalogue.

Key references: [90, 93].

12. Central Southern Alps Matched-Filter Microseismicity Catalogue (Michailos 2009-2020)

background · geographical · template-matching · reviewed

- **Also known as:** SAMBA matched-filter catalogue; Southern Alps/Ka Tiritiri o te Moana matched-filter catalog
- **Coverage:** 2009 — 2020
- **Region:** Central Southern Alps / Alpine Fault (Ka Tiritiri o te Moana)
- **Bounding box:** Central Southern Alps / Alpine Fault, South Island, New Zealand
- **Magnitude:** Microseismicity; local magnitudes provided per event
- **Content:** hypocentral locations; relative relocations; phase arrival-time picks; quality information; other
- **Producer / contact:** Konstantinos Michailos with Calum J. Chamberlain, Simon C. Cox, John Townend et al. (Victoria University of Wellington / Australian National University)
- **Data source:** Templates from the 2008-2017 SAMBA hypoDD catalogue applied via matched-filter (template-matching) detection to continuous GeoNet + Southern Alps array waveforms (2009-2020); QuakeML + CSV catalogue on Zenodo.
- **Availability:** available (DOI) · DOI [10.5281/zenodo.15002912](https://doi.org/10.5281/zenodo.15002912)

A template-matching (matched-filter) enhanced microseismicity catalogue for the central Southern Alps / Alpine Fault (2009-2020) that substantially densifies the 2019 SAMBA catalogue with many additional small events. Produced to study temporal seismicity variations, including rainfall/snowmelt-triggered seismicity. A distinct child of the Michailos 2008-2017 catalogue rather than a duplicate.

Key references: [91, 94].

13. ISC Bulletin / Reviewed ISC Bulletin (New Zealand component)

compilation · geographical · manual · reviewed

- **Also known as:** International Seismological Centre On-line Bulletin; ISC Reviewed Bulletin; reporting agency code ISC (NZ contributing agencies include GNS/WEL and historically DSIR)
- **Coverage:** 1900 — ongoing
- **Region:** Global; New Zealand component covers onshore NZ, Hikurangi-Kermadec subduction margin, and subantarctic region
- **Bounding box:** Global (NZ component covers approx. -50 to -33 lat, 165 to -176 lon, including Hikurangi/Kermadec margin and subantarctic region)
- **Depth range:** 0 to ~700 km (full crustal-to-deep subduction range; global product)
- **Magnitude:** Global product; Reviewed Bulletin applies manual review to all events with $M \geq 3.5$ (M = maximum reported network magnitude), and selectively for $2.5 \leq M < 3.5$. No single M_c defined for the NZ component (M_c varies with era and reporting completeness).
- **Content:** hypocentral locations, phase arrival-time picks, amplitude picks, focal mechanisms, quality information, other
- **Producer / contact:** International Seismological Centre (ISC), Thatcham, United Kingdom (Director: D.A. Storchak). NZ data contributed primarily by GNS Science / GeoNet (and historically DSIR Geophysics Division).
- **Data source:** Parametric earthquake data (hypocentres, phase arrival-time picks, amplitude/magnitude readings, collected focal-mechanism solutions) reported to the ISC by ~150 seismological agencies and networks worldwide, grouped and relocated by the ISC. The NZ component derives mainly from GNS Science / GeoNet (and historically DSIR) station and bulletin contributions, supplemented by globally collected phases.
- **Availability:** available (DOI) · DOI <https://doi.org/10.31905/D808B830>

The ISC Bulletin is regarded as the definitive long-term parametric record of global seismicity, compiling hypocentres, phase arrival times, amplitudes and magnitudes, and collected focal-mechanism solutions. The 'Reviewed' product applies manual analyst checking and ISC relocation to all events with $M \geq 3.5$ (and selectively for $2.5 \leq M < 3.5$), producing the authoritative reviewed location/magnitude record for moderate-to-large NZ events. It is distinct from the ISC-EHB and ISC-GEM derivative products and is widely cited for NZ regional/teleseismic locations.

Key references: [19, 74, 126, 127].

14. USGS NEIC Preliminary Determination of Epicenters (PDE) / ANSS Comprehensive Catalog (ComCat) — New Zealand component

compilation · geographical · manual · reviewed

- **Also known as:** USGS NEIC PDE Bulletin; ANSS ComCat; reporting agency code US / NEIC / PDE
- **Coverage:** 1973 — ongoing
- **Region:** Global; New Zealand component covers onshore NZ, Hikurangi-Kermadec subduction margin, and subantarctic region
- **Bounding box:** Global (NZ component covers approx. -50 to -33 lat, 165 to -176 lon)
- **Depth range:** 0 to ~700 km (global product)
- **Magnitude:** Global bulletin; NZ component is dominated by larger/teleseismically detected events (roughly $M \geq 4.5$ -5 regionally complete, decreasing over time). No formally published M_c for the NZ subset.
- **Content:** hypocentral locations, phase arrival-time picks, amplitude picks, focal mechanisms, quality information, other
- **Producer / contact:** U.S. Geological Survey, Earthquake Hazards Program / National Earthquake Information Center (NEIC), Golden, Colorado, USA
- **Data source:** Global earthquake bulletin compiled by the USGS NEIC from worldwide and US network phase/parametric contributions; the PDE is one discrete product within the ANSS Comprehensive Catalog (ComCat), which aggregates ~26 contributing networks. The NZ component is the subset of US/NEIC-located events located within/around New Zealand (predominantly the larger, teleseismically recorded events).
- **Availability:** available (DOI) · DOI <https://doi.org/10.5066/F74T6GJC>

The USGS NEIC PDE is the agency's global earthquake bulletin and a core contributing product within the

ANSS Comprehensive Catalog (ComCat), providing near-global hypocentres, magnitudes, and associated products (moment tensors, ShakeMap, PAGER, DYFI). Its NZ component captures predominantly the larger/teleseismically recorded events along the Hikurangi-Kermadec margin and crustal NZ. It is a distinct global compilation, separate from regional GeoNet/NEID products and from the ISC and GCMT compilations.

Key references: [138, 139].

7.3 Historical and macroseismic (felt-report) catalogues

Pre-instrumental and felt-report catalogues that extend the record through the nineteenth century and quantify shaking via Modified Mercalli / macroseismic intensity.

15. Atlas of Iseismatic Maps of New Zealand Earthquakes (1st edition, Downes 1995)

compilation · geographical · felt-reports · reviewed

- **Also known as:** Downes 1995; IGNS Monograph 11; EQC Paper 7 (EQC 91/40)
- **Coverage:** 1848 — 1990
- **Region:** New Zealand (national)
- **Bounding box:** New Zealand mainland and near-offshore; one event south of the Kermadec Islands
- **Magnitude:** Smallest event ML 3.5 (1963 Mangonui); largest the 1855 Wairarapa earthquake (~Mw 8.0–8.2); intensities in NZ-adapted Modified Mercalli (MM)
- **Events:** 123 earthquakes / 133 isoseismal maps
- **Content:** historical observations; hypocentral locations; quality information; other
- **Producer / contact:** G.L. Downes, Institute of Geological & Nuclear Sciences (IGNS, now GNS Science); funded by EQC (Earthquake Commission, now Natural Hazards Commission Toka Tu Ake)
- **Data source:** Compilation of historical felt/macroseismic observations, newspaper and documentary records, and prior catalogues (incl. Eiby) reassessed into NZ-adapted Modified Mercalli intensities; isoseismal maps drawn for each event.
- **Availability:** available (DOI) · DOI <https://www.naturalhazards.govt.nz/assets/Publications-Resources/7-Atlas-of-iseismatic-maps-of-New-Zealand-earthquakes-compressed-v2.pdf>

Foundational national atlas of isoseismal (felt-intensity) maps for New Zealand earthquakes, produced to systematise macroseismic intensity data for hazard, engineering and historical seismicity studies using the NZ-adapted Modified Mercalli scale. Provides 133 isoseismal maps for the felt distributions of 123 earthquakes (122 in/near the NZ mainland plus one Mw~6.7 event south of the Kermadecs), each with revised hypocentral parameters and a brief descriptive account. Free PDF via the Natural Hazards Commission/EQC; superseded and extended by the 2014 2nd edition (Monograph 25).

Key references: [32].

16. Atlas of Iseismatic Maps of New Zealand Earthquakes 1843–2003 (2nd edition, Downes & Dowrick 2014)

compilation · geographical · felt-reports · reviewed

- **Also known as:** Downes & Dowrick 2014; GNS Science Monograph 25; ISBN 978-0-478-19663-4
- **Coverage:** 1843 — 2003
- **Region:** New Zealand (national)
- **Bounding box:** New Zealand mainland and near-offshore margins
- **Magnitude:** Spans small felt events through the largest historical NZ shocks (incl. 1855 Wairarapa ~Mw 8.0–8.2); intensities in NZ-adapted Modified Mercalli (MM)
- **Content:** historical observations; hypocentral locations; quality information; other
- **Producer / contact:** G.L. Downes & D.J. Dowrick, GNS Science, Lower Hutt
- **Data source:** Revision and extension of the 1995 1st edition (Monograph 11), incorporating reassessed Modified Mercalli intensities, magnitudes and hypocentral locations from additional documentary and instrumental sources.

- **Availability:** available on request

Revised and expanded national isoseismal atlas (797 pp.) covering NZ earthquakes 1843–2003, extending the 1995 1st edition with reassessed MM intensities, magnitudes and locations and additional events/maps. Produced to provide an updated authoritative macroseismic reference for seismic hazard, engineering and historical-seismicity work in New Zealand. A distinct successor edition (GNS Science Monograph 25), originally distributed on CD and available on request rather than as an open download.

Key references: [33].

17. A Descriptive Catalogue of New Zealand Earthquakes (Eiby 1968/1973)

historical · geographical · felt-reports · reviewed

- **Also known as:** Eiby 1968 Part I; Eiby 1973 Part II
- **Coverage:** pre-1845 (Part I); 1846 (Part II) — 1854
- **Region:** New Zealand (national)
- **Bounding box:** New Zealand (national)
- **Magnitude:** No instrumental magnitudes; events assigned to four felt-based magnitude classes
- **Events:** >80 (Part I, pre-1845); ~300 further entries (Part II, 1846–1854)
- **Content:** historical observations; quality information; other
- **Producer / contact:** G.A. Eiby, Geophysics Division, DSIR (New Zealand)
- **Data source:** Documentary/historical records: published and previously unpublished accounts, newspapers, settlers' and observers' reports of felt earthquakes; full source references given per event.
- **Availability:** available (DOI) · DOI [10.1080/00288306.1968.10423671](https://doi.org/10.1080/00288306.1968.10423671)

Foundational documentary chronological catalogue of pre-instrumental New Zealand earthquakes compiled from felt/historical evidence, with events assigned to four felt-based magnitude classes and full source references. Part I (1968) lists >80 events felt before the end of 1845; Part II (1973) adds ~300 further entries for shocks felt 1846–1854 (incl. correcting the mis-dated/mis-located 1848 Marlborough event) with isoseismal map(s). Underpins later Downes historical-macroscopic work and W.D. Smith's 1976 computer file.

Key references: [43, 45].

18. An Annotated List of New Zealand Earthquakes, 1460–1965 (Eiby 1968)

compilation · geographical · felt-reports · reviewed

- **Also known as:** Eiby 1968 annotated list
- **Coverage:** 1460 — 1965
- **Region:** New Zealand (national)
- **Bounding box:** New Zealand (national)
- **Magnitude:** Mix of felt-based magnitude classes (early/pre-instrumental) and early instrumental magnitude estimates
- **Content:** historical observations; hypocentral locations; quality information; other
- **Producer / contact:** G.A. Eiby, Geophysics Division, DSIR (New Zealand)
- **Data source:** Merged pre-instrumental documentary/felt records and early instrumental observatory data into a single long-span annotated chronological list.
- **Availability:** available (DOI) · DOI [10.1080/00288306.1968.10420275](https://doi.org/10.1080/00288306.1968.10420275)

Long-span (1460–1965) annotated chronological list of New Zealand earthquakes that merges pre-instrumental documentary/felt records with early instrumental observatory determinations, serving as a companion to Eiby's descriptive catalogue and a precursor to W.D. Smith's 1976 computer file. Distinct in scope and span from the descriptive catalogue (which stops at 1854). Published in NZJGG 11(3):630–647.

Key references: [44].

19. GeoNet Felt Classic / Community Modified Mercalli Intensity (CMMI) Dataset

compilation · parameter-testing · felt-reports · reviewed

- **Also known as:** Felt Classic; CMMI; Goded et al. 2018 method dataset
- **Coverage:** 2004 — 2016
- **Region:** New Zealand (national)
- **Bounding box:** New Zealand (national)
- **Magnitude:** CMMI test set spans Mw 5.7–7.8; intensities expressed as community Modified Mercalli intensity (CMMI)
- **Events:** ~43,000 felt reports across 9 earthquakes (CMMI test set); broader database ~914,000 reports / 27,688 earthquakes (2004–2016)
- **Content:** historical observations; quality information; other
- **Producer / contact:** Tatiana Goded et al., GNS Science / GeoNet
- **Data source:** Crowd-sourced online 'Felt Classic' citizen felt reports collected via the GeoNet website (2004–2016), converted by an expert/matrix method into community Modified Mercalli intensities (CMMI).
- **Availability:** available on request · DOI [10.1785/0220170163](https://doi.org/10.1785/0220170163)

Citizen-science macroseismic dataset built from GeoNet's 'Felt Classic' online felt reports (2004–2016), converted by an expert (matrix) method into community Modified Mercalli intensities (CMMI) to provide consistent intensity data for NZ earthquakes. The method paper (Goded et al. 2018) implements and tests it on ~43,000 reports from nine moderate-to-large (Mw 5.7–7.8) earthquakes (incl. 2010 Darfield, 2011 Christchurch, 2016 Kaikoura), and reports a first CMMI database of ~914,000 reports from 27,688 earthquakes (2004–2016). Superseded by Felt Rapid (from 2015) and complemented by Felt Detailed.

Key references: [61].

20. GeoNet Aotearoa New Zealand Felt Rapid Dataset

compilation · geographical · felt-reports · non-reviewed

- **Also known as:** Felt Rapid; GeoNet reported intensity
- **Coverage:** 2015 — ongoing
- **Region:** New Zealand (national)
- **Bounding box:** New Zealand (national)
- **Magnitude:** All felt earthquakes; reported MM-style (Modified Mercalli) intensities derived from cartoon shaking selections
- **Content:** historical observations; other
- **Producer / contact:** GNS Science / GeoNet
- **Data source:** Ongoing crowd-sourced rapid citizen felt reports submitted via the GeoNet website/app using a cartoon shaking-selection interface; served through the GeoNet API (intensity?type=reported); geohashed for privacy.
- **Availability:** available (DOI) · DOI [10.21420/RS7F-VE53](https://doi.org/10.21420/RS7F-VE53)

Ongoing citizen-science dataset of rapid felt reports collected via GeoNet's web/app cartoon shaking-selection tool from 2015 to present, providing reported MM-style intensities for all felt NZ earthquakes in near-real time. Served through the GeoNet API and geohashed to protect reporter privacy. Successor to Felt Classic and complements the long-form Felt Detailed product; feeds intensity inputs used in Shaking Layers.

Key references: [56].

21. GeoNet Felt Detailed Macroseismic Intensity Dataset

compilation · geographical · felt-reports · reviewed

- **Also known as:** Felt Detailed; Goded et al. 2017 Kaikoura felt intensities
- **Coverage:** 2016 — ongoing
- **Region:** New Zealand (national); demonstrated for Kaikoura/upper South Island & lower North Island

- **Bounding box:** New Zealand (national)
- **Magnitude:** Issued for very large earthquakes (demonstration event M7.8 2016 Kaikoura); community Modified Mercalli intensities
- **Events:** ~3,500 Felt Detailed reports for the 2016 Kaikoura demonstration (plus ~15,000 Felt Rapid)
- **Content:** historical observations; quality information; other
- **Producer / contact:** Tatiana Goded et al., GNS Science / GeoNet
- **Data source:** Community/MM intensities derived from GeoNet’s long-form (~40-question) ‘Felt Detailed’ citizen questionnaires (issued after very large earthquakes from ~2016), demonstrated for the 2016 Kaikoura earthquake.
- **Availability:** available on request · DOI [10.5459/bnzsee.50.2.352-362](https://doi.org/10.5459/bnzsee.50.2.352-362)

Macroseismic dataset of community/Modified Mercalli intensities derived from GeoNet’s long-form ‘Felt Detailed’ citizen questionnaires (issued after very large earthquakes from ~2016) to capture richer damage/shaking detail than the rapid product. Demonstrated for the M7.8 14 November 2016 Kaikoura earthquake using ~3,500 Felt Detailed plus ~15,000 Felt Rapid reports (Goded et al. 2017). Available on request; intensities feed the GeoNet Shaking Layers product.

Key references: [60].

22. GeoNet Aotearoa New Zealand Shaking Layers Dataset

compilation · geographical · other · non-reviewed

- **Also known as:** Shaking Layers; ShakeMapNZ
- **Coverage:** unknown — not documented — ongoing
- **Region:** New Zealand (national)
- **Bounding box:** New Zealand (national)
- **Magnitude:** Earthquakes $M \geq 3.5$; outputs include modelled MMI plus PGA/PGV/spectral acceleration
- **Content:** quality information; other
- **Producer / contact:** GNS Science / GeoNet (now Earth Sciences New Zealand)
- **Data source:** Near-real-time modelled ground-shaking layers combining strong-motion station recordings with ground-motion modelling and felt/citizen (Felt Rapid/Detailed) observations.
- **Availability:** available (DOI) · DOI [10.21420/J856-2J84](https://doi.org/10.21420/J856-2J84)

Near-real-time modelled ground-shaking dataset producing layers of macroseismic intensity (MMI), peak ground acceleration (PGA), peak ground velocity (PGV) and spectral acceleration for NZ earthquakes $M \geq 3.5$, combining strong-motion data, ground-motion modelling and felt/citizen observations. Borderline for a macroseismic catalogue because outputs are modelled estimates rather than felt-only intensities, but it is a documented, DOI-assigned NZ shaking/intensity product. Outputs available as dynamic/static maps, JSON and GIS layers.

Key references: [2].

23. Historical Incidence of Modified Mercalli Intensity in New Zealand (1840-1997)

compilation · geographical · felt-reports · reviewed

- **Also known as:** Dowrick & Cousins 2003 MM-incidence dataset (NOT ‘Dowrick & Rhoades’)
- **Coverage:** 1840 — 1997
- **Region:** New Zealand (national)
- **Bounding box:** 47 locations across New Zealand (national)
- **Depth range:** 0-100 km (events with depth ≤ 100 km)
- **Magnitude:** Earthquakes $M_w \geq 5.25$; tabulates Modified Mercalli intensities MM4-MM7 incidence/return periods
- **Content:** historical observations; quality information; other
- **Producer / contact:** D.J. Dowrick & W.J. Cousins, GNS Science (Institute of Geological & Nuclear Sciences)

- **Data source:** Derived from the Downes isoseismal atlas / MM intensity database, selecting earthquakes $M_w \geq 5.25$ and $\text{depth} \leq 100$ km and tabulating MM intensities experienced at 47 NZ locations.
- **Availability:** available (DOI) · DOI [10.5459/bnzsee.36.1.1-24](https://doi.org/10.5459/bnzsee.36.1.1-24)

Derived macroseismic dataset giving the historical incidence and return periods of Modified Mercalli intensities (MM4–MM7) from earthquakes $M_w \geq 5.25$ and $\text{depth} \leq 100$ km at 47 New Zealand locations over 1840–1997, built largely on the Downes isoseismal atlas, and compared with seismic-hazard models. Provides an empirical intensity-occurrence record for hazard evaluation. Open via the Bulletin of the NZSEE.

Key references: [34].

24. Patterns of Earthquake-Related Mortality in New Zealand, 1840–2017

historical · particular-event-type · felt-reports · reviewed

- **Also known as:** Abeling et al. 2020 NZ earthquake mortality dataset
- **Coverage:** 1840 — 2017
- **Region:** New Zealand (national)
- **Bounding box:** New Zealand (national)
- **Magnitude:** 21 damaging earthquakes with felt intensity $\text{MMI} \geq \text{VII}$ (Modified Mercalli); event magnitudes not the catalogue’s primary axis
- **Events:** 21 damaging earthquakes; ≥ 489 primary earthquake-related deaths
- **Content:** historical observations; other
- **Producer / contact:** Shannon Abeling (University of Auckland) and co-authors incl. N. Horspool, D. Johnston, D. Dizhur, N. Wilson, C. Clement, J. Ingham
- **Data source:** Historical records of earthquake-related deaths in New Zealand 1840–2017 (coroners’/historical/medical and documentary sources), each death classified by cause across 21 damaging ($\text{MMI} \geq \text{VII}$) earthquakes.
- **Availability:** available (DOI) · DOI [10.1177/8755293019878190](https://doi.org/10.1177/8755293019878190)

National compilation of earthquake-related fatalities in New Zealand over 1840–2017, identifying and classifying each death by cause (e.g. building collapse, landslide, tsunami) across 21 damaging earthquakes ($\text{MMI} \geq \text{VII}$), totalling at least 489 primary deaths. It is a deaths/damage catalogue tied to historical events rather than a hypocentre list, but a documented historical event compilation relevant to macroseismic impact. Published in Earthquake Spectra.

Key references: [5].

25. Magnitudes of New Zealand Earthquakes, 1901–1993 (Dowrick & Rhoades 1998)

large-events-only · parameter-testing · other · reviewed

- **Also known as:** Dowrick & Rhoades 1998 M_s/M_w catalogue
- **Coverage:** 1901 — 1993
- **Region:** New Zealand (national)
- **Bounding box:** New Zealand and near-offshore
- **Magnitude:** 202 larger earthquakes; M_s determinations with M_w and M_L comparisons (M_w -from- M_s standard error ~ 0.15)
- **Events:** 202
- **Content:** hypocentral locations; quality information; other
- **Producer / contact:** D.J. Dowrick & D.A. Rhoades, GNS Science (Institute of Geological & Nuclear Sciences)
- **Data source:** Reassessment of surface-wave (M_s) and other magnitudes for larger NZ earthquakes from observatory/bulletin amplitude data, with revision of earlier observatory magnitudes and locations; conversions to M_w .
- **Availability:** available (DOI) · DOI [10.5459/bnzsee.31.4.260-280](https://doi.org/10.5459/bnzsee.31.4.260-280)

Reassessment catalogue of magnitudes for 202 larger New Zealand earthquakes over 1901–1993, deriving

surface-wave magnitudes (Ms) and conversions to moment magnitude (Mw) and ML, and revising earlier observatory magnitudes and locations. Provides a consistent magnitude backbone for NZ historical seismicity alongside the Downes isoseismal atlas. Published in the Bulletin of the NZSEE.

Key references: [35].

26. Pre-2010 Historical Seismicity near Christchurch (1869 Christchurch & 1870 Lake Ellesmere)

historical · geographical · felt-reports · reviewed

- **Also known as:** Downes & Yetton 2012 Canterbury historical reassessment
- **Coverage:** 1869 — 1870
- **Region:** Canterbury
- **Bounding box:** Canterbury — Christchurch and Lake Ellesmere (Te Waihora) area, South Island NZ
- **Magnitude:** 1869 Christchurch Mw 4.7-4.9; 1870 Lake Ellesmere Mw 5.6-5.8 (estimated from MM intensities)
- **Events:** 2
- **Content:** historical observations; hypocentral locations; quality information; other
- **Producer / contact:** G. Downes (GNS Science) & M. Yetton (Geotech Consulting)
- **Data source:** Documentary felt/MM evidence (newspaper and historical accounts) reassessed to relocate and re-magnitude two 19th-century Canterbury earthquakes, produced after the 2010-2011 Canterbury sequence.
- **Availability:** available (DOI) · DOI [10.1080/00288306.2012.690767](https://doi.org/10.1080/00288306.2012.690767)

Focused regional historical reassessment relocating and re-magnituting the 1869 Christchurch (Mw 4.7-4.9) and 1870 Lake Ellesmere (Mw 5.6-5.8) earthquakes from documentary felt/MM evidence, undertaken after the 2010-2011 Canterbury earthquake sequence to clarify pre-2010 Canterbury seismicity. A focused two-event historical study rather than a broad catalogue. Published in NZJGG 55(3):199-205.

Key references: [31].

7.4 Aftershock-sequence rapid-response catalogues

Sequence-specific catalogues, often built from rapid-response temporary instrumentation deployed around a significant mainshock.

27. 1987 Edgecumbe Earthquake Sequence (ML 6.3, Bay of Plenty)

aftershock · geographical · manual · reviewed

- **Also known as:** Edgecumbe aftershock sequence; 1987 Edgecumbe earthquake
- **Coverage:** 1987-02 — 1987-03
- **Region:** Bay of Plenty (Rangitaiki Plains / Whakatane Graben)
- **Bounding box:** Rangitaiki Plains / Whakatane Graben, Bay of Plenty, North Island, New Zealand
- **Depth range:** ~4-8
- **Magnitude:** Mainshock ML ~6.3 (Mw ~6.5); largest aftershock ~ML 5.6; sequence reported to include hundreds of shocks ML ≥ 3.0
- **Content:** hypocentral locations; phase arrival-time picks; fault-geometry
- **Producer / contact:** Euan G. C. Smith & Clive M. M. Oppenheimer (Victoria University of Wellington) / DSIR Geophysics Division (now GNS Science)
- **Data source:** New Zealand National Seismograph Network (NZNSN) plus rapidly deployed portable seismographs on the Rangitaiki Plains
- **Availability:** available on request · DOI [10.1080/00288306.1989.10421386](https://doi.org/10.1080/00288306.1989.10421386)

Foreshock-mainshock-aftershock catalogue of the 2 March 1987 Edgecumbe earthquake (NZDT; 1 March UTC), which produced surface rupture on the Edgecumbe Fault across the Rangitaiki Plains. Produced to

characterise the spatial and temporal evolution of the sequence and the geometry of the causative faulting in the Whakatane Graben. The defining paper is Smith & Oppenheimer (1989), covering 1987 February 21 to March 18.

Key references: [123].

28. 2003 Mw 7.2 Fiordland (Secretary Island) Aftershock Relocations

aftershock · geographical · manual · reviewed

- **Also known as:** 2003 Fiordland earthquake; Secretary Island / Te Anau 2003
- **Coverage:** 2003-08 — unknown — not documented
- **Region:** Fiordland (subduction interface; overlying Alpine Fault)
- **Bounding box:** Fiordland / Secretary Island region, southwest South Island, New Zealand
- **Magnitude:** Aftershocks ML 1.0-6.1; mainshock ML 7.0 / Mw 7.2
- **Events:** ~8,025 recorded; 1,771 relocated (377 best-quality)
- **Content:** hypocentral locations; relative relocations; phase arrival-time picks; fault-geometry
- **Producer / contact:** Peter McGinty & Russell Robinson (GNS Science)
- **Data source:** NZNSN plus ~10 helicopter-deployed portable seismographs deployed within 3 days of the mainshock
- **Availability:** available on request · DOI [10.1111/j.1365-246X.2007.03336.x](https://doi.org/10.1111/j.1365-246X.2007.03336.x)

Double-difference relocated aftershock catalogue of the 21 August 2003 Mw 7.2 (ML 7.0) Fiordland subduction-interface earthquake. Produced to map the aftershock distribution, constrain the main shock fault plane, and assess static stress changes on the overlying Alpine Fault. ~8,025 aftershocks recorded (ML 1.0-6.1), 1,771 relocated by double-difference, 377 best-quality.

Key references: [86].

29. 2010 Darfield (Canterbury) Mw 7.1 High-Resolution Aftershock Relocations

aftershock · geographical · manual · reviewed

- **Also known as:** Darfield aftershock relocations; Syracuse et al. 2013; D-RAD / FDSN 4A deployment
- **Coverage:** 2010-09 — 2011
- **Region:** Canterbury (Greendale Fault / Darfield)
- **Bounding box:** Canterbury Plains west of Christchurch, South Island, New Zealand
- **Magnitude:** Mainshock Mw 7.1
- **Content:** hypocentral locations; relative relocations; phase arrival-time picks; fault-geometry
- **Producer / contact:** Ellen M. Syracuse & Clifford H. Thurber (University of Wisconsin-Madison); co-authors C. J. Rawles, M. K. Savage (VUW), S. Bannister (GNS Science)
- **Data source:** Rapid-response 10-station PASSCAL/IRIS RAMP array (Darfield RAMP Aftershock Deployment, FDSN 4A_2010, DOI [10.7914/SN/4A_2010](https://doi.org/10.7914/SN/4A_2010), operated by University of Wisconsin-Madison) plus temporary GNS and GeoNet stations
- **Availability:** available on request · DOI [10.1002/jgrb.50301](https://doi.org/10.1002/jgrb.50301)

High-resolution double-difference tomography relocation of Darfield aftershocks following the 4 September 2010 Mw 7.1 earthquake. Produced to delineate buried/blind faults (including the Greendale Fault) and derive a 3-D velocity model, clarifying fault activity in the Canterbury sequence. Distinct from the later Christchurch and Pegasus Bay sub-sequence relocations.

Key references: [29, 130].

30. 2011 Christchurch (Mw 6.2) Fine-Scale Aftershock Relocations

aftershock · geographical · manual · reviewed

- **Also known as:** Christchurch 22 February 2011 aftershock relocations; Bannister et al. 2011
- **Coverage:** 2011-02 — unknown — not documented
- **Region:** Canterbury (Christchurch / Port Hills fault)
- **Bounding box:** Christchurch / Port Hills, Canterbury, South Island, New Zealand
- **Magnitude:** Mainshock Mw 6.2
- **Content:** hypocentral locations; relative relocations; phase arrival-time picks; fault-geometry
- **Producer / contact:** Stephen Bannister, Bill Fry, Martin Reyners, John Ristau, Haijiang Zhang (GNS Science; Zhang at MIT/USTC)
- **Data source:** Dense GeoNet permanent network plus CanNet and temporary portable recorders around Christchurch
- **Availability:** available on request · DOI [10.1785/gssr1.82.6.839](https://doi.org/10.1785/gssr1.82.6.839)

Fine-scale double-difference tomography relocation of aftershocks of the 22 February 2011 Mw 6.2 Christchurch earthquake. Produced to resolve the near-surface (Port Hills) fault responsible for the high ground motions that caused widespread damage and fatalities in Christchurch. Part of the broader Canterbury earthquake sequence work, distinct from the Darfield (Syracuse 2013) and Pegasus Bay (Ristau 2013) relocations.

Key references: [11].

31. Pegasus Bay Aftershock Sequence (2011-2012, Canterbury)

aftershock · geographical · manual · reviewed

- **Also known as:** Pegasus Bay sequence; offshore Christchurch 2011-2012 aftershocks; Ristau et al. 2013
- **Coverage:** 2011-12 — 2012
- **Region:** Canterbury (offshore Pegasus Bay)
- **Bounding box:** Pegasus Bay, offshore east of Christchurch, Canterbury, New Zealand
- **Magnitude:** Mainshock (parent) Mw 7.1 Darfield; Pegasus Bay sub-sequence aftershock magnitude range not verified
- **Events:** 1,075 events located; 53 regional moment tensors
- **Content:** hypocentral locations; relative relocations; focal mechanisms; phase arrival-time picks
- **Producer / contact:** John Ristau, Caroline Holden, Anna Kaiser, Charles Williams, Stephen Bannister, Bill Fry (GNS Science)
- **Data source:** GeoNet network plus temporary low-cost accelerometers; regional moment tensor analysis
- **Availability:** available on request · DOI [10.1093/gji/ggt222](https://doi.org/10.1093/gji/ggt222)

Relocated double-difference tomography catalogue of the offshore Pegasus Bay sub-sequence of the Mw 7.1 Darfield (Canterbury) earthquake, which began ~December 2011 east of Christchurch in Pegasus Bay. Produced to locate the offshore aftershock activity and characterise its faulting style; 1,075 events located and 53 regional moment tensors derived (predominantly reverse mechanisms).

Key references: [108].

32. 2015 ML 6 Wanaka Aftershock Matched-Filter Catalogue

aftershock · geographical · template-matching · reviewed

- **Also known as:** 4 May 2015 Wanaka earthquake aftershock catalogue; Warren-Smith et al. 2017
- **Coverage:** 2015-05 — 2015-05
- **Region:** Central Otago / Southern Alps (Wanaka)
- **Bounding box:** Wanaka / Central Otago, Southern Alps, South Island, New Zealand
- **Magnitude:** Mainshock ML 6.0; aftershock catalogue extends below GeoNet Mc via matched-filter detection
- **Events:** 2,544 aftershocks detected (over ~26 days)
- **Content:** hypocentral locations; relative relocations; phase arrival-time picks; fault-geometry
- **Producer / contact:** Emily Warren-Smith, Calum J. Chamberlain, Simon Lamb, John Townend (Victoria University of Wellington; Warren-Smith later GNS Science)

- **Data source:** Temporary broadband network plus GeoNet national network; matched-filter (template-matching) detection using mainshock plus aftershock templates
- **Availability:** available on request · DOI [10.1785/0220170016](https://doi.org/10.1785/0220170016)

High-precision matched-filter detected and relocated aftershock catalogue of the 4 May 2015 ML 6.0 Wanaka earthquake, the largest shallow Central Otago / Southern Alps event in the record. Produced to characterise the aftershock sequence and identify a NW-dipping blind fault using template-matching detection and precise relative relocation. Detected 2,544 aftershocks over the ~26 days following the mainshock.

Key references: [148].

33. 2016 Kaikoura (Mw 7.8) Aftershock Relocations (Lanza et al. 2019, STREWN + GeoNet)

aftershock · geographical · manual · reviewed

- **Also known as:** Kaikoura aftershock relocations; Lanza et al. 2019; STREWN FDSN Z1 (2016-2017)
- **Coverage:** 2016-11 — 2017-05
- **Region:** North Canterbury / Marlborough (Kaikoura rupture zone)
- **Bounding box:** Northeast South Island / Marlborough to Kaikoura / Cook Strait, New Zealand
- **Magnitude:** ~2,700 events $ML \geq 3$; mainshock Mw 7.8
- **Events:** ~2,700 ($ML \geq 3$)
- **Content:** hypocentral locations; relative relocations; phase arrival-time picks; fault-geometry
- **Producer / contact:** Federica Lanza (lead; later ETH Zurich) with C. J. Chamberlain, K. Jacobs, E. Warren-Smith, H. J. Godfrey, M. Kortink, C. H. Thurber, M. K. Savage, J. Townend, S. Roecker, D. Eberhart-Phillips
- **Data source:** Rapid-response STREWN temporary network (Seismic Triggering Response for Earthquakes around Wellington NZ; FDSN Z1_2016, DOI 10.7914/SN/Z1_2016, University of Wisconsin-Madison) plus GeoNet
- **Availability:** available on request · DOI [10.1029/2019GL082780](https://doi.org/10.1029/2019GL082780)

Relocated aftershock catalogue of the 14 November 2016 Mw 7.8 Kaikoura earthquake (~2,700 $ML \geq 3$ events, Nov 2016-May 2017), repicked and relocated in the Eberhart-Phillips NZ-Wide 3-D velocity model. Produced to constrain crustal fault connectivity across the >20 faults that ruptured in the Kaikoura earthquake. Distinct from the longer Chamberlain et al. 2021 10-year matched-filter catalogue.

Key references: [82, 129].

34. 2016 Kaikoura 10-Year Matched-Filter Seismicity Catalogue (Chamberlain et al. 2021)

background · geographical · template-matching · reviewed

- **Also known as:** Chamberlain et al. 2021 Kaikoura catalogue; pre/co/post-seismic Kaikoura catalogue
- **Coverage:** 2009 — 2020
- **Region:** North Canterbury / Marlborough (Kaikoura rupture faults)
- **Bounding box:** Northeast South Island / Marlborough to Kaikoura rupture zone, New Zealand
- **Magnitude:** Extends well below GeoNet M_c via matched-filter detection; mainshock Mw 7.8
- **Events:** 33,328 events; 1,755 focal mechanisms / template events
- **Content:** hypocentral locations; relative relocations; phase arrival-time picks; focal mechanisms
- **Producer / contact:** Calum J. Chamberlain (Victoria University of Wellington) with W. B. Frank (MIT), F. Lanza, J. Townend, E. Warren-Smith
- **Data source:** GeoNet network plus temporary deployments; matched-filter (template-matching) detection (EQcorrscan) using 1,755 template events
- **Availability:** available on request · DOI [10.1029/2021JB022304](https://doi.org/10.1029/2021JB022304)

Dense matched-filter catalogue of 33,328 earthquakes (2009-2020) on and adjacent to the faults that ruptured in the 2016 Mw 7.8 Kaikoura earthquake, spanning the pre-seismic, co-seismic and post-seismic phases and extending well below the GeoNet magnitude of completeness. Focal mechanisms were computed for 1,755 template events. Produced to illuminate the full seismic cycle around the Kaikoura rupture; distinct from

the Lanza et al. 2019 aftershock-only relocation catalogue.

Key references: [25].

35. Southern Cook Strait / Seddon-Lake Grassmere 2013 Earthquake Response Deployment

aftershock · geographical · manual · partially reviewed

- **Also known as:** Cook Strait 2013 sequence; Seddon earthquake; 2013 Lake Grassmere earthquake response array
- **Coverage:** 2013-09 — 2014-01
- **Region:** Cook Strait / Marlborough (Seddon, Lake Grassmere)
- **Bounding box:** Cook Strait / Seddon-Lake Grassmere, Marlborough, northeast South Island, New Zealand
- **Magnitude:** 246 events $M > 3$ during deployment; mainshocks Mw 6.5 (Seddon) and Mw 6.6 (Lake Grassmere)
- **Events:** 246 ($M > 3$ during deployment)
- **Content:** hypocentral locations; phase arrival-time picks; other
- **Producer / contact:** Martha K. Savage, John Townend, Katrina Jacobs, Tim Stern (Victoria University of Wellington); John Louie (University of Nevada, Reno)
- **Data source:** Rapid-response temporary array of 21 seismometers (2 Massey University broadband + 14 US PASSCAL short-period 2 Hz + 5 Kyoto University instruments from Y. Iio's group) plus GeoNet
- **Availability:** available on request

Rapid-response temporary seismometer deployment recording aftershocks of the 21 July 2013 Mw 6.5 Seddon and 16 August 2013 Mw 6.6 Lake Grassmere earthquakes across Cook Strait, operating 6 September 2013 to 21 January 2014. Recorded 246 events with $M > 3$ and was used primarily for structural imaging parallel to the subduction interface. More a response/imaging deployment than a standalone published relocated earthquake catalogue; reported in EQC report 13/U654.

Key references: [114].

36. Kermadec 2021 Refined Earthquake Catalogue and Slip Models (Lythgoe et al.)

aftershock · particular-event-type · manual · reviewed

- **Also known as:** Kermadec 2021 refined earthquake catalogue and slip models; Lythgoe et al. 2023 Kermadec doublet dataset; Edinburgh DataShare 10283/8556
- **Coverage:** 2021-03 — 2021
- **Region:** Kermadec subduction zone (New Zealand offshore northern margin / Kermadec-Tonga)
- **Magnitude:** Mainshocks Mw 7.4 and Mw 8.1 (5 March 2021); compared to 1976 Mw ~7.9 doublet.
- **Content:** hypocentral locations, relative relocations, focal mechanisms, fault-geometry, other (finite-fault slip models)
- **Producer / contact:** Karen H. Lythgoe (Earth Observatory of Singapore, Nanyang Technological University; previously associated via University of Edinburgh DataShare deposit). Co-authors: K. Bradley, H. Zeng, S. Wei.
- **Data source:** Refined relocations, focal mechanisms and finite-fault slip models derived for the 5 March 2021 Kermadec Mw 7.4 and Mw 8.1 doublet sequence (and re-analysis of the 1976 doublet at the same location). Based on regional/teleseismic and aftershock data analysed in Lythgoe et al. (2023, EPSL). Dataset deposited on University of Edinburgh DataShare.
- **Availability:** available (DOI) · DOI [10.7488/ds/7534](https://doi.org/10.7488/ds/7534)

A standalone dataset accompanying Lythgoe et al. (2023) that provides a refined earthquake catalogue, focal mechanisms and slip models for the 2021 Kermadec Mw 7.4 + Mw 8.1 megathrust doublet, comparing it to the 1976 doublet that ruptured the same asperity. Produced to investigate why large earthquakes recur on the same plate-boundary segment with differing slip distributions, attributing persistent asperities to forearc structure and an isolated sedimentary basin. Focus is the Kermadec subduction zone along New Zealand's offshore northern margin.

Key references: [84, 85].

37. 2016 Mw 7.1 Te Araroa (East Cape) Matched-Filter Foreshock & Aftershock Catalogue (Warren-Smith et al. 2018)

aftershock · particular-event-type · template-matching · non-reviewed

- **Also known as:** Te Araroa 2016 matched-filter catalogue; East Cape 2 September 2016 Mw 7.1 foreshock/aftershock catalogue; Warren-Smith et al. 2018 EQcorrscan catalogue
- **Coverage:** 2016-08 — 2016-11
- **Region:** Offshore East Cape / northern Hikurangi margin, North Island, New Zealand
- **Magnitude:** Mainshock Mw 7.1; largest aftershock ~M6.2; foreshock Mw 5.7. Matched-filter detections down to ~ML 2.5–3.5 reported.
- **Events:** ~8,000+ earthquakes over ~66 days (from 582 template events)
- **Content:** hypocentral locations, phase arrival-time picks, other (template/matched-filter detections; relative timing). Relative relocation not confirmed.
- **Producer / contact:** Emily Warren-Smith (lead; GNS Science). Co-authors: Bill Fry (GNS Science), Yoshihiro Kaneko, Calum J. Chamberlain (Victoria University of Wellington).
- **Data source:** Matched-filter (template-matching) detection applied to GeoNet continuous waveforms using 582 well-located GeoNet template events; recorded on broadband and short-period sensors (~27 stations). Built on the GeoNet catalogue and continuous archive.
- **Availability:** available on request

A dense matched-filter foreshock and aftershock catalogue for the 2 September 2016 Mw 7.1 Te Araroa earthquake offshore East Cape, North Island. Produced to study precursory/foreshock slip, delayed triggering, and the dynamic reinvigoration of the aftershock sequence by the November 2016 Mw 7.8 Kaikoura earthquake. Detects highly correlated events beginning ~12 minutes after a Mw 5.7 foreshock and documents bilateral migration near the eventual mainshock hypocentre.

Key references: [151].

38. 1994 Mw 6.7 Arthur’s Pass Earthquake Finely Relocated Aftershock Catalogue (Bannister et al. 2006)

aftershock · particular-event-type · manual · reviewed

- **Also known as:** Arthur’s Pass 1994 finely relocated aftershocks; Bannister, Thurber & Louie 2006 double-difference tomography catalogue
- **Coverage:** 1994-06 — unknown — not documented
- **Region:** Arthur’s Pass / central Southern Alps, South Island, New Zealand
- **Depth range:** ~0–10 km (>98% of relocated events above ~10 km; brittle-ductile transition near 10 km)
- **Magnitude:** Mainshock Mw 6.7.
- **Events:** >3,500 relocated (of >6,000 recorded aftershocks)
- **Content:** hypocentral locations, relative relocations, phase arrival-time picks, fault-geometry, other (3-D Vp and Vp/Vs velocity model)
- **Producer / contact:** Stephen Bannister (lead; GNS Science). Co-authors: Clifford Thurber (University of Wisconsin-Madison), John Louie (University of Nevada, Reno).
- **Data source:** Aftershock arrival-time and waveform data following the 18 June 1994 Arthur’s Pass Mw 6.7 earthquake (Southern Alps), relocated with double-difference tomography (tomoDD) using waveform cross-correlation and catalogue differential times.
- **Availability:** available on request

A finely relocated aftershock catalogue for the 18 June 1994 Mw 6.7 Arthur’s Pass earthquake in the Southern Alps, South Island. More than 6,000 aftershocks were recorded and over 3,500 relocated, resolving two parallel ~N60E (+/-10 deg), 70–80 deg-dipping fault clusters and a 3-D Vp / Vp/Vs model around the aftershock volume. Produced to illuminate the detailed fault structure of an unusual, complex earthquake sequence.

Key references: [10].

39. 2009 Mw 7.8 Dusky Sound (Fiordland/Puysegur) Aftershock Deployment Relocations

aftershock · particular-event-type · manual · partially reviewed

- **Also known as:** Dusky Sound 2009 aftershocks; Fiordland 2009 Mw 7.8 earthquake response; Puysegur subduction 2009 sequence
- **Coverage:** 2009-07 — unknown — not documented (aftershock deployment ran from 17 July 2009; broader host catalogue spans 2001-2011)
- **Region:** Dusky Sound / SW Fiordland, Puysegur subduction zone, South Island, New Zealand
- **Content:** hypocentral locations, relative relocations (within the broader 2001-2011 relocated catalogue); historical observations / response reports
- **Producer / contact:** Response/deployment: Bill Fry, Stephen Bannister, John Beavan et al. (GNS Science). Broader relocated catalogue that subsumes the aftershocks: Donna Eberhart-Phillips & Martin Reyners (GNS Science).
- **Data source:** GeoNet permanent network densified by a portable seismograph deployment (6-7 instruments installed from 17 July 2009) after the 15 July 2009 Mw 7.8 Dusky Sound earthquake on the Puysegur subduction interface. The aftershocks are captured within a national 2001-2011 catalogue relocated with a 3-D seismic velocity model.
- **Availability:** available on request

The 15 July 2009 Mw 7.8 Dusky Sound (Fiordland) earthquake ruptured the Puysegur subduction interface and was the largest NZ earthquake in ~80 years; a portable deployment densified GeoNet to better locate its rich aftershock sequence. There is NO verifiable dedicated standalone relocated-aftershock catalogue product with its own DOI: the response is documented in preliminary reports (Fry et al. 2010, Bull. NZSEE) and GPS/InSAR source papers (Beavan et al. 2010, GJI), while the relocated aftershocks are subsumed within the broader Eberhart-Phillips & Reyners (2023) 2001-2011 New Zealand relocated catalogue.

Key references: [16, 40, 49].

7.5 Temporary onshore passive-seismic deployments

Time-limited onshore passive-seismic experiments and dense arrays, frequently registered under temporary FDSN network codes; the largest and least-discoverable group in the inventory.

40. SAPSE — Southern Alps Passive Seismic Experiment (1995-96)

background · geographical · manual · reviewed

- **Also known as:** SAPSE; FDSN XC (1995-1996); Cooperative Project on South Island New Zealand (NZ Passive)
- **Coverage:** 1995-11 — 1996-04
- **Region:** Southern Alps / central Alpine Fault
- **Bounding box:** Central South Island / Southern Alps, New Zealand (Alpine Fault region around Mt Cook / Aoraki)
- **Depth range:** 0~12 km seismogenic (relocated events)
- **Magnitude:** Local microseismicity; ~5491 events triggered during the deployment, 195 selected and precisely relocated. Mc not formally documented in sources read.
- **Events:** ~5491 recorded; 195 precisely relocated
- **Content:** hypocentral locations; phase arrival-time picks; focal mechanisms; quality information
- **Producer / contact:** IRIS/PASSCAL with Institute of Geological & Nuclear Sciences (GNS) and Oregon State University; lead analysis Beate Leitner / Donna Eberhart-Phillips
- **Data source:** Temporary US-NZ passive deployment of ~26 portable broadband/short-period seismographs across the central South Island (IRIS/PASSCAL + IGNS + Oregon State); passive partner to the active-source SIGHT transect. Network XC continuous waveforms.
- **Availability:** available (DOI) · DOI [10.7914/SN/XC_1995](https://doi.org/10.7914/SN/XC_1995)

A focused dense temporary deployment to image microseismicity, focal mechanisms and stress along the central Alpine Fault and Southern Alps, Nov 1995–Apr 1996. Produced an early high-resolution Alpine Fault microseismicity and stress dataset and constrained the seismogenic thickness (~12 km). Served as the passive seismology component complementing the SIGHT active-source crustal transect.

Key references: [75, 83].

41. SAMBA — Southern Alps Microearthquake Borehole Array (network deployment)

background · geographical · automatic · partially reviewed

- **Also known as:** SAMBA; FDSN 9F (2008–2020); FDSN 9K (2021–2030)
- **Coverage:** 2008 — ongoing
- **Region:** Central Alpine Fault / Southern Alps
- **Bounding box:** Central Alpine Fault, Southern Alps, South Island, New Zealand (~Whataroa–Franz Josef–Fox area)
- **Magnitude:** Microseismicity (down to ~M0); magnitude completeness depends on derived catalogue (Michailos et al. 2019).
- **Content:** hypocentral locations; relative relocations; phase arrival-time picks; other
- **Producer / contact:** Victoria University of Wellington (VUW); John Townend / Martha Savage; relocated catalogue by Konstantinos Michailos
- **Data source:** Semi-permanent VUW borehole/short-period seismometer array (~10–13 sites) on the central Alpine Fault, installed 2008 as FDSN 9F and continued as 9K from 2021. Continuous waveform network.
- **Availability:** available (DOI) · DOI [10.7914/SN/9F_2008](https://doi.org/10.7914/SN/9F_2008)

A long-running near-fault waveform array densifying coverage of the central Alpine Fault for microseismicity, tremor and low-frequency-earthquake studies. It is the recording network underpinning derived event catalogues (Michailos relocated/matched-filter catalogue, Alpine Fault LFE catalogues) and tomography/thermal studies. This record describes the waveform network itself, distinct from the derived event catalogues.

Key references: [92, 143].

42. DFDP-2 Real-Time / Matched-Filter Earthquake Catalogue, Whataroa (2014–2015)

background · particular-event-type · template-matching · reviewed

- **Also known as:** DFDP-2 monitoring catalogue; Deep Fault Drilling Project Phase 2
- **Coverage:** 2014-08 — 2015-01
- **Region:** Central Alpine Fault / Whataroa, Westland
- **Bounding box:** Within ~30 km of Whataroa, central Alpine Fault, Westland, South Island, New Zealand
- **Magnitude:** ML 0.6–4.2; 493 located earthquakes
- **Events:** 493
- **Content:** hypocentral locations; phase arrival-time picks; quality information; other
- **Producer / contact:** Calum J. Chamberlain & co-authors (VUW, GNS, Univ. Auckland, Univ. Wisconsin, Tohoku Univ.)
- **Data source:** Real-time monitoring network of borehole + surface seismometers within ~30 km of the DFDP-2 drill site at Whataroa, central Alpine Fault, operated around-the-clock Aug 2014–Jan 2015; real-time picking plus offline matched-filter detection.
- **Availability:** available on request

An earthquake catalogue produced from real-time seismic monitoring and a traffic-light response protocol during DFDP-2 scientific drilling on the central Alpine Fault. The network detected and located 493 earthquakes (ML 0.6–4.2) near Whataroa; no event occurred within 3 km of the drill site and none required changes to drilling. Demonstrates real-time + matched-filter monitoring around an active scientific borehole.

Key references: [24].

43. DFDP10 / Whataroa-Wanganui Passive Seismology Experiment 2010 (FDSN XO)*background · geographical · automatic · non-reviewed*

- **Also known as:** FDSN XO (2010); DFDP10/VicU; Whataroa-Wanganui Passive Seismology Experiment
- **Coverage:** 2010 — 2010
- **Region:** Central Alpine Fault / Whataroa, Westland
- **Bounding box:** Whataroa-Wanganui (Whataroa River) area, central Alpine Fault, Westland, South Island, New Zealand
- **Magnitude:** Local microseismicity; no published Mc or event count in sources read.
- **Content:** hypocentral locations; phase arrival-time picks; other
- **Producer / contact:** Victoria University of Wellington (VUW)
- **Data source:** VUW temporary passive seismometer deployment (FDSN XO) in the Whataroa-Wanganui river area around the planned Alpine Fault (DFDP) drilling target; continuous waveform network operated during 2010.
- **Availability:** available (DOI) · DOI [10.7914/SN/XO_2010](https://doi.org/10.7914/SN/XO_2010)

An early VUW passive deployment to characterise background microseismicity around the future Deep Fault Drilling Project target on the central Alpine Fault. It contributes waveform data feeding the broader central Alpine Fault microseismicity catalogues; primarily a waveform dataset rather than a standalone published event catalogue.

Key references: [145].

44. ALFA08 / ALFA09 Alpine Fault Arrays (FDSN YR)*background · geographical · manual · partially reviewed*

- **Also known as:** ALFA08; ALFA09; FDSN YR (2008-2010)
- **Coverage:** 2008 — 2010
- **Region:** Central Alpine Fault (Harihari-Ross-Arthur's Pass)
- **Bounding box:** Central Alpine Fault between Harihari and Arthur's Pass, Westland/Canterbury, South Island, New Zealand
- **Magnitude:** Local microseismicity; event count not separately published in sources read.
- **Content:** hypocentral locations; phase arrival-time picks; other
- **Producer / contact:** GNS Science (IGNS); analysis Sandra Bourguignon, Stephen Bannister, John Townend, Haijiang Zhang
- **Data source:** GNS Science temporary short-period arrays (FDSN YR) between Harihari and Arthur's Pass on the central Alpine Fault: ALFA08 (~8 sensors, 2008-09) and ALFA09 (~10 short-period, 2009-10). Continuous waveform network plus located local seismicity.
- **Availability:** available (DOI) · DOI [10.7914/SN/YR_2008](https://doi.org/10.7914/SN/YR_2008)

Temporary Alpine Fault arrays deployed to record and locate local seismicity and characterise current fault activity between Harihari and Arthur's Pass. The recorded seismicity and travel-time data underpin midcrustal seismic tomography of the central Alpine Fault (Bourguignon et al. 2015) and multi-network microseismicity studies.

Key references: [20, 54].

45. WIZARD Array — Wisconsin, New Zealand And Rensselaer Deployment (FDSN ZT / 6F)*background · geographical · automatic · non-reviewed*

- **Also known as:** WIZARD; FDSN ZT (2012-2014); FDSN 6F (2012-2020) DFDP13 tight array
- **Coverage:** 2012 — 2020
- **Region:** Central Alpine Fault / Whataroa, Westland
- **Bounding box:** Whataroa / central Alpine Fault drilling target area, Westland, South Island, New Zealand

- **Magnitude:** Local/regional microseismicity; no standalone event count in sources read.
- **Content:** hypocentral locations; phase arrival-time picks; other
- **Producer / contact:** University of Wisconsin-Madison (Clifford Thurber) with Rensselaer Polytechnic Institute and VUW/GNS
- **Data source:** US-NZ temporary networks around the DFDP-2 Alpine Fault drilling target at Whataroa: ZT = broader WIZARD short-period array (2012–2014); 6F = tight DFDP13 supplementary array (2012–2020). Recorded local/regional seismicity, ambient noise and active-source data.
- **Availability:** available (DOI) · DOI [10.7914/SN/ZT_2012](https://doi.org/10.7914/SN/ZT_2012)

Temporary deployments installed to record seismicity, ambient noise and active-source data around the second DFDP Alpine Fault drilling site, including potential drilling-induced activity. Data feed central Alpine Fault microseismicity, tomography and attenuation models. Two FDSN codes correspond to the broader WIZARD (ZT) and the tight DFDP13 array (6F).

Key references: [136, 137].

46. DWARFS — Dense Westland Arrays Researching Fault Segmentation (FDSN 7S/8N)

background · geographical · automatic · non-reviewed

- **Also known as:** DWARFS; FDSN 7S (2019–2021)
- **Coverage:** 2019 — 2021
- **Region:** Westland / Alpine Fault rupture barriers
- **Bounding box:** Westland, central-south Alpine Fault, South Island, New Zealand
- **Magnitude:** Small near-fault earthquakes; no standalone event count in sources read.
- **Content:** hypocentral locations; phase arrival-time picks; fault-geometry; other
- **Producer / contact:** GNS Science (IGNS)
- **Data source:** GNS Science dense temporary seismometer arrays (FDSN code 7S; dataset DOI suffix 8N) deployed near persistent Alpine Fault rupture barriers in Westland, 2019–2021. Continuous waveform arrays recording small earthquakes.
- **Availability:** available (DOI) · DOI [10.7914/SN/8N_2019](https://doi.org/10.7914/SN/8N_2019)

Dense temporary arrays installed near two major Alpine Fault rupture barriers in Westland to investigate earthquake segmentation, fault geometry and near-fault source properties (e.g. stress drops). The recordings map barrier geometry and mechanical properties and feed Alpine Fault segmentation/source studies and VUW theses.

Key references: [58].

47. SALSA — Southern Alps Long Skinny Array (FDSN ZX)

background · parameter-testing · automatic · non-reviewed

- **Also known as:** SALSA; FDSN ZX (2021–2023)
- **Coverage:** 2021 — 2023
- **Region:** Alpine Fault (onshore, Fiordland to Maruia)
- **Bounding box:** Onshore Alpine Fault, Milford Sound to Maruia, South Island, New Zealand
- **Magnitude:** Microseismicity component; no published event count in sources read.
- **Content:** other
- **Producer / contact:** Victoria University of Wellington (VUW); PASSCAL/Harvard instrument support; Marsden-funded
- **Data source:** VUW temporary array (FDSN ZX): ~45 seismometers at ~10 km spacing along the entire onshore Alpine Fault (Milford Sound to Maruia), plus microseismicity stations, 2021–2023. Continuous waveform dataset.
- **Availability:** available (DOI) · DOI [10.7914/SN/ZX_2021](https://doi.org/10.7914/SN/ZX_2021)

A long, fault-parallel array spanning the onshore Alpine Fault for virtual-earthquake and ground-motion analysis along the fault, supplemented by stations recording local microseismicity. Primarily a waveform

dataset supporting along-fault ground-motion prediction; no published standalone event-catalogue count.

Key references: [146].

48. SOSA — Southland Otago Seismic Array Catalogue (2022-2023)

background · geographical · automatic · reviewed

- **Also known as:** SOSA; FDSN 6Y (2022-2023)
- **Coverage:** 2022-10 — 2023-10
- **Region:** Southland / Otago / SE Fiordland
- **Bounding box:** Western Catlins to eastern Fiordland, Southland/Otago, South Island, New Zealand
- **Depth range:** 0-40 km (microseismicity nucleation depths reported)
- **Magnitude:** ML ~0.2-3.1; 85 located earthquakes (~37% in swarms)
- **Events:** 85
- **Content:** hypocentral locations; relative relocations; phase arrival-time picks; quality information
- **Producer / contact:** University of Otago; Jack N. Williams, Mark W. Stirling et al.
- **Data source:** University of Otago temporary array (FDSN 6Y): 19 short-period stations between the western Catlins and eastern Fiordland, Oct 2022-Oct 2023, densifying sparse GeoNet coverage (~10-30 km spacing vs ~100 km).
- **Availability:** available (DOI) · DOI [10.7914/jr68-qq17](https://doi.org/10.7914/jr68-qq17)

A temporary dense array deployed to image microseismicity of the low-coverage southeastern South Island near the transpressive Australian-Pacific plate boundary. The resulting catalogue of 85 earthquakes reveals deeper-than-expected and clustered microseismicity (depths 0-40 km), including Fiordland swarms. A priority temporary deployment improving on sparse permanent coverage.

Key references: [134, 157].

49. Otago Temporary Broadband Network (FDSN 6K, 2014-2015)

background · geographical · automatic · non-reviewed

- **Also known as:** FDSN 6K (2014-2015) Otago temporary broadband network
- **Coverage:** 2014-03 — 2015-04
- **Region:** Otago / SE South Island
- **Bounding box:** Otago, southeastern South Island, New Zealand
- **Magnitude:** Low-seismicity region; no published Mc or event count in sources read.
- **Content:** hypocentral locations; phase arrival-time picks; other
- **Producer / contact:** GNS Science (IGNS); Marsden-funded
- **Data source:** GNS Science temporary broadband deployment (FDSN 6K): two consecutive six-month deployments of 20 Guralp CMG-40T sensors across Otago, Mar 2014-Apr 2015. Continuous waveform dataset.
- **Availability:** available (DOI) · DOI [10.7914/SN/6K_2014](https://doi.org/10.7914/SN/6K_2014)

A temporary broadband network deployed to study seismicity and crustal structure of the low-seismicity, reverse-faulting southeastern South Island (Otago). Primarily a waveform dataset; no standalone published event-catalogue count located in sources read.

Key references: [55].

50. Marlborough Fault Zone Anisotropy/Converted-Phase Array (FDSN XB, 2000-2002)

background · geographical · manual · partially reviewed

- **Also known as:** FDSN XB (2000-2002) Marlborough Fault Zone array
- **Coverage:** 2000-12 — 2002-05

- **Region:** Marlborough Fault System
- **Bounding box:** Marlborough Fault System (Wairau/Awatere/Clarence faults), northern South Island, New Zealand
- **Magnitude:** Local seismicity recorded; no standalone event count in sources read.
- **Content:** hypocentral locations; phase arrival-time picks; quality information; other
- **Producer / contact:** IRIS/PASSCAL; tomographic analysis Donna Eberhart-Phillips & Stephen Bannister (GNS)
- **Data source:** IRIS/PASSCAL temporary deployment (FDSN XB): five small arrays along a profile crossing the Wairau (Alpine), Awatere and Clarence faults in Marlborough, Dec 2000-May 2002. Continuous waveforms; recorded local seismicity.
- **Availability:** available (DOI) · DOI [10.7914/SN/XB_2000](https://doi.org/10.7914/SN/XB_2000)

A temporary array profile primarily designed for S-wave anisotropy and converted-phase imaging of strain distribution beneath the Marlborough Fault System, while also recording local earthquakes. The recorded seismicity contributed to 3-D velocity/hypocentre inversions of the Marlborough strike-slip and subducted-plate system (Eberhart-Phillips & Bannister 2010). Borderline as an event catalogue (mainly a structure-imaging deployment) but a recorded-seismicity temporary array.

Key references: [39, 76].

51. SAHKE — Seismic Array HiKurangi Experiment (Wellington Geophysical Transect, passive)

background · geographical · manual · partially reviewed

- **Also known as:** SAHKE; FDSN X2 (2009-2010); Seismic Analysis of the HiKurangi Experiment
- **Coverage:** 2009-11 — 2010-04
- **Region:** Southern Hikurangi margin / Wellington transect
- **Bounding box:** Lower North Island transect, Kapiti coast to east coast (Wairarapa), New Zealand
- **Magnitude:** Local and teleseismic events recorded.
- **Content:** hypocentral locations; phase arrival-time picks; other
- **Producer / contact:** Victoria University of Wellington / GNS Science with ERI (Univ. Tokyo, Japan) and US partners; Stuart Henrys (GNS) lead author
- **Data source:** VUW/GNS/Japan-ERI/US passive deployment (FDSN X2): ~10 broadband + short-period sensors plus active-source shots along a lower North Island transect (Kapiti coast to east coast), Nov 2009-Apr 2010. Continuous waveforms + active-source data.
- **Availability:** available (DOI) · DOI [10.7914/SN/X2_2009](https://doi.org/10.7914/SN/X2_2009)

A passive (and active-source) deployment along a Wellington-region geophysical transect to image the locked southern Hikurangi subduction megathrust beneath the lower North Island. Recorded local and teleseismic events used with active-source data for tomography and reflection imaging of the subduction interface and overlying crust.

Key references: [62, 144].

52. CNIPSE / NIGHT Passive — Central North Island Passive Seismic Experiment (FDSN XQ)

background · geographical · manual · reviewed

- **Also known as:** CNIPSE; NIGHT Passive; FDSN XQ (2001-2002); North Island Geophysical Transect Passive
- **Coverage:** 2001-01 — 2001-06
- **Region:** Central North Island / Hikurangi subduction zone
- **Bounding box:** Central North Island, ~Bennydale to Tarawera (Hikurangi subduction system), New Zealand
- **Depth range:** 0~300 km (subduction zone to ~300 km depth imaged)

- **Magnitude:** Local-to-regional seismicity; >4700 events located, 1239 used in tomography. Mc not separately documented in sources read.
- **Events:** >4700 located (1239 used in tomography)
- **Content:** hypocentral locations; phase arrival-time picks; other
- **Producer / contact:** IRIS/PASSCAL with GNS Science and VUW; analysis Martin Reyners, Donna Eberhart-Phillips et al.
- **Data source:** Dense temporary deployment (FDSN XQ) of ~74 portable seismographs (32 broadband) across the central North Island, ~Jan-Jun 2001 (IRIS/PASSCAL + GNS + VUW). Continuous waveforms; >4700 located earthquakes.
- **Availability:** available (DOI) · DOI [10.7914/SN/XQ_2001](https://doi.org/10.7914/SN/XQ_2001)

A dense ~6-month passive deployment to produce a high-resolution snapshot of Hikurangi subduction-zone seismicity beneath the central North Island and to support 3-D Vp/Vp-Vs tomography from the trench to ~300 km depth. Over 4700 earthquakes were located within or near the network; a subset (1239 events) was used in the defining tomographic inversion.

Key references: [77, 103].

53. HADES — Deep Geothermal HADES Seismic Array (Taupo, FDSN Z8)

background · geographical · automatic · partially reviewed

- **Also known as:** HADES; FDSN Z8 (2009-2011)
- **Coverage:** 2009 — 2011
- **Region:** Taupo Volcanic Zone (TVZ)
- **Bounding box:** Central/northern Taupo Volcanic Zone (Rotorua-Rotomahana-Tarawera), ~25 x 35 km, North Island, New Zealand
- **Depth range:** 3-10 km (target mid-crustal depth range)
- **Magnitude:** Crustal/geothermal microseismicity; event count for the HADES deployment itself not separately documented in sources read.
- **Content:** hypocentral locations; phase arrival-time picks; other
- **Producer / contact:** GNS Science (IGNS); Stephen Bannister
- **Data source:** GNS Science temporary array (FDSN Z8) of ~39 surface seismometers at ~5 km spacing over the central/northern Taupo Volcanic Zone geothermal fields (Rotorua-Rotomahana-Tarawera), ~25 x 35 km, 2009-2011. Continuous waveforms supplementing GeoNet.
- **Availability:** available (DOI) · DOI [10.7914/SN/Z8_2009](https://doi.org/10.7914/SN/Z8_2009)

A temporary dense array deployed to image mid-crustal (3-10 km) seismic properties and crustal seismicity of the central/northern TVZ geothermal region, supplementing the permanent GeoNet network. Its recordings feed into high-precision TVZ earthquake catalogues and crustal-structure studies.

Key references: [9, 68].

54. Taranaki Dense Temporary Network Crustal Seismicity Catalogue (Sherburn & White 2005)

background · geographical · manual · reviewed

- **Also known as:** Taranaki dense network; Cape Egmont fault zone deployment
- **Coverage:** 2001-12 — 2002-09
- **Region:** Taranaki / Cape Egmont Fault Zone
- **Bounding box:** Taranaki region around Mt Taranaki/Egmont and Cape Egmont Fault Zone, North Island, New Zealand
- **Magnitude:** ML 1.4-4.0; 389 located earthquakes; >15,000 phase picks
- **Events:** 389
- **Content:** hypocentral locations; phase arrival-time picks; quality information
- **Producer / contact:** Steven Sherburn (GNS Science) & Robert S. White (University of Cambridge)

- **Data source:** Dense temporary deployment of 75 three-component broadband seismographs around Mt Taranaki/Egmont, Dec 2001-Sep 2002 (GNS Science + University of Cambridge). >15,000 P/S phase picks; 389 located local earthquakes.
- **Availability:** available on request

A dense temporary network deployed to obtain accurate hypocentres of crustal seismicity in the Taranaki region. It located 389 local earthquakes (ML 1.4–4.0), found the Cape Egmont Fault Zone most active, and notably found no volcanic earthquakes beneath Mt Taranaki. A focused regional crustal-seismicity catalogue.

Key references: [121].

55. Tongariro Volcanic Centre Local Earthquake Tomography Catalogue (Rowlands et al. 2005)

background · particular-event-type · manual · reviewed

- **Also known as:** TgVC tomography; Rowlands, White & Haines 2005
- **Coverage:** 2001-01 — 2001-06
- **Region:** Tongariro Volcanic Centre (TVZ)
- **Bounding box:** Tongariro Volcanic Centre, central North Island, New Zealand
- **Magnitude:** Local volcanic-tectonic earthquakes; 155 selected events. Mc not separately documented in sources read.
- **Events:** 155 (selected for tomography)
- **Content:** hypocentral locations; phase arrival-time picks; quality information
- **Producer / contact:** D. P. Rowlands, R. S. White (University of Cambridge) & A. J. Haines (GNS Science)
- **Data source:** Temporary deployment of 23 broadband seismometers in the Tongariro Volcanic Centre, Jan-Jun 2001 (University of Cambridge / GNS), part of the same 2001 central North Island campaign as CNIPSE. ~3000 P + ~1100 S picks; 155 selected local earthquakes.
- **Availability:** available on request

A temporary local network deployed to obtain accurate earthquake depths and 3-D Vp / Vp-Vs structure of the Tongariro Volcanic Centre. The best 155 local earthquakes (with quality criteria on picks and azimuthal gap) were selected for local-earthquake tomography of the volcanic system. A focused volcanic-zone seismicity/tomography catalogue.

Key references: [113].

56. Local Earthquake and Focal Mechanism Catalogue for the Northern Hikurangi Margin, December 2017–October 2018 (NZ3D onshore network)

background · geographical · manual · reviewed

- **Also known as:** NZ3D onshore earthquake/focal-mechanism catalogue; Woodward et al. (2026) Northern Hikurangi catalogue; FDSN network 3C (2017–2018) 'NZ3D FWI'
- **Coverage:** 2017-12 — 2018-10
- **Region:** Northern Hikurangi subduction margin, near Gisborne, eastern North Island, New Zealand
- **Events:** 3071 high-quality local earthquakes (reported in the defining study; plus associated focal mechanisms)
- **Content:** hypocentral locations, phase arrival-time picks, focal mechanisms, quality information
- **Producer / contact:** Amy Woodward (lead author), Ian D. Bastow, Rebecca Bell — Imperial College London; deployment PI Rebecca Bell (Imperial College London)
- **Data source:** Onshore temporary NZ3D-FWI seismograph network (FDSN code 3C, 2017–2018): 49 Guralp CMG-6TD broadband instruments (NERC GEF) deployed Dec 2017–Oct 2018, supplemented by ~119 short-period DATACUBE3 (GIPP Potsdam) and 25 GSX3 (ERI Tokyo) short-period instruments (Dec 2017–Feb 2018), in a ~30 km x 15 km array near Gisborne above the 10–30 km portion of the northern Hikurangi slow-slip region. Hypocentres located and focal mechanisms derived from these continuous recordings; v2 of the dataset removed the active-source airgun shots present in v1.

- **Availability:** available (DOI) · DOI [10.5281/zenodo.17063327](https://doi.org/10.5281/zenodo.17063327)

Produced to seismologically characterise the deeper (10–30 km) portion of the northern Hikurangi margin slow-slip region near Gisborne, associating microseismicity and faulting with normal faults, subducting seamounts and fluid seep sites. The catalogue resolves well-located earthquakes and focal mechanisms (including deep normal-faulting events within the subducting slab) to trace fluid pathways rising from the slab mantle to surface seeps. It fills a gap left by the offshore HOBITSS catalogues and the NZ3D-FWI active-source velocity-model work, which did not include an onshore-network-derived hypocentre+focal-mechanism event catalogue.

Key references: [71, 158, 159].

57. Southern Lakes / Northern Fiordland High-Resolution Microseismicity Catalogue (Central Otago Seismic Array, COSA)

background · geographical · manual · reviewed

- **Also known as:** COSA microearthquake catalogue; Warren-Smith et al. (2017) Southern South Island microseismicity catalogue; FDSN network 6S (Central Otago Seismic Array)
- **Coverage:** 2012-06 — 2013-03
- **Region:** Southern Lakes / northern Fiordland, southern South Island, New Zealand (zone adjacent to the southern Alpine Fault)
- **Content:** hypocentral locations, relative relocations, phase arrival-time picks, focal mechanisms, quality information
- **Producer / contact:** Emily Warren-Smith (lead author); co-authors Simon Lamb, Tim A. Stern, Euan Smith — Victoria University of Wellington (VUW). COSA network operated by Victoria University of Wellington.
- **Data source:** Central Otago Seismic Array (COSA), FDSN network code 6S, ~8 temporary broadband stations operated by Victoria University of Wellington supplementing the permanent GeoNet network; stations installed mid-2012 to early 2013 (~June 2012–March 2013). A new 15-month high-resolution microearthquake catalogue derived from these recordings, including relocations and (in the companion paper) focal mechanisms and stress inversions.
- **Availability:** available on request

Built to image diffuse crustal-to-mantle microseismicity in the Southern Lakes and northern Fiordland region adjacent to the southern Alpine Fault, and to test mechanisms of crustal deformation in an obliquely convergent (transpressive) Australian-Pacific plate boundary zone. The catalogue documents shallow (<25 km) distributed crustal seismicity across a ~150 km-wide zone plus deeper/upper-mantle events, interpreted as distributed shear and crustal thickening. A companion paper (Warren-Smith et al., 2017) derives the stress field and kinematics from the same dataset.

Key references: [3, 149, 150].

58. Dunedin Region Short-Term Broadband Array Microseismicity Catalogue (Akatore Fault)

background · geographical · template-matching · reviewed

- **Also known as:** Todd et al. (2020) Dunedin microseismicity catalogue; Dunedin short-term broadband seismic array catalogue; Akatore Fault microseismicity cluster
- **Coverage:** unknown — not documented (short-term deployments; exact array dates not extracted from accessible sources) — unknown — not documented
- **Region:** Dunedin region / Akatore Fault, eastern Otago, southern South Island, New Zealand
- **Content:** hypocentral locations, phase arrival-time picks, quality information, other (template/matched-filter detections)
- **Producer / contact:** Erin K. Todd (lead author); co-authors Mark W. Stirling, Bill Fry, Jerome Salichon, Pilar Villamor — University of Otago (Stirling) / GNS Science (Fry, Salichon, Villamor)
- **Data source:** Short-term temporary broadband seismic array deployments in/around Dunedin (a low-seismicity region), supplementing the permanent GeoNet national network. Matched-filter (template-

matching) analysis using a master event detected events absent from the GeoNet national catalogue, including a microearthquake cluster at the northern end of the Akatore Fault (<15 km from Dunedin city centre).

- **Availability:** available on request

Produced to characterise microseismicity in a low-seismicity region (Dunedin) where the sparse permanent network under-detects small events, and to demonstrate the value of short-term dense broadband arrays for hazard assessment. Using template/matched-filter detection it revealed otherwise-undetected microseismicity, notably a cluster at the northern Akatore Fault — consistent with paleoseismic evidence that the fault is in an elevated-activity state after ~110 ka of quiescence.

Key references: [132].

59. Hawke’s Bay Region Microearthquake Catalogue (Bannister 1988; subducting Pacific plate fine structure)

background · geographical · manual · reviewed

- **Also known as:** Bannister (1988) Hawke’s Bay microseismicity catalogue; Hawke’s Bay 11-station temporary network microearthquake dataset
- **Coverage:** unknown — not documented (mid-1980s; deployment year not stated in accessible sources) — unknown — not documented (two-month deployment)
- **Region:** Hawke’s Bay, eastern North Island, New Zealand (Hikurangi subduction forearc)
- **Events:** more than 350 micro-earthquakes (recorded over two months)
- **Content:** hypocentral locations, phase arrival-time picks, other (velocity-structure inversion)
- **Data source:** Temporary dense onshore network of 11 seismograph stations in the Hawke’s Bay region; >350 micro-earthquakes recorded over a two-month deployment (mid-1980s). Hypocentres located and the data inverted for P-wave velocity/structure of the shallow subducting Pacific plate.
- **Availability:** not available

Produced to image the fine structure of the shallow subducting Pacific plate beneath Hawke’s Bay using a short, dense temporary seismograph deployment. From >350 microearthquakes the study resolved a thin (~11–12 km) low P-velocity band interpreted as subducted sediments, with the slab dipping ~6° beneath Hawke Bay and steepening (>20°) westward. It provides an early, citable regional microseismicity product for a North Island east-coast area lacking inventoried 1980s dense-array catalogues.

Key references: [14].

7.6 Temporary offshore / ocean-bottom (OBS) deployments

Ocean-bottom seismometer deployments along the offshore Hikurangi margin and the Kermadec volcanic arc.

60. HOBITSS Microearthquake Catalogue (2014–2015, northern Hikurangi, manual)

background · particular-event-type · manual · reviewed

- **Also known as:** HOBITSS; Hikurangi Ocean Bottom Investigation of Tremor and Slow Slip; YH (2014); Yarce et al. 2019 catalogue
- **Coverage:** 2014-06 — 2015-05
- **Region:** Northern Hikurangi subduction margin (offshore Gisborne/Poverty Bay)
- **Bounding box:** approx 37.5S–39.3S, 176.5E–179.5E (offshore Gisborne / northern Hikurangi margin, North Island)
- **Magnitude:** approx M0.5–4.7 (local magnitude)
- **Events:** approx 2,300 (approximate)
- **Content:** hypocentral locations; phase arrival-time picks
- **Producer / contact:** Jefferson Yarce / Anne F. Sheehan (CIRES, University of Colorado Boulder); HOBITSS PIs incl. Laura Wallace, Susan Schwartz, Spahr Webb, Kimihiro Mochizuki, Stuart Henrys; network operated by University of Texas at Austin

- **Data source:** HOBITSS ocean-bottom seismometer array (FDSN network YH, 2014) deployed on the northern Hikurangi margin seafloor, supplemented by onshore GeoNet permanent stations; manual P- and S-wave picking and hypocentre location
- **Availability:** available (DOI) · DOI [10.5281/zenodo.2022405](https://doi.org/10.5281/zenodo.2022405)

Built to investigate whether a spatial/temporal relationship exists between microseismicity and the shallow ~2-week September–October 2014 slow slip event (SSE) targeted by the year-long HOBITSS deployment above the northern Hikurangi plate interface. Earthquakes were detected and located by hand-picking P/S arrivals on OBS plus onshore GeoNet stations. The catalogue reveals a ‘microseismicity gap’ bordering the 2014 slow-slip patch, later reused by Yarce et al. (2023).

Key references: [135, 160, 161].

61. HOBITSS Automatic-Detection Earthquake Catalogue (2014 SSE, REST detector)

background · particular-event-type · automatic · non-reviewed

- **Also known as:** HOBITSS auto-detection catalogue; Yarce et al. 2023; REST (Regressive ESTimator) catalogue
- **Coverage:** 2014-09 — 2014-10
- **Region:** Northern Hikurangi subduction margin (offshore Gisborne)
- **Bounding box:** approx 37.5S–39.3S, 176.5E–179.5E (offshore northern Hikurangi margin; 3D velocity model box of Yarce et al. 2021 down to ~94 km)
- **Depth range:** approx 0–15 km plate interface focus; full catalogue depth extent not separately stated (velocity model to ~94 km)
- **Magnitude:** Magnitude of completeness $M_c = 1.7$ and Gutenberg-Richter b -value = 1.3, derived from the 451 events with REST-based magnitude estimates (subset of catalogue); local magnitude M_L scale
- **Events:** 854 events (per Zenodo dataset record); paper text reports >3x the manual analyst count and 451 events with magnitude estimates
- **Content:** hypocentral locations; phase arrival-time picks; quality information
- **Producer / contact:** Jefferson Yarce / Anne F. Sheehan (CIRES & Dept. of Geological Sciences, University of Colorado Boulder); Steven Roecker (Rensselaer Polytechnic Institute)
- **Data source:** Continuous waveform data from the HOBITSS ocean-bottom seismometers (FDSN network YH, 2014) plus onshore GeoNet land stations; events detected and located automatically using the REST (Regressive ESTimator) autoregressive phase-arrival-time detector and onset estimator
- **Availability:** available (DOI) · DOI [10.5281/zenodo.7274399](https://doi.org/10.5281/zenodo.7274399)

An automatically generated earthquake catalogue for the two months around the September–October 2014 northern Hikurangi slow slip event, created to test whether automated detection increases catalogue completeness and to examine the temporal relationship between SSEs and microseismicity. REST applied to HOBITSS OBS + GeoNet data found more than three times as many events as manual analysis, and shows seismicity increasing during and remaining elevated for ~2 weeks after the 2014 SSE. A larger, distinct product from the same OBS deployment as the manual Yarce et al. (2019) catalogue.

Key references: [135, 162, 163].

62. Brothers Volcano Ocean-Bottom Hydrophone Seismicity Catalogue (south Kermadec arc)

background · geographical · other · reviewed

- **Also known as:** Brothers OBH array; Kermadec arc OBH T-wave catalogue
- **Coverage:** 2004 — 2005
- **Region:** Southern Kermadec arc (Brothers submarine volcano)
- **Bounding box:** Brothers volcano caldera, southern Kermadec arc, approx 34.9S, 179.1E (~350 km NE of New Zealand / ~340 km NE of Whakaari/White Island)
- **Events:** 964 regional earthquakes (plus local harmonic tremor)
- **Content:** hypocentral locations; other
- **Producer / contact:** Robert P. Dziak / Joseph H. Haxel / Haru Matsumoto (NOAA Pacific Marine

Environmental Laboratory & Oregon State University, Hatfield Marine Science Center); with Cornel E. J. de Ronde and collaborators (GNS Science / NOAA Kermadec-arc program)

- **Data source:** Four ocean-bottom hydrophones (OBH) deployed on the caldera floor of Brothers submarine volcano for ~7 months (2004–2005); low-frequency (0.5–110 Hz) hydroacoustic T-wave arrivals recorded on three of the four OBH used to locate earthquakes and characterize harmonic tremor
- **Availability:** not available

A hydroacoustic (T-wave) catalogue produced to characterize regional and local seismicity and volcanic harmonic tremor at Brothers volcano in the southern Kermadec intraoceanic arc (~350 km NE of New Zealand). T-wave-derived locations for 964 regional earthquakes show clustering beneath the dacite cone and along the east flank of the caldera, while local harmonic tremor reflects subseafloor hydrothermal/magmatic processes. It is the only OBS/OBH-based seismicity catalogue identified for the NZ Kermadec arc.

Key references: [37].

63. Monowai Submarine Volcano Long-Range Hydroacoustic (T-phase) Activity Catalogue

induced · particular-event-type · other · reviewed

- **Also known as:** Metz et al. (2018) Monowai T-phase catalogue; Tracking Submarine Volcanic Activity at Monowai
- **Coverage:** 2003-07 — 2017-01
- **Region:** Northern Kermadec Arc (Monowai volcanic centre), offshore New Zealand / Kermadec territory, SW Pacific
- **Bounding box:** Point source: Monowai volcanic centre ~25.9 deg S, 177.2 deg W (northern Kermadec Arc, SW Pacific). Detected via IMS hydrophone array near Juan Fernandez Islands.
- **Magnitude:** Detection threshold reported as ~2.2 mb in the Kermadec Arc region (about one order of magnitude better than regional land networks). The '1.4–4.2 mb' arrival range from the proposal brief was NOT corroborated in the sources accessed; treat as unverified.
- **Events:** 82 discrete activity episodes (Jul 2003–Mar 2004 and Apr 2014–Jan 2017)
- **Content:** other (long-range hydroacoustic T-phase detections / activity episodes), quality information, amplitude picks
- **Producer / contact:** D. Metz (University of Oxford); A. B. Watts (University of Oxford); I. Grevemeyer (GEOMAR Helmholtz Centre for Ocean Research Kiel)
- **Data source:** Long-range hydroacoustic (T-phase) recordings from a CTBTO International Monitoring System (IMS) hydrophone array located near the Juan Fernandez Islands (SE Pacific, ~17,000 km distant); direction-of-arrival and density-based spatial clustering analysis of 3.5 years of recordings to track Monowai volcanic centre.
- **Availability:** available (DOI) · DOI [10.1029/2018JB015888](https://doi.org/10.1029/2018JB015888)

A hydroacoustic event catalogue tracking eruptive/seismic activity at the Monowai submarine volcanic centre, northern Kermadec Arc (within New Zealand's Kermadec territory). Produced to demonstrate that ultra-long-range SOFAR-channel T-phase recordings from distant IMS hydrophone arrays can resolve submarine volcanic activity roughly one order of magnitude smaller than land-based seismic networks. 82 discrete episodes of activity were identified across two recording windows. Distinct from the Brothers volcano local OBH T-wave catalogue (different volcano; remote IMS hydrophone array rather than local ocean-bottom instruments).

Key references: [88, 89].

64. Dataset of Earthquakes in the Northern Hikurangi Subduction Zone During the 2019 Slow Slip Event (2018–2019 OBS deployment)

background · particular-event-type · template-matching · reviewed

- **Also known as:** Iwasaki et al. (2025) northern Hikurangi 2019 SSE OBS catalogue; Zenodo 15713529
- **Coverage:** 2018 — 2019
- **Region:** Northern Hikurangi subduction zone, offshore Gisborne, Aotearoa New Zealand

- **Events:** 964 located events
- **Content:** hypocentral locations, phase arrival-time picks, other (event waveforms, instrument responses, station coordinates)
- **Producer / contact:** Y. (Yuriko) Iwasaki (lead author); co-authors S. Schwartz, K. Mochizuki, H. Crume, D. Bassett, T. Yamada, D. Morad
- **Data source:** Ocean-bottom seismometer (OBS) network deployed 2018-2019 directly above the main slip region of the 2019 northern Hikurangi slow slip event, offshore Gisborne; high-resolution catalogue built with a detailed velocity model plus machine-learning and template-matching detection. Archive includes the earthquake catalogue, three-component OBS event waveforms, instrument (pole-zero) responses, and station coordinates.
- **Availability:** available (DOI) · DOI [10.5281/zenodo.15713529](https://doi.org/10.5281/zenodo.15713529)

An OBS earthquake catalogue from a deployment directly above the main slip region of the large 2019 northern Hikurangi slow slip event, offshore Gisborne, New Zealand. Produced to characterise how the SSE modulated nearby seismicity, revealing intraslab seismicity near subducted seamounts reactivated by the slow slip and potentially influenced by fluid migration. Distinct from the earlier HOBITSS 2014-2015 manual (Yarce et al. 2019) and auto-detection (Yarce et al. 2023) catalogues - this covers the later 2018-2019 OBS deployment and the 2019 SSE.

Key references: [78, 79].

7.7 Volcano-seismic, geothermal and induced catalogues

Volcano-monitoring, magmatic-unrest, and geothermal injection-induced microseismicity catalogues, typically from dense local or borehole networks.

65. Okataina Volcanic Centre Double-Difference Relocated Catalogue (2010-2021)

background · geographical · manual · reviewed

- **Also known as:** OVC double-difference catalogue; Bannister et al. 2022 Okataina tomography catalogue
- **Coverage:** 2010 — 2021
- **Region:** Okataina Volcanic Centre, Taupo Volcanic Zone
- **Bounding box:** Okataina Volcanic Centre / eastern Taupo Volcanic Zone, Rotorua region, North Island, New Zealand
- **Depth range:** low-velocity magmatic zone resolved at ~5-8 km depth
- **Events:** 6989
- **Content:** hypocentral locations; relative relocations; phase arrival-time picks; other
- **Producer / contact:** Stephen Bannister, GNS Science
- **Data source:** Temporary array of broadband and short-period seismometers (~4+ years) supplementing the permanent GeoNet New Zealand National Seismograph Network; double-difference tomography with waveform cross-correlation
- **Availability:** available on request · DOI [10.1016/j.jvolgeores.2022.107653](https://doi.org/10.1016/j.jvolgeores.2022.107653)

Produced to image sub-caldera structure and seismicity beneath the Okataina Volcanic Centre (TVZ, Rotorua region) using a 3-D P-wave velocity model derived by double-difference seismic tomography. The study relocated 6989 earthquakes (2010-2021) to resolve spatial clusters (e.g. west of Lake Rotomahana, beneath Haroharo/Makatiti) and a ~5-8 km low-velocity zone interpreted as partial melt and/or fluid-volatile pathways.

Key references: [13].

66. Taupo Volcano Magmatic-Seismicity Catalogue (2019-2022 ECLIPSE temporary deployment)

background · geographical · manual · reviewed

- **Also known as:** Mestel et al. 2025 Taupo catalogue; ECLIPSE Taupo deployment

- **Coverage:** 2019 — 2022
- **Region:** Taupo Volcano / Taupo Volcanic Zone
- **Bounding box:** Taupo caldera / Lake Taupo, central Taupo Volcanic Zone, North Island, New Zealand
- **Depth range:** arcuate ring-fault structure resolved at ~6 km depth
- **Content:** hypocentral locations; phase arrival-time picks; fault-geometry; other
- **Producer / contact:** Eleanor R. H. Mestel / Finnigan Illsley-Kemp, Victoria University of Wellington (ECLIPSE programme)
- **Data source:** Dedicated temporary broadband deployment of ~13 seismometers (ECLIPSE programme) supplementing the GeoNet permanent network around Lake Taupo
- **Availability:** available on request · DOI [10.1029/2025JB031644](https://doi.org/10.1029/2025JB031644)

Built from a dedicated temporary broadband deployment (ECLIPSE programme) to improve location control under Lake Taupo where the permanent network gives poor coverage, and to separate magmatic from tectonic seismicity through the 2022–2023 unrest. Resolves an arcuate caldera-ring-fault structure at roughly 6 km depth. A priority temporary-deployment catalogue.

Key references: [87].

67. Taupo 2019 Volcanic Unrest Earthquake Catalogue (Illsley-Kemp et al. 2021)

background · particular-event-type · template-matching · reviewed

- **Also known as:** Illsley-Kemp et al. 2021 Taupo unrest catalogue
- **Coverage:** 2019 — 2019
- **Region:** Taupo Volcano / Taupo Volcanic Zone
- **Bounding box:** Taupo caldera / Lake Taupo, central Taupo Volcanic Zone, North Island, New Zealand
- **Magnitude:** up to ~M5.2 (largest event of the 2019 unrest)
- **Events:** >1,100 (caldera earthquakes)
- **Content:** hypocentral locations; phase arrival-time picks; other
- **Producer / contact:** Finnigan Illsley-Kemp, Victoria University of Wellington (VUW / EQcorrscan team)
- **Data source:** GeoNet network data analysed by the VUW group; integrated with deformation (GNSS/InSAR) and gas observations
- **Availability:** available on request · DOI [10.1029/2021GC009803](https://doi.org/10.1029/2021GC009803)

Catalogue of the first instrumentally recorded Taupo caldera unrest episode (2019), documenting >1,100 caldera earthquakes culminating in an M5.2 event, interpreted as related to a magma-mush margin. Integrates seismicity with ground deformation and gas data to assess causes, mechanisms and implications of the unrest.

Key references: [70].

68. Taupo 2022–2023 Volcanic Unrest Seismicity Catalogue (Lamb et al. 2024)

background · particular-event-type · manual · reviewed

- **Also known as:** Lamb et al. 2024 Taupo unrest catalogue
- **Coverage:** 2022-05 — 2023-05
- **Region:** Taupo Volcano / Taupo Volcanic Zone
- **Bounding box:** Taupo caldera / Lake Taupo, central Taupo Volcanic Zone, North Island, New Zealand
- **Magnitude:** largest event M5.7 (largest beneath Lake Taupo in ~50 years)
- **Content:** hypocentral locations; phase arrival-time picks; focal mechanisms; other
- **Producer / contact:** Oliver Lamb / Stephen Bannister, GNS Science / GeoNet
- **Data source:** GeoNet network data; moment-tensor and velocity-model analysis by GNS Science
- **Availability:** available (DOI) · DOI [10.26443/seismica.v3i2.1125](https://doi.org/10.26443/seismica.v3i2.1125)

Characterises the May 2022–May 2023 Taupo unrest episode, which produced the highest earthquake count of any instrumented Taupo unrest and the largest earthquake beneath the lake in ~50 years (M5.7). Moment-tensor analysis of the largest event shows an inflationary isotropic component, leading the authors to infer

magmatic intrusion at depth triggering fault reactivation.

Key references: [81].

69. Cascading Earthquake Swarms, Northern Taupo Volcanic Zone (2019, Aber et al. 2025)

background · geographical · other · reviewed

- **Also known as:** Aber et al. 2025 cascading swarms catalogue
- **Coverage:** 2019-03 — 2019-09
- **Region:** Northern/central Taupo Volcanic Zone (Okataina, Rotorua, toward Whakaari)
- **Bounding box:** ~175 km of the northern and central Taupo Volcanic Zone, North Island, New Zealand
- **Events:** 10 swarms (number of individual events not verified)
- **Content:** hypocentral locations; other
- **Producer / contact:** S. Aber / C. J. Ebinger (with VUW and GNS co-authors)
- **Data source:** GeoNet network data and temporary deployments across the northern/central TVZ; swarm detection and relocation
- **Availability:** available on request · DOI [10.1029/2024GC012079](https://doi.org/10.1029/2024GC012079)

Documents and analyses 10 spatio-temporal earthquake swarms cascading along ~175 km of the northern and central Taupo Volcanic Zone (March–September 2019), near Okataina and Rotorua and extending toward Whakaari/White Island, interpreted as fault-magma interaction. Treats the same 2019 episode as Illsley-Kemp et al. 2021 but focuses on the broader northern-TVZ cascading sequence rather than the Taupo caldera.

Key references: [6].

70. Tarawera 2019 Dyke-Intrusion Earthquake Catalogue (Puhipuhi, Okataina; Benson et al. 2021)

background · particular-event-type · template-matching · reviewed

- **Also known as:** Benson et al. 2021 Tarawera/Puhipuhi swarm catalogue
- **Coverage:** 2019-03 — 2019-03
- **Region:** Tarawera / Okataina Volcanic Centre, Taupo Volcanic Zone
- **Bounding box:** Puhipuhi embayment, NE of Mt Tarawera, Okataina Volcanic Centre, North Island, New Zealand
- **Depth range:** ~8–10.5 km
- **Magnitude:** ML -1.12 to 2.54
- **Events:** 348 detected; 94 relocated
- **Content:** hypocentral locations; relative relocations; phase arrival-time picks; other
- **Producer / contact:** Thomas W. Benson / Finnigan Illsley-Kemp, Victoria University of Wellington
- **Data source:** GeoNet network data, analysed with ObsPy and GrowClust (matched-filter detection + relative relocation)
- **Availability:** available (DOI) · DOI [10.5281/zenodo.4035171](https://doi.org/10.5281/zenodo.4035171)

Catalogue of a March 2019 earthquake swarm in the Puhipuhi embayment NE of Tarawera (Okataina Volcanic Centre), interpreted as a dyke intrusion. Detected 348 events and relocated 94 of them (ML -1.12 to 2.54, ~8–10.5 km depth), demonstrating magmatic intrusion during the 2019 activity.

Key references: [17, 18].

71. Waimangu-Rotomahana-Mt Tarawera Geothermal Field Earthquake Swarm Catalogue (2004–2015)

background · geographical · manual · reviewed

- **Also known as:** Bannister et al. 2016 WRT swarm catalogue
- **Coverage:** 2004 — 2015
- **Region:** Waimangu-Rotomahana-Tarawera, Okataina / Taupo Volcanic Zone
- **Bounding box:** Waimangu-Rotomahana-Mt Tarawera geothermal field, SW Okataina Volcanic Centre, North Island, New Zealand
- **Events:** 582 (relocated)
- **Content:** hypocentral locations; relative relocations; phase arrival-time picks
- **Producer / contact:** Stephen Bannister, GNS Science
- **Data source:** GeoNet / GNS network data; earthquake relocation of swarm seismicity
- **Availability:** not available · DOI [10.1016/j.jvolgeores.2015.07.024](https://doi.org/10.1016/j.jvolgeores.2015.07.024)

Catalogue of 582 relocated earthquakes (2004–2015) in the Waimangu-Rotomahana-Mt Tarawera (WRT) geothermal field on the SW flank of the Okataina Volcanic Centre. The swarm activity highlights crustal faulting associated with the geothermal/volcanic system in the Taupo Volcanic Zone.

Key references: [12].

72. Mt Ruapehu Earthquake Swarm Catalogue (1990–2023)

background · geographical · machine-learning · reviewed

- **Also known as:** Mitchinson et al. 2024 Ruapehu ML catalogue
- **Coverage:** 1990 — 2023
- **Region:** Mt Ruapehu, Taupo Volcanic Zone
- **Bounding box:** Mt Ruapehu region, southern Taupo Volcanic Zone, North Island, New Zealand
- **Magnitude:** ML ~1.58–4.21; Mc ~1.6
- **Events:** 2,795 swarm events (from 28,522 catalogued)
- **Content:** hypocentral locations; other
- **Producer / contact:** Sam Mitchinson / Jessica H. Johnson, University of East Anglia
- **Data source:** GeoNet QuakeSearch earthquake catalogue (33 years of locations); unsupervised machine-learning clustering (HDBSCAN/DBSCAN)
- **Availability:** available (DOI) · DOI [10.3389/feart.2024.1343874](https://doi.org/10.3389/feart.2024.1343874)

A derived catalogue isolating earthquake swarms at Mt Ruapehu from decades of GeoNet QuakeSearch data using unsupervised machine-learning clustering (HDBSCAN/DBSCAN). From 28,522 catalogued events it identifies 2,795 swarm-related events in 8 sequences (ML ~1.58–4.21, Mc ~1.6) to test whether the 'Erua' swarm was distinctive in space, time and magnitude.

Key references: [96].

73. Whakaari/White Island Very-Long-Period (VLP) Earthquake Catalogue (2007–2019)

background · particular-event-type · other · reviewed

- **Also known as:** Park et al. 2020 Whakaari VLP families F1/F2
- **Coverage:** 2007 — 2019
- **Region:** Whakaari/White Island, Taupo Volcanic Zone (offshore)
- **Bounding box:** Whakaari/White Island, Bay of Plenty, offshore North Island, New Zealand
- **Content:** hypocentral locations; quality information; other
- **Producer / contact:** Iseul Park / Arthur Jolly, University of Canterbury / GNS Science
- **Data source:** GeoNet seismic stations on Whakaari/White Island; waveform-semblance detection and clustering of VLP events
- **Availability:** available on request · DOI [10.1186/s40623-020-01224-z](https://doi.org/10.1186/s40623-020-01224-z)

A long-term classification catalogue of very-long-period (VLP) volcanic earthquakes at Whakaari/White Island over 2007 to end-2019, detected by waveform semblance and grouped into two families (F1, F2). F2 swarms mark the onset of unrest, making the catalogue relevant to eruption forecasting at the volcano.

Key references: [99].

74. Ngatamariki Geothermal Injection-Induced Microseismicity Catalogue (2012-2015)

induced · particular-event-type · template-matching · reviewed

- **Also known as:** Hopp et al. 2019 Ngatamariki matched-filter catalogue
- **Coverage:** 2012 — 2015
- **Region:** Ngatamariki geothermal field, Taupo Volcanic Zone
- **Bounding box:** Ngatamariki geothermal field, central Taupo Volcanic Zone, North Island, New Zealand
- **Depth range:** ~1-3 km (reservoir)
- **Magnitude:** ML ~0.26-3.17
- **Events:** ~77,457 total (templates + detections); ~41,114 located (approximate)
- **Content:** hypocentral locations; relative relocations; phase arrival-time picks; focal mechanisms; other
- **Producer / contact:** Chet Hopp / John Townend & Martha Savage, Victoria University of Wellington (with Mercury NZ)
- **Data source:** Mercury NZ operator temporary seismic network at Ngatamariki plus GeoNet; matched-filter (template-matching) detection
- **Availability:** available on request · DOI [10.1029/2019GC008243](https://doi.org/10.1029/2019GC008243)

Catalogue of injection/stimulation-induced microseismicity at the Ngatamariki geothermal field tied to 2012-2015 well stimulation and injection, built by matched-filter detection. Reports very large numbers of detections (templates plus matched-filter detections), a located subset in a ~1-3 km reservoir (ML ~0.26-3.17), and includes focal mechanisms, examining the relationship between injection and induced seismicity.

Key references: [63].

75. Rotokawa Geothermal Field Induced Microseismicity Catalogue (2012-2015)

induced · particular-event-type · template-matching · reviewed

- **Also known as:** Hopp et al. 2020 Rotokawa matched-filter catalogue
- **Coverage:** 2012 — 2015
- **Region:** Rotokawa geothermal field, Taupo Volcanic Zone
- **Bounding box:** Rotokawa geothermal field, central Taupo Volcanic Zone, North Island, New Zealand
- **Depth range:** ~1-3 km (reservoir)
- **Magnitude:** ML ~0.37-3.52; Mc ~0.74
- **Events:** ~25,148 events (2,665 templates); ~6,500 relocated (approximate)
- **Content:** hypocentral locations; relative relocations; phase arrival-time picks; other
- **Producer / contact:** Chet Hopp / John Townend & Martha Savage, Victoria University of Wellington (with Mercury NZ)
- **Data source:** Mercury NZ operator temporary seismic network at Rotokawa plus GeoNet; matched-filter (template-matching) detection and double-difference relocation
- **Availability:** available on request · DOI [10.1016/j.geothermics.2019.101750](https://doi.org/10.1016/j.geothermics.2019.101750)

A ~four-year matched-filter catalogue of induced/produced seismicity at the Rotokawa geothermal field, examining the seismic response to evolving injection. Reports thousands of templates yielding tens of thousands of detected events, with a relocated subset in a ~1-3 km reservoir (ML ~0.37-3.52, Mc ~0.74).

Key references: [64].

76. Rotokawa & Ngatamariki Combined Matched-Filter Catalogue (2012-2015, anisotropy)

induced · parameter-testing · template-matching · reviewed

- **Also known as:** Mroczek et al. 2020 combined geothermal catalogue

- **Coverage:** 2012 — 2015
- **Region:** Rotokawa & Ngatamariki geothermal fields, Taupo Volcanic Zone
- **Bounding box:** Rotokawa and Ngatamariki geothermal fields, central Taupo Volcanic Zone, North Island, New Zealand
- **Depth range:** ~1-3 km (reservoirs)
- **Events:** ~160,000 matched-filter detections; ~17,702 used for shear-wave splitting (approximate)
- **Content:** hypocentral locations; phase arrival-time picks; other
- **Producer / contact:** Stefan Mroczek / Martha K. Savage, Victoria University of Wellington (with Mercury NZ)
- **Data source:** Combined Rotokawa + Ngatamariki matched-filter microseismicity (Mercury NZ networks + GeoNet); shear-wave splitting analysis
- **Availability:** available on request · DOI [10.1093/gji/ggz400](https://doi.org/10.1093/gji/ggz400)

A combined Rotokawa + Ngatamariki induced-microseismicity catalogue (~2012-2015) used to measure shear-wave splitting and track reservoir stress and fracture changes with production/injection. Reports on the order of ~160,000 matched-filter detections, of which ~17,702 events were used for the shear-wave-splitting analysis. Distinct from the single-field Hopp 2019/2020 catalogues by combining both fields and adding SWS.

Key references: [97].

77. Rotorua and Kawerau Geothermal Systems Relocated Seismicity Catalogue (1984-2004; Clarke et al. 2009)

background · geographical · manual · reviewed

- **Also known as:** Clarke et al. 2009 Rotorua-Kawerau relocated seismicity; Rotorua and Kawerau geothermal seismicity (TVZ)
- **Coverage:** 1984 — 2004
- **Region:** Rotorua and Kawerau geothermal fields, central/eastern Taupo Volcanic Zone, North Island, New Zealand
- **Bounding box:** Two sub-areas in the central/eastern Taupo Volcanic Zone, North Island, New Zealand: Rotorua geothermal system (~38.1S, 176.25E) and Kawerau geothermal field (~38.07S, 176.74E).
- **Depth range:** Shallow crustal, approximately 0-20 km (datasets defined as events <=20 km depth)
- **Magnitude:** Local magnitude (ML); low-magnitude microseismicity to small earthquakes.
- **Events:** Input datasets: 504 shallow earthquakes around Rotorua and 1875 around Kawerau (1984-2004); relocated hypocentres computed for 155 (Rotorua) and 400 (Kawerau) earthquakes using combined cross-correlation and catalogue arrival times.
- **Content:** hypocentral locations, relative relocations, phase arrival-time picks, quality information, focal mechanisms
- **Producer / contact:** D. Clarke (lead author); J. Townend, M.K. Savage (Victoria University of Wellington); S. Bannister (GNS Science). Underlying picks/waveforms: GeoNet / GNS Science.
- **Data source:** Earthquakes recorded by the New Zealand national seismograph network and local stations around the Rotorua and Kawerau geothermal areas; parent catalogue arrival-time picks plus newly computed cross-correlation differential travel times. New 1-D P/S velocity models and station corrections derived with VELEST; relocation with HypoDD double-difference algorithm.
- **Availability:** available on request · DOI [10.1016/j.jvolgeores.2008.11.004](https://doi.org/10.1016/j.jvolgeores.2008.11.004)

A dedicated double-difference (HypoDD) relocated earthquake catalogue for two Taupo Volcanic Zone geothermal systems (Rotorua city / Mt Ngongotaha and the Kawerau geothermal field), produced to clarify the nature of faulting and the relationship between seismicity and known/inferred geological structures in these fields. New 1-D velocity models and station corrections were derived and a subset of events relocated using a combination of catalogue and cross-correlation arrival times. It is distinct from the inventoried TVZ-wide high-precision 2007-2023 catalogue: an earlier period, field-specific focus, and different methods.

Key references: [28].

78. Rotokawa Geothermal Field Microseismicity Catalogue 2008-2012 (Sherburn et al. 2015)*induced · geographical · manual · reviewed*

- **Also known as:** Sherburn et al. 2015 Rotokawa microseismicity; Microseismicity at Rotokawa geothermal field 2008-2012
- **Coverage:** 2008 — 2012
- **Region:** Rotokawa geothermal field, central Taupo Volcanic Zone, North Island, New Zealand
- **Bounding box:** Rotokawa geothermal field, central Taupo Volcanic Zone, North Island, New Zealand (~38.62S, 176.20E); most events confined to a ~1 km² zone between reinjection and production wells.
- **Depth range:** Shallow reservoir depths; >70% of events concentrated ~1.5-3 km depth
- **Magnitude:** ML; >1000 events ML≥0.8, ~50 events ML≥2, 2 events ML≥3 (largest M3.1). Stated completeness around ML~0.8.
- **Events:** >1000 events of ML≥0.8 located mid-2008 to end-2012 (50 of ML≥2; 2 of ML≥3)
- **Content:** hypocentral locations, phase arrival-time picks, quality information
- **Producer / contact:** S. Sherburn (lead author, GNS Science) and co-authors S.M. Sewell, S. Bourguignon, W. Cumming, S. Bannister, C. Bardsley, J. Winick, J. Quinao, I.C. Wallis (GNS Science / Mighty River Power / Contact / consultants).
- **Data source:** Events located by the local Rotokawa geothermal-field seismic network (operator monitoring), 2008-2012; located using local-network arrival-time picks. Interpreted in the context of deep fluid reinjection (e.g. Nga Awa Purua power station) and production records.
- **Availability:** available (DOI) · DOI [10.1016/j.geothermics.2014.11.001](https://doi.org/10.1016/j.geothermics.2014.11.001)

A field-specific induced-microseismicity catalogue for the Rotokawa geothermal field (TVZ) covering mid-2008 to end-2012, interpreted as injection-induced via cooling-driven reservoir contraction (reinjecting fluid ~200°C cooler than the reservoir). It documents the spatial-temporal variation of microseismicity and its close correlation with deep reinjection rate. It is distinct from the inventoried Rotokawa matched-filter catalogue (Hopp et al. 2020, 2012-2015) and the Mroczek et al. 2020 anisotropy catalogue (2012-2015): different time window and detection approach (local-network locations rather than 2012-2015 array template matching).

Key references: [122].

79. Wairakei Geothermal Field Microseismicity Catalogue (Wairakei Seismic Network; Sepulveda et al. 2015)*induced · geographical · manual · partially reviewed*

- **Also known as:** Wairakei Seismic Network (WSN) microseismicity; Spatial-temporal Characteristics of Microseismicity (2009-2014) of the Wairakei Geothermal Field (Sepulveda et al. 2015, WGC); Wairakei-Tauhara downhole microseismic catalogue
- **Coverage:** 2009-03 — ongoing
- **Region:** Wairakei-Tauhara geothermal field, central Taupo Volcanic Zone, North Island, New Zealand
- **Bounding box:** Wairakei-Tauhara geothermal field, central Taupo Volcanic Zone, North Island, New Zealand (~38.62S, 176.10E).
- **Depth range:** Shallow crustal; ~0 to ~8.5 km, with 95% of seismicity reported above ~6.5 km and 99% above ~8.5 km, constraining the reservoir base
- **Magnitude:** Microseismicity (low magnitude)
- **Events:** 5,649 located events (March 2009 - April 2013), per Sepulveda et al. (2015) and Sepulveda et al. (2013); spatial-temporal characterization extended to 2009-2014
- **Content:** hypocentral locations, quality information
- **Producer / contact:** Fabian R. Sepulveda (Contact Energy Ltd, Wairakei) and co-authors (J. Andrews/GNS Science; J. Kim; C. Siega; S.F. Milloy). Operator: Contact Energy Ltd.
- **Data source:** Local Wairakei Seismic Network of 9-13 downhole/borehole seismometers (depths ~65 m to ~1400 m) operated by Contact Energy, commissioned from 2009; events located from local-network arrival times.
- **Availability:** available on request

A dedicated operator-run downhole borehole microseismic monitoring catalogue for the Wairakei (Wairakei-Tauhara) geothermal field, used to constrain the deep field structure, reservoir base, fluid-circulation depth

and the modern (microseismicity-constrained) field boundary at depth. It is not represented in the existing inventory. Recorded by a 9-13 station borehole network commissioned from early 2009.

Key references: [115, 116].

80. Mt Ruapehu 2022 Volcanic Unrest Drumbeat / Tremor Seismicity Catalogue (Bramwell et al. 2025)

background · particular-event-type · manual · reviewed

- **Also known as:** Drumbeat and low frequency catalogue for 2022 unrest at Ruapehu volcano (Zenodo 13910455); Bramwell et al. 2025 Ruapehu drumbeat/tremor catalogue
- **Coverage:** 2022-03 — 2022-07
- **Region:** Mt Ruapehu, Tongariro National Park, southern Taupo Volcanic Zone, North Island, New Zealand
- **Bounding box:** Mt Ruapehu crater/summit area, southern Taupo Volcanic Zone, North Island, New Zealand (~39.28S, 175.57E)
- **Depth range:** Shallow volcanic / crater-lake hydrothermal system
- **Magnitude:** Low-frequency drumbeat earthquakes and tremor (not characterized by standard ML magnitude/Mc); analysed in time, amplitude and frequency domains
- **Events:** 42,951 LF drumbeats; ~144 h of overlapping drumbeats; >2000 h of tremor over the 121-day unrest period (7 March - 6 July 2022)
- **Content:** other (low-frequency drumbeat earthquakes and volcanic tremor; source dynamics), quality information
- **Producer / contact:** Liam A. Bramwell (lead author, Victoria University of Wellington); co-authors Finnigan Illsley-Kemp, Ery C. Hughes, Sophie Butcher, Oliver D. Lamb, Yannik Behr. Waveforms: GeoNet / GNS Science.
- **Data source:** Continuous seismic waveforms from the GeoNet permanent network around Ruapehu (19 stations within 26 km; primary station MAVZ ~1.45 km N of the crater lake). Events manually picked using the Pyrocko Snuffler toolbox. Full catalogue archived on Zenodo.
- **Availability:** available (DOI) · DOI [10.5281/zenodo.13910455](https://doi.org/10.5281/zenodo.13910455)

A manually picked catalogue of low-frequency (LF) drumbeat earthquakes and volcanic tremor produced for Mt Ruapehu’s 2022 volcanic unrest, used to characterize the source dynamics (magma ascent into the shallow system) alongside crater-lake temperature and gas data. It is a distinct event-type catalogue (LF drumbeats + tremor) not captured by the inventoried Mitchinson et al. 2024 Ruapehu ML swarm catalogue (which clusters GeoNet’s standard volcano-tectonic catalogue, 1990-2023).

Key references: [21, 22].

81. Auckland Volcanic Field Matched-Filter Earthquake Catalogue (van Wijk et al. 2021, COVID-19 lockdown study)

background · geographical · template-matching · reviewed

- **Also known as:** AVF COVID-19 lockdown matched-filter catalogue; van Wijk et al. 2021 Auckland Volcanic Field seismicity; Auckland Volcanic Seismic Network (AVSN) detections
- **Coverage:** 2019-11 — 2020-06
- **Region:** Auckland Volcanic Field, North Island, New Zealand
- **Bounding box:** Auckland Volcanic Field, greater Auckland region, North Island, New Zealand (~36.9S, 174.8E)
- **Depth range:** Shallow crustal
- **Magnitude:** Small local earthquakes
- **Events:** 35 local earthquakes (1 Nov 2019 - 15 Jun 2020), expanded from 5 known GeoNet events (30 new detections)
- **Content:** hypocentral locations, phase arrival-time picks
- **Producer / contact:** Kasper van Wijk (lead author, University of Auckland); co-authors Calum

J. Chamberlain (VUW), Thomas Lecocq (Royal Observatory of Belgium), Koen Van Noten (Royal Observatory of Belgium). Waveforms: GeoNet FDSN.

- **Data source:** Continuous seismic data from the Auckland Volcanic Seismic Network (AVSN), available via the GeoNet FDSN service. 59 known-earthquake templates used in a matched-filtering (template matching) scheme with EQcorrscan to detect additional local events; starting GeoNet catalogue of 5 known local events.
- **Availability:** available on request · DOI [10.5194/se-12-363-2021](https://doi.org/10.5194/se-12-363-2021)

A small but genuinely distinct local catalogue for the Auckland Volcanic Field (a monogenetic volcanic field not otherwise represented in the inventory). Template matched-filtering on the AVSN identified 35 local earthquakes for 1 Nov 2019–15 Jun 2020 (30 new detections beyond the 5 GeoNet events), exploiting reduced urban seismic noise during the COVID-19 lockdown. Produced to test detection capability and seismic monitoring of the AVF during the lockdown 'quieting'.

Key references: [141, 142].

7.8 Slow-slip, tremor, LFE and moment-tensor catalogues

Catalogues of the subduction slow-earthquake family (slow-slip events, tectonic tremor, low-frequency and repeating earthquakes) together with regional moment-tensor solutions.

82. GeoNet New Zealand Earthquake Moment Tensor Solutions (MwNZ)

background · geographical · automatic · partially reviewed

- **Also known as:** MwNZ; GeoNet CMT; regional moment tensor (RMT) catalogue; Ristau RMT catalogue
- **Coverage:** 2003–08 — ongoing
- **Region:** New Zealand and adjacent offshore regions
- **Bounding box:** New Zealand and adjacent offshore regions (approx. NZ landmass plus Hikurangi/Kermadec and Puysegur margins)
- **Magnitude:** Approximately Mw 3.0–8.0; solutions routinely computed for moderate ($M > \sim 4$) earthquakes
- **Events:** ~1900 (grows monthly; exact count not stated on landing pages)
- **Content:** focal mechanisms; hypocentral locations; quality information
- **Producer / contact:** GNS Science / GeoNet (John Ristau, GNS Science)
- **Data source:** Regional broadband seismic waveform data from the GeoNet national network (and adjacent stations), inverted for regional centroid moment tensor solutions; routine production since August 2003.
- **Availability:** available (DOI) · DOI [10.21420/MMJ9-CZ67](https://doi.org/10.21420/MMJ9-CZ67)

National regional centroid moment tensor (focal mechanism) catalogue for New Zealand and adjacent offshore earthquakes, produced routinely since August 2003 to provide consistent Mw and fault-geometry information for moderate-to-large NZ earthquakes. It established the MwNZ regional moment magnitude scale used across NZ seismology and hazard work. Two inversion approaches have been used over time (Dreger/Berkeley code 2003–2020; a Penn State inversion code from June 2020).

Key references: [53, 106, 107].

83. Hikurangi Slow Slip Event Inventory (Wallace & Beavan 2010)

compilation · particular-event-type · other · reviewed

- **Also known as:** Hikurangi SSE catalogue; Hikurangi slow-slip inventory; Wallace & Beavan 2010
- **Coverage:** 2002 — 2010
- **Region:** Hikurangi margin
- **Bounding box:** North Island, New Zealand / Hikurangi subduction margin (eastern North Island)
- **Depth range:** shallow (north/central) to >25–30 km (deep southern SSEs)
- **Magnitude:** Equivalent moment magnitudes approximately Mw 6.3–7.2
- **Events:** ~15 slow slip events

- **Content:** fault-geometry; other
- **Producer / contact:** Laura M. Wallace and John Beavan, GNS Science
- **Data source:** Continuous GPS (cGPS) time series from the GeoNet continuous GPS network across the North Island / Hikurangi margin; elastic dislocation modelling of surface displacements.
- **Availability:** available on request · DOI [10.1029/2010JB007717](https://doi.org/10.1029/2010JB007717)

Foundational inventory of Hikurangi slow slip events documented with continuous GPS, revealing marked diversity in SSE behaviour: shallow, frequent events in the north and central margin versus deep (>25–30 km), roughly 5-yearly events in the south. It underpins nearly all subsequent NZ SSE studies and characterised durations (days to >1 year), equivalent moment release (Mw ~6.3–7.2) and recurrence intervals (2 to >5 years).

Key references: [147].

84. 14-Year GNSS Slow Slip Event Catalogue for the Hikurangi Subduction Zone (Michel et al. 2025)

compilation · particular-event-type · other · reviewed

- **Also known as:** Hikurangi 14-year SSE catalogue; Michel et al. 2025; '14 Years of Slip on the Hikurangi Subduction Zone'
- **Coverage:** 2009 — 2023
- **Region:** Hikurangi margin
- **Bounding box:** Hikurangi subduction zone, North Island, New Zealand
- **Events:** 27 slow slip events
- **Content:** fault-geometry; other
- **Producer / contact:** Sylvain Michel et al. (corresponding institutions per the JGR Solid Earth paper)
- **Data source:** 14 years (2009–2023) of GeoNet continuous GNSS time series; Independent Component Analysis decomposition combined with an inversion scheme to derive the slip history on the Hikurangi megathrust.
- **Availability:** available on request · DOI [10.1029/2024JB030865](https://doi.org/10.1029/2024JB030865)

Catalogue of 27 slow slip events extracted from a 14-year (2009–2023) model of aseismic slip on the Hikurangi subduction megathrust derived from GNSS time series. It is the modern successor to the Wallace & Beavan (2010) inventory, providing a continuous spatio-temporal history of accelerations and slowdowns of fault slip. The methodology combines Independent Component Analysis with a slip-inversion scheme.

Key references: [95].

85. Shallow Hikurangi Slow Slip Event Catalogue (Perez-Silva et al. 2025)

compilation · particular-event-type · other · reviewed

- **Also known as:** Perez-Silva et al. 2025 shallow Hikurangi SSE catalogue; SSEcatalog.mat
- **Coverage:** 2006 — 2024
- **Region:** Hikurangi margin
- **Bounding box:** Northern/central Hikurangi subduction margin, North Island, New Zealand
- **Depth range:** shallow (Hikurangi megathrust)
- **Content:** other; quality information
- **Producer / contact:** Andrea Perez-Silva (with Ting Wang, Laura Wallace, Mark Bebbington, Paul Denys)
- **Data source:** GeoNet continuous GPS (cGPS) time series; wavelet-based detection of shallow SSEs, producing start/end times and detecting-station lists.
- **Availability:** available (DOI) · DOI [10.5281/zenodo.14848128](https://doi.org/10.5281/zenodo.14848128)

Catalogue of shallow Hikurangi slow slip events (start/end times and detecting stations) compiled to study SSE occurrence statistics using renewal processes. Built from GeoNet cGPS data with wavelet detection over roughly 2006–2024, it is published as data products (e.g. SSEcatalog.mat plus R/MATLAB analysis

scripts) on Zenodo. It supports a companion GRL paper on assessing recurrence/forecast patterns of shallow Hikurangi SSEs.

Key references: [100, 101].

86. Northern Hikurangi Tectonic Tremor Catalogue 2010-2015 (Todd & Schwartz 2016)

compilation · particular-event-type · other · reviewed

- **Also known as:** Todd & Schwartz 2016 tremor catalogue; northern Hikurangi tremor catalogue
- **Coverage:** 2010 — 2015
- **Region:** Hikurangi margin (Raukumara Peninsula)
- **Bounding box:** Northern Hikurangi margin / Raukumara Peninsula, North Island, New Zealand
- **Content:** other
- **Producer / contact:** Erin K. Todd and Susan Y. Schwartz (University of California, Santa Cruz)
- **Data source:** Densified onshore GeoNet seismic network (northern Hikurangi / Raukumara Peninsula); envelope cross-correlation of continuous seismic data to detect and locate tectonic tremor.
- **Availability:** available on request · DOI [10.1002/2016JB013480](https://doi.org/10.1002/2016JB013480)

First comprehensive tectonic tremor catalogue for the northern Hikurangi margin (2010-2015), associating tremor with geodetic transients off the Raukumara Peninsula and helping identify slow slip events. It used envelope cross-correlation on the densified onshore GeoNet network. This is a tremor catalogue rather than a hypocentral earthquake catalogue.

Key references: [131].

87. Ambient Tectonic Tremor Catalogue: Manawatu, Cape Turnagain, Marlborough & Puysegur (Romanet & Ide 2019)

compilation · particular-event-type · other · reviewed

- **Also known as:** Romanet & Ide 2019 ambient tremor catalogue
- **Coverage:** 2005 — 2016
- **Region:** Manawatu, Cape Turnagain, Marlborough, Puysegur (Fiordland)
- **Bounding box:** Four NZ regions: Manawatu, Cape Turnagain (lower North Island), Marlborough (northern South Island), Puysegur/Fiordland (SW South Island)
- **Content:** other
- **Producer / contact:** Pierre Romanet and Satoshi Ide (University of Tokyo)
- **Data source:** GeoNet continuous seismic data across four NZ regions; tremor detection using duration-energy scaling combined with visual inspection.
- **Availability:** available on request · DOI [10.1186/s40623-019-1039-1](https://doi.org/10.1186/s40623-019-1039-1)

Multi-region ambient tectonic tremor catalogue across four New Zealand areas (Manawatu, Cape Turnagain, Marlborough and Puysegur), notably documenting tremor in the Puysegur/Fiordland subduction system in addition to the Hikurangi-related regions. Tremor was detected using a duration-energy scaling approach plus visual inspection of GeoNet data over roughly 2005-2016. It is a tremor catalogue rather than a hypocentral earthquake catalogue.

Key references: [112].

88. Alpine Fault Low-Frequency Earthquake Catalogue (Chamberlain et al. 2014)

background · particular-event-type · template-matching · reviewed

- **Also known as:** Alpine Fault LFE catalogue 2014; Chamberlain et al. 2014
- **Coverage:** 2009 — 2012
- **Region:** Alpine Fault / Southern Alps

- **Bounding box:** Central Alpine Fault, Southern Alps, South Island, New Zealand
- **Depth range:** ~20-30 km (deep extent of the Alpine Fault)
- **Events:** ~8,760 LFEs in 14 families
- **Content:** hypocentral locations; phase arrival-time picks; relative relocations; other
- **Producer / contact:** Calum J. Chamberlain (with David R. Shelly, John Townend, Tim A. Stern), Victoria University of Wellington / USGS
- **Data source:** GeoNet plus SAMBA (Southern Alps Microearthquake Borehole Array) continuous seismic data; matched-filter (template-matching) detection of low-frequency earthquakes within tremor.
- **Availability:** available on request · DOI [10.1002/2014GC005436](https://doi.org/10.1002/2014GC005436)

First low-frequency earthquake (LFE) catalogue for the deep Alpine Fault: approximately 8,760 LFEs in 14 families over 36 months (2009-2012), revealing punctuated slow slip at roughly 20-30 km depth. It was built from GeoNet and SAMBA continuous data using matched-filter detection within tremor and is the earliest NZ LFE catalogue, serving as a precursor to Baratin et al. (2018).

Key references: [23].

89. Alpine Fault Low-Frequency Earthquake Catalogue with Focal Mechanisms (Baratin et al. 2018)

background · particular-event-type · template-matching · reviewed

- **Also known as:** Baratin et al. 2018 Alpine Fault LFE catalogue
- **Coverage:** 2009 — 2017
- **Region:** Alpine Fault / Southern Alps
- **Bounding box:** Central Alpine Fault, Southern Alps, South Island, New Zealand
- **Depth range:** ~20-30 km (deep extent of the Alpine Fault)
- **Content:** hypocentral locations; phase arrival-time picks; relative relocations; focal mechanisms; other
- **Producer / contact:** Laura-May Baratin (with Calum J. Chamberlain, John Townend, Martha K. Savage), Victoria University of Wellington
- **Data source:** GeoNet and SAMBA continuous seismic data; matched-filter (template-matching) detection of LFEs, extended templates relative to Chamberlain et al. (2014), with LFE focal mechanism determination.
- **Availability:** available on request · DOI [10.1016/j.epsl.2017.12.021](https://doi.org/10.1016/j.epsl.2017.12.021)

Extended, longer-duration deep Alpine Fault LFE catalogue (2009-2017) that adds the first LFE focal mechanisms for the region. It shows that increased LFE rates coincide with tremor and regional earthquakes, interpreted as quasi-continuous deformation and triggered slow slip on the deep Alpine Fault. It builds on and extends the Chamberlain et al. (2014) template set.

Key references: [15].

90. Kaimanawa (Deep Hikurangi) Low-Frequency Earthquake Catalogue (Aden-Antoniow et al. 2024)

background · particular-event-type · template-matching · reviewed

- **Also known as:** Aden-Antoniow et al. 2024 Hikurangi LFE catalogue; Kaimanawa LFE catalogue
- **Coverage:** 2010 — 2022
- **Region:** Hikurangi margin (Kaimanawa Range)
- **Bounding box:** Beneath the Kaimanawa Range, central North Island, New Zealand (Hikurangi subduction zone)
- **Depth range:** ~50 km
- **Events:** ~21,000 events from 36 LFE sources
- **Content:** hypocentral locations; relative relocations; other
- **Producer / contact:** Florent Aden-Antoniow (with William B. Frank, Calum J. Chamberlain, John Townend, Laura M. Wallace, Stephen Bannister)

- **Data source:** GeoNet regional continuous seismic data; two-iteration matched-filter (template-matching) detection using templates derived from prior tectonic tremor observations.
- **Availability:** available on request · DOI [10.1029/2023JB027971](https://doi.org/10.1029/2023JB027971)

First low-frequency earthquake catalogue for the Hikurangi subduction zone proper, beneath the Kaimanawa Range at about 50 km depth, downdip of regularly recurring deep ~M7 slow slip events. It comprises 36 LFE sources producing almost 21,000 events over more than a decade (2010–2022), detected with a two-iteration matched-filter on GeoNet data. It is distinct from the Alpine Fault LFE catalogues.

Key references: [7].

91. Raukumara Peninsula Repeating Earthquake Catalogue 2003–2020 (Hughes et al. 2021)

background · particular-event-type · template-matching · reviewed

- **Also known as:** Hughes et al. 2021 Raukumara repeating-earthquake catalogue
- **Coverage:** 2003 — 2020
- **Region:** Hikurangi margin (Raukumara Peninsula)
- **Bounding box:** Raukumara Peninsula, northern Hikurangi subduction margin, North Island, New Zealand
- **Events:** 61 repeating-earthquake families
- **Content:** hypocentral locations; relative relocations; focal mechanisms; other
- **Producer / contact:** Laura Hughes (with Calum J. Chamberlain, John Townend, Amanda M. Thomas), Victoria University of Wellington
- **Data source:** GeoNet continuous seismic data, northern Hikurangi / Raukumara Peninsula; waveform cross-correlation to identify repeating-earthquake families.
- **Availability:** available on request · DOI [10.1029/2021GC009670](https://doi.org/10.1029/2021GC009670)

Catalogue of 61 repeating-earthquake families on the northern Hikurangi margin (Raukumara Peninsula, 2003–2020) used as an interface ‘creepmeter’. The families coincide spatially with slow-slip patches and include both interplate and intraplate focal mechanisms. It was built from GeoNet continuous data using waveform cross-correlation.

Key references: [65].

92. New Zealand-Wide Earthquake Focal Mechanism Catalogue for Tectonic Stress (Townend et al. 2012)

compilation · geographical · other · reviewed

- **Also known as:** Townend et al. (2012) NZ focal mechanism compilation; central NZ stress-field focal mechanism dataset
- **Coverage:** 2004-01 — 2011-02
- **Region:** Central New Zealand — Hikurangi subduction margin (North Island, including Hawke’s Bay area) and South Island, along the Australia-Pacific plate boundary
- **Events:** ~3424 focal mechanisms
- **Content:** focal mechanisms; hypocentral locations; quality information
- **Producer / contact:** John Townend (lead author), Victoria University of Wellington / GNS Science; co-authors Steven Sherburn (GNS Science), Richard Arnold (VUW), Carolin Boese, Louise Woods
- **Data source:** Compiled first-motion / polarity focal mechanisms derived from earthquakes recorded by the GeoNet national seismograph network (and related temporary deployments); a purpose-built NZ-wide focal-mechanism catalogue assembled for stress-tensor inversion. Distinct from the GeoNet moment-tensor (MwNZ / Ristau RMT) product.
- **Availability:** available on request · DOI [10.1016/j.epsl.2012.08.003](https://doi.org/10.1016/j.epsl.2012.08.003)

A New Zealand-wide compilation of ~3424 earthquake focal mechanisms assembled to estimate the present-day tectonic stress field at ~100 locations across central New Zealand and map the orientation of maximum horizontal compressive stress (SHmax) along the Australia-Pacific plate boundary. Produced to characterise

three-dimensional variations in stress along the Hikurangi subduction margin and South Island, and to relate stress orientation to along-strike variations in subduction-thrust coupling. It was the largest study of the NZ plate-boundary stress field at the time of publication.

Key references: [133].

93. Southeastern South Island / Otago Earthquake Focal Mechanism Catalogue (Williams et al. 2026)

compilation · geographical · other · reviewed

- **Also known as:** Williams et al. (2026) southeastern South Island focal mechanisms; Otago/Southland focal-mechanism catalogue (OtagoNet + SOSA + COSA)
- **Coverage:** unknown — not documented — unknown — not documented
- **Region:** Southeastern South Island, Aotearoa New Zealand — Otago and Southland
- **Magnitude:** MLV 1.3–4.3
- **Events:** 126 combined focal mechanisms (91 rated C-quality or better, of which 57 strike-slip); constituent solutions: 96 OtagoNet + 30 SOSA + 11 COSA (167 array solutions plus GeoNet RMT entries)
- **Content:** focal mechanisms; hypocentral locations; quality information
- **Producer / contact:** Jack Williams (lead author); co-authors Donna Eberhart-Phillips, Sandra Bourguignon, Mark Stirling, Martin Reyners, Phaedra Upton (University of Otago / GNS Science)
- **Data source:** First-motion / polarity focal mechanisms computed from earthquakes recorded by temporary regional networks: OtagoNet (Otago temporary broadband network, Reyners 2014), the Southland Otago Seismic Array (SOSA, Williams et al. 2022), and the Central Otago Seismic Array (COSA); supplemented by GeoNet regional moment-tensor (RMT) catalogue entries.
- **Availability:** available (DOI) · DOI [10.5281/zenodo.17228270](https://doi.org/10.5281/zenodo.17228270)

A regional focal-mechanism catalogue for the southeastern South Island (Otago/Southland) of Aotearoa New Zealand, compiling 126 combined nodal-plane solutions for small earthquakes (MLV 1.3–4.3) to study how oblique Australia-Pacific plate motion is accommodated. The mechanisms reveal a strike-slip stress state with a WNW-trending maximum compressive stress and are used to argue for scale-dependent partitioning of transpressional strain controlled by inherited structures. Data are deposited on Zenodo under CC-BY-4.0.

Key references: [155, 156].

94. HOBITSS Ocean-Bottom Focal-Mechanism / Stress-Tensor Cycling Catalogue, Northern Hikurangi (Warren-Smith et al. 2019)

background · particular-event-type · other · reviewed

- **Also known as:** Warren-Smith et al. (2019) HOBITSS focal mechanisms; northern Hikurangi intraslab focal-mechanism / stress-ratio cycling dataset
- **Coverage:** 2014-05 — 2015-06
- **Region:** Northern Hikurangi subduction margin, offshore Gisborne, North Island, New Zealand
- **Depth range:** ~13–15 km thick intraslab seismicity zone within the subducting Pacific oceanic crust; subduction interface seismogenic depths roughly 15–35 km (per-event depth range not tabulated in the paper)
- **Magnitude:** Earthquakes with $M_w > 1.8$ (lower magnitude threshold for mechanism calculation); upper range and M_c not separately documented
- **Events:** 278 intraslab earthquakes analysed (subset of $M_w > 1.8$ GeoNet events); ~140 focal mechanisms at SSE initiation and ~80 at SSE shutdown used for stress inversions
- **Content:** focal mechanisms; hypocentral locations; quality information; other
- **Producer / contact:** Emily Warren-Smith (lead author), GNS Science, Lower Hutt; co-authors B. Fry, L. Wallace, E. Chon, S. Henrys, A. Sheehan, K. Mochizuki, S. Schwartz, S. Webb, S. Lebedev
- **Data source:** Earthquake first-motion polarity focal mechanisms computed for $M_w > 1.8$ events from the GeoNet national seismicity catalogue, using the 2014–2015 HOBITSS (Hikurangi Ocean Bottom Investigation of Tremor and Slow Slip) ocean-bottom seismometer network (10 broadband + 5 short-period

OBS, plus 24 absolute pressure gauges) combined with onshore GeoNet stations, offshore Gisborne.

- **Availability:** available on request · DOI [10.1038/s41561-019-0367-x](https://doi.org/10.1038/s41561-019-0367-x)

A focal-mechanism catalogue for microearthquakes in the subducting Pacific oceanic crust at the northern Hikurangi margin offshore Gisborne, derived from the 2014-2015 HOBITSS ocean-bottom seismometer deployment. Used to perform time-varying stress-tensor inversions and track the stress shape ratio (R) as a proxy for fluid pressure cycling before, during, and after four shallow slow slip events. The product is distinct in content (focal mechanisms plus a stress-ratio time series) from the HOBITSS hypocentral microearthquake catalogues (Yarce et al. 2019, 2023).

Key references: [152].

95. Manawatu Deep Short-Term Slow-Slip and Tremor Catalogue (Fasola et al. 2023)

background · particular-event-type · other · reviewed

- **Also known as:** Fasola et al. (2023) Manawatu deep tremor / short-term ETS-like SSE; Manawatu/Kaimanawa deep tremor-associated slip model
- **Coverage:** 2005 — 2016
- **Region:** Manawatu and Kaimanawa region, deep central Hikurangi subduction zone, lower North Island, Aotearoa New Zealand
- **Events:** ~354 tremor events (input Romanet & Ide 2019 Manawatu catalogue), grouped into 26 tremor clusters (durations 1-80 days, ~75% of tremor); no separate event count published for the derived slip catalogue
- **Content:** hypocentral locations, fault-geometry, other
- **Producer / contact:** Shannon L. Fasola (lead author; Univ. of Kansas / KU Crustal Deformation group) — co-authors Noel M. Jackson (formerly Noel M. Bartlow, Univ. of Kansas) and Charles A. Williams (GNS Science, New Zealand)
- **Data source:** Tremor locations/times reused from the Romanet & Ide (2019) ambient-tremor catalogue (envelope cross-correlation on GeoNet permanent seismometers; ~354 Manawatu tremor events, 2005-2016). Continuous GNSS daily time-series from the GeoNet Aotearoa NZ Continuous GNSS Network (GNS Science). New products derived in this paper: time-averaged tremor-associated surface displacements/velocities and a static slip inversion (tremor-associated slip distribution) on the Hikurangi plate interface, plus identification of deep short-term ETS-like slow slip.
- **Availability:** available on request · DOI [10.1029/2023GL105428](https://doi.org/10.1029/2023GL105428)

Investigates whether small, short-term ETS-like slow-slip events occur at the deep Manawatu/Kaimanawa tremor source, where tectonic tremor lies adjacent to (not co-located with) the deep long-term Manawatu and Kaimanawa slow-slip events on the central Hikurangi subduction interface. The authors cluster existing tremor, decompose GNSS at tremor times to recover tremor-associated displacement offsets, and invert these for slip on the plate interface, concluding the interface below deep long-term SSEs likely slips frequently in small ETS-like SSEs too small to detect individually with geodesy. Distinct in focus (deep Manawatu tremor-associated slip modeling) from the northern Hikurangi tremor catalogue (Todd & Schwartz 2016) and the source ambient-tremor catalogue (Romanet & Ide 2019).

Key references: [48, 112].

96. Global Centroid Moment Tensor (GCMT) catalogue (New Zealand component)

compilation · particular-event-type · other · reviewed

- **Also known as:** Global CMT; GCMT; formerly Harvard CMT (HRV) catalog
- **Coverage:** 1976 — ongoing
- **Region:** Global; New Zealand component covers onshore NZ, Hikurangi-Kermadec subduction margin, and subantarctic region
- **Bounding box:** Global (NZ component covers approx. -50 to -33 lat, 165 to -176 lon)
- **Depth range:** 0 to ~700 km (centroid depths; global product spanning shallow crustal to deep subduction)
- **Magnitude:** Mw >= ~5.0 globally (systematic threshold); the NZ component contains moderate-to-large

events at or above this threshold

- **Content:** focal mechanisms, hypocentral locations, other
- **Producer / contact:** Global CMT Project, Lamont-Doherty Earth Observatory (LDEO), Columbia University (PI: Göran Ekström; Co-PI: Meredith Nettles). Originally founded at Harvard University (1982); relocated to LDEO in 2006. NSF-funded.
- **Data source:** Long-period and broadband digital seismic waveform data from global networks (e.g. GSN/FDSN), inverted independently at LDEO/Columbia to derive centroid moment tensor solutions. Not a regional NZ network product; the NZ component is the subset of GCMT solutions located within/around New Zealand.
- **Availability:** available (DOI) · DOI <https://doi.org/10.1016/j.pepi.2012.04.002>

The GCMT (formerly Harvard CMT) Project systematically determines centroid moment tensors and focal mechanisms for global earthquakes of $M \geq 5.0$ since 1976, with more than 25,000 solutions. Its NZ component provides an independent, long-standing record of focal mechanisms and centroid locations for moderate-to-large events along the Hikurangi-Kermadec margin, crustal NZ, and the subantarctic region. It is distinct from the regional GeoNet/Ristau MwNZ moment-tensor catalogue, which is NZ-operated and extends to smaller events.

Key references: [38, 46].

7.9 Derived, homogenised and synthetic / hazard-model catalogues

Magnitude-homogenised, declustered, and physics-based synthetic catalogues prepared for the National Seismic Hazard Model.

97. NSHM 2010 Earthquake Catalogue (Stirling et al. 2012)

background · geographical · other · reviewed

- **Also known as:** NSHM 2010 catalogue; Stirling et al. 2012 catalogue
- **Coverage:** 1840 — 2009
- **Region:** New Zealand (nationwide)
- **Bounding box:** New Zealand and its offshore margins (approx. 165E-180E / 178W, 48S-33S)
- **Content:** hypocentral locations; quality information; historical observations; other
- **Producer / contact:** GNS Science (Mark W. Stirling et al.); parent network GeoNet
- **Data source:** Derived backbone catalogue built from the GeoNet/national instrumental earthquake catalogue plus the ~170-year historical NZ earthquake record; homogenised to Mw and declustered, with duplicates, explosions and mining-induced events removed.
- **Availability:** available on request

Homogenised (Mw), declustered backbone earthquake catalogue underpinning the distributed-seismicity component of the 2010 New Zealand National Seismic Hazard Model. It was produced to give a temporally consistent magnitude basis (instrumental + ~170-yr historical record) for probabilistic seismic hazard analysis across New Zealand. The catalogue is documented within and alongside the NSHM 2010 update.

Key references: [125].

98. Mw-Homogenised NZ Earthquake Catalogue for NSHM 2022 (Christophersen et al. 2022)

compilation · parameter-testing · other · reviewed

- **Also known as:** Christophersen et al. 2022 'consistent magnitudes' catalogue; GNS Science report 2021/42
- **Coverage:** 1931 — 2020
- **Region:** New Zealand (nationwide)
- **Bounding box:** New Zealand and offshore margins
- **Magnitude:** Recomputes magnitudes and converts to a standardized Mw proxy (MwNZ); introduces MLNZ20 local-magnitude scale; standardization roughly halves the count of events with $M \geq 4.95$ and

corrects 2011–2012 step changes.

- **Events:** >250,000 earthquakes (magnitudes recomputed for >99% of catalogue)
- **Content:** quality information; other
- **Producer / contact:** GNS Science (Annemarie Christophersen et al.); parent network GeoNet
- **Data source:** GNS reprocessing of >250,000 NZ earthquakes using >2 million individual GeoNet waveforms; derives a new local magnitude scale (MLNZ20) and regressions to convert historical magnitudes to a Mw proxy for consistency over time.
- **Availability:** available (DOI) · DOI [10.21420/A2SN-XM76](https://doi.org/10.21420/A2SN-XM76)

A GNS Science reprocessing that recomputes earthquake magnitudes for the NZ catalogue and standardises them to moment magnitude (Mw) so that earthquake size is consistent over the full catalogue period. It was produced to provide a temporally consistent magnitude backbone for the 2022 National Seismic Hazard Model, correcting step changes (e.g. the 2012 SeisComP transition) and the systematic ML-vs-Mw offset. It is the magnitude basis later folded into the Rollins integrated catalogue and the NSHM 2022 seismicity rate model.

Key references: [26, 27].

99. Integrated / Augmented Earthquake Catalogue for Aotearoa New Zealand (NSHM 2022; Rollins et al.)

compilation · geographical · other · reviewed

- **Also known as:** Augmented NZ earthquake catalogue; Rollins et al. catalogue; GNS Science report 2021/58; Integrated Earthquake Catalog Version 1; NSHM22 Mw-standardized catalogue
- **Coverage:** 1840 — 2020
- **Region:** New Zealand (nationwide) incl. Hikurangi-Kermadec and Puysegur margins
- **Bounding box:** Greater New Zealand region incl. Hikurangi-Kermadec and Puysegur subduction zones
- **Magnitude:** Uses Christophersen et al. 2022 standardized Mw (MwNZ proxy) magnitudes; provides regional magnitude-frequency distributions and Mc in the companion paper (Rollins et al. 2024).
- **Content:** hypocentral locations; relative relocations; focal mechanisms; quality information; historical observations; other
- **Producer / contact:** GNS Science (Chris Rollins, Annemarie Christophersen, Kiran Kumar S. Thingbaijam, Matthew C. Gerstenberger et al.); parent network GeoNet
- **Data source:** GeoNet national operational seismic catalogue, with event parameters (depths, uncertainties, focal mechanisms, locations) overwritten using refined estimates from regional relocation studies, the literature, the NZ Centroid Moment Tensor (CMT) database and global catalogues; magnitudes from the Christophersen et al. 2022 Mw standardization.
- **Availability:** available (DOI) · DOI [10.21420/XT4Y-WY45](https://doi.org/10.21420/XT4Y-WY45)

An augmented, integrated NZ earthquake catalogue built for the 2022 National Seismic Hazard Model. It merges higher-quality depths, uncertainties, focal mechanisms and relocated hypocentres into the GeoNet catalogue, applies the Christophersen 2022 standardized Mw magnitudes, and classifies events as upper-plate (crustal), subduction-interface, or intraslab to feed the separate components of the Seismicity Rate Model. Revised parameters were imported for ~60% of events (incl. ~92% of 2000–2020 events).

Key references: [109–111].

100. NSHM 2022 Seismicity Rate Model Distributed-Seismicity / Declustered Catalogue (Gerstenberger et al. 2024)

background · parameter-testing · other · reviewed

- **Also known as:** NZ NSHM 2022 SRM; Distributed Seismicity Model (DSM)
- **Coverage:** 1840 — 2020
- **Region:** New Zealand (nationwide)
- **Bounding box:** New Zealand and offshore margins incl. Hikurangi-Kermadec and Puysegur subduction zones

- **Magnitude:** Uses Mw-homogenised magnitudes (Christophersen 2022); gridded occurrence rates as a function of magnitude rather than a fixed magnitude range.
- **Content:** other
- **Producer / contact:** GNS Science (Matthew C. Gerstenberger, Russell J. Van Dissen, Chris Rollins, Chris DiCaprio et al.)
- **Data source:** Derived from the declustered, Mw-homogenised NZ catalogue (the Rollins integrated catalogue with Christophersen 2022 standardized magnitudes); declustering sensitivity tested across multiple algorithms. Combines a multi-dataset hybrid model with a non-Poisson uniform-rate-zone model for low-seismicity regions.
- **Availability:** available (DOI) · DOI [10.1785/0120230165](https://doi.org/10.1785/0120230165)

The seismicity rate model (SRM) for the 2022 NZ NSHM, comprising an inversion fault model (IFM) and a distributed seismicity model (DSM). The DSM provides gridded background earthquake rates derived from the declustered, Mw-homogenised national catalogue. It is a gridded-rate hazard product (forecast rates), not a raw event list, produced to forecast distributed seismicity for probabilistic seismic hazard.

Key references: [51].

101. RSQSim Synthetic Earthquake Catalogue for New Zealand (NSHM2012 fault system, 276 kyr)

background · parameter-testing · other · non-reviewed

- **Also known as:** RSQSim Simulated Earthquake Catalog 5091; rundir5091; Shaw et al. 2022
- **Coverage:** synthetic (276,000-year simulated span; published 2021) — synthetic (276,000-year simulated span)
- **Region:** New Zealand (nationwide fault system)
- **Bounding box:** New Zealand (NSHM 2012 fault model domain)
- **Magnitude:** Simulated event magnitudes derived from rupture areas/slip on the fault model; full magnitude distribution over the 276 kyr simulation (no observational M_c).
- **Content:** hypocentral locations; fault-geometry; other
- **Producer / contact:** Bruce E. Shaw (Lamont-Doherty Earth Observatory, Columbia University); collaborators Bill Fry, Andrew Nicol, Andrew Howell, Matthew Gerstenberger (GNS Science)
- **Data source:** Synthetic catalogue generated by the RSQSim rate-and-state earthquake simulator run on the New Zealand NSHM 2012 fault model (fault geometries, slip rates, rupture parameters); not based on observed events.
- **Availability:** available (DOI) · DOI [10.5281/zenodo.5534462](https://doi.org/10.5281/zenodo.5534462)

A ~276,000-year synthetic earthquake catalogue produced by the RSQSim rate-and-state simulator on the NZ NSHM 2012 fault system, generating deterministic finite-fault ruptures for physics-based seismic hazard. It was produced to test physics-based earthquake simulation as an alternative/complement to empirical hazard models for New Zealand. Distinct from the later NZNSHM22-fault RSQSim version (rundir5566).

Key references: [117, 119].

102. RSQSim Synthetic Earthquake Catalogue + Ground Motions for New Zealand (NZNSHM22 fault system)

background · parameter-testing · other · non-reviewed

- **Also known as:** RSQSim rundir5566; Shaw & Milner 2025 NZ catalog
- **Coverage:** synthetic (RSQSim simulated span; published 2025) — synthetic (RSQSim simulated span)
- **Region:** New Zealand (nationwide fault system); plus California in the bundled record
- **Bounding box:** New Zealand (NZNSHM22 fault model domain); record also includes California
- **Magnitude:** Simulated event magnitudes from rupture area/slip on 2 km triangular fault patches; full simulated magnitude distribution.
- **Content:** hypocentral locations; fault-geometry; other

- **Producer / contact:** Bruce E. Shaw (Columbia University) & Kevin R. Milner (USGS); Christine Goulet (USGS) co-author on the companion paper
- **Data source:** Synthetic catalogue, rupture slip-time histories and simulated broadband ground motions generated by the RSQSim simulator on the 2022 NZ NSHM (NZNSHM22) fault system discretized into 2 km triangular patches; bundled in the same Zenodo record as a California (UCERF3, rundir5652) catalogue.
- **Availability:** available (DOI) · DOI [10.5281/zenodo.14532399](https://doi.org/10.5281/zenodo.14532399)

An updated physics-based synthetic earthquake catalogue with rupture slip-time histories and simulated broadband ground motions on the 2022 NZ NSHM fault system (NZ rundir5566). It extends the earlier NSHM2012-fault RSQSim NZ catalogue to the current hazard-model fault geometry and adds ground-motion simulations, supporting comparison of physics-based simulators against empirical ground-motion models. The Zenodo record bundles both the NZ (rundir5566) and a California (rundir5652) catalogue.

Key references: [118, 120].

103. Hikurangi Trench Earthquake-Swarm Catalogue (ETAS-based, Nishikawa et al. 2021)

compilation · particular-event-type · other · non-reviewed

- **Also known as:** Nishikawa Hikurangi swarm catalog; ETAS swarm catalogue, Hikurangi Trench (1997-2015)
- **Coverage:** 1997 — 2015
- **Region:** Hikurangi Trench / Hikurangi subduction margin, North Island, New Zealand (offshore eastern margin)
- **Magnitude:** $M \geq 3$; magnitude of completeness (M_c) estimated to vary in time: 3.0 (1997-2001), 2.7 (2002-2006), 2.5 (2007-2011), 3.1 (2012-2015), attributed to changes in the GeoNet location algorithm/procedure in 2012
- **Content:** hypocentral locations; quality information; other (statistical swarm classification / background-rate anomalies)
- **Producer / contact:** Tomoaki Nishikawa (lead author); at time of publication Graduate School of Science / Disaster Prevention Research Institute, Kyoto University, Japan. Co-authors Takuya Nishimura and Yutaro Okada.
- **Data source:** Derived by applying the epidemic-type aftershock-sequence (ETAS) model to the GeoNet New Zealand earthquake catalogue ($M \geq 3$, 1997-2015) to statistically separate background/swarm seismicity from aftershock sequences along the Hikurangi Trench.
- **Availability:** DOI [10.1029/2020JB020618](https://doi.org/10.1029/2020JB020618)

A derived, declustering-type catalogue produced to study the spatiotemporal relationship between seismicity and slow slip events (SSEs) along the Hikurangi subduction margin. The authors fitted the ETAS model to GeoNet data to identify anomalous earthquake swarms (statistically elevated background rate) and built a new swarm catalogue, finding many swarms occur as intraplate events lagging SSEs by days or more. The focus is on isolating SSE-associated swarm activity rather than cataloguing all seismicity.

Key references: [98].

104. EEPAS Lag-Compensated Earthquake Forecast for Central New Zealand (Rhoades et al. 2022)

compilation · parameter-testing · other · reviewed

- **Also known as:** EEPAS (Every Earthquake a Precursor According to Scale) lag-compensated medium-term forecast, central New Zealand; EEPAS-TLC
- **Coverage:** 2019 — 2030
- **Region:** Central New Zealand (Wellington / Cook Strait region); broader EEPAS reviewed for NZ, California, Japan, Italy, Greece and synthetic catalogues
- **Bounding box:** Central New Zealand; a 2-degree rectangle centred on Wellington ~173-177 E, 39-43 S (per the 2019 central-NZ forecast paper; the 2022 review is global in scope)

- **Magnitude:** Forecast target magnitudes $M \geq 6.0, 6.5, 7.0, 7.5, 8.0$ (central-NZ lag-compensated forecast); model fitted using NZ catalogue data with lower magnitude thresholds for precursory seismicity
- **Content:** other (forecast occurrence-rate / probability grids; not observed events)
- **Producer / contact:** David A. Rhoades (lead author), GNS Science (Te Pu Ao / Earth Sciences New Zealand), Lower Hutt; co-authors Sepideh J. Rastin and Annemarie Christophersen (2022 review). The central-NZ-to-2030 forecast product itself is Rhoades & Christophersen, GNS Science.
- **Data source:** Derived/forecasting product: the EEPAS model fitted to the GeoNet New Zealand earthquake catalogue, producing gridded medium-term forecast occurrence-rate / probability estimates rather than observed events.
- **Availability:** available on request · DOI [10.3390/geosciences12090349](https://doi.org/10.3390/geosciences12090349)

A derived forecasting product, not a discrete event catalogue. EEPAS treats every earthquake as a precursor contributing a transient increment to the expected rate of larger events via empirical predictive scaling relations; the lag-compensated (time-lag) variant corrects the model's tendency to underpredict when forecasting periods beginning several years ahead. It was produced to forecast medium-term $M \geq 6.0$ –8.0 occurrence in central New Zealand and is the EEPAS medium-term component of NZ public/operational earthquake forecasts.

Key references: [104, 105].

105. New Zealand Earthquake Forecast Testing Centre (CSEP-NZ) Authorized Test Catalogue

compilation · parameter-testing · other · reviewed

- **Also known as:** CSEP-NZ testing catalogue; NZ Earthquake Forecast Testing Centre authorized catalogue; CSEP New Zealand evaluation catalogue
- **Coverage:** 2008 — ongoing
- **Region:** New Zealand (CSEP-NZ collection/testing region)
- **Depth range:** 0–40 (epicentral depths $h \leq 40$ km included in the test region)
- **Magnitude:** Short-term models tested at $M \geq 4$ (24-hour bins); intermediate-term and long-term models tested at $M \geq 5$ (3-month, 6-month and 5-year bins); depth cutoff $h \leq 40$ km
- **Content:** hypocentral locations; quality information; other (authorized/binned test catalogue for forecast scoring)
- **Producer / contact:** Matthew C. Gerstenberger & David A. Rhoades, GNS Science (Te Pu Ao / Earth Sciences New Zealand), Lower Hutt; operated under the Collaboratory for the Study of Earthquake Predictability (CSEP).
- **Data source:** Processed/filtered subset of the GeoNet New Zealand earthquake catalogue, authorized and binned (time/magnitude/location) for prospective scoring of forecast models within the CSEP testing infrastructure.
- **Availability:** available on request

A derived 'authorized' evaluation catalogue: the GeoNet NZ catalogue filtered to defined magnitude/depth thresholds and binned in time-magnitude-space cells, used as the observation set against which prospective earthquake-forecast models are scored (N, L and R tests) in the CSEP New Zealand testing centre. It was created as forecast-testing infrastructure to enable verifiable prospective evaluation of time-varying earthquake-occurrence models for the NZ region, with tests running from the start of 2008.

Key references: [50].

7.10 Strong-motion and ground-motion databases

Instrumental strong-motion and ground-motion databases that pair earthquake parameters with recorded waveforms and intensity measures for engineering and ground-motion-model use.

106. New Zealand Strong Motion Database (NZSMDB)

compilation · particular-event-type · other · reviewed

- **Also known as:** NZSMDB; NZ Strong Motion Database; GeoNet/GNS Science strong-motion database; Van Houtte et al. (2017) database
- **Coverage:** 1968 — ongoing
- **Region:** New Zealand (incl. Hikurangi and Fiordland subduction margins; crustal and subducting-plate seismicity)
- **Events:** 276 earthquakes (4,148 uniformly-processed recordings; 598 of 756 three-component accelerograms identified as suitable for time-domain structural response analyses)
- **Content:** hypocentral locations, fault-geometry, quality information, other
- **Producer / contact:** Chris Van Houtte (lead author); GNS Science / GeoNet (producing institution). Co-authors: Stephen Bannister, Caroline Holden, Sandra Bourguignon, Graeme McVerry.
- **Data source:** Strong-motion (accelerograph) recordings from the GeoNet network and earlier NZ strong-motion network operators, compiled and uniformly reprocessed by GNS Science. Event source parameters (hypocentres, magnitudes) and rupture models derived from seismic and geodetic data; site metadata characterised separately for GeoNet stations.
- **Availability:** available (DOI) · DOI <https://doi.org/10.5459/bnzsee.50.1.1-20>

GNS Science/GeoNet engineering strong-motion database compiling 276 NZ earthquakes (Mw 3.5–7.8) recorded by strong-motion instruments and uniformly processed into 4,148 recordings, to provide a consistent, high-quality dataset for ground-motion modelling and seismic-hazard/engineering applications. Beyond the ground-motion products it carries an explicit event component: recomputed source parameters (hypocentres, magnitudes), rupture models for selected large events in a common format, and associated site metadata. It is the foundational engineering strong-motion event compilation for NZ, distinct from the routine GeoNet earthquake catalogue.

Key references: [57, 80, 140].

107. New Zealand Ground-Motion Database (NZGMDB)

compilation · parameter-testing · manual · reviewed

- **Also known as:** NZGMDB; 2023 NZGMDB; 2021 New Zealand Ground-Motion Database (GNS Science Report 2021/56); NZ Ground-Motion Database 2024 Update
- **Coverage:** 2000 — 2022-12
- **Region:** New Zealand (including crustal, subduction interface, outer-rise and inslab/intraslab events of the Hikurangi-Kermadec and Puysegur margins)
- **Magnitude:** Moment magnitude Mw ~3.0 to ~7.8 (2023 version; abstract states events with Mw greater than ~3.0 to end-2022). Predecessor 2021 version used a threshold of Mw $\geq$ 4.0.
- **Events:** ~2,725 events reported for the 2023 version in the source (breakdown given as ~1,598 crustal, 432 interface, 98 outer-rise, 597 inslab)
- **Content:** hypocentral locations, quality information, other (recomputed moment magnitudes, event tectonic classifications, ground-motion intensity measures: PGA / pseudo-acceleration response spectra / smoothed Fourier amplitude spectra / cumulative & duration metrics, source-receiver path and site information)
- **Producer / contact:** Jesse A. Hutchinson, Robin L. Lee, Brendon A. Bradley and colleagues (University of Canterbury / QuakeCoRE and GNS Science). Producing institutions: University of Canterbury and GNS Science.
- **Data source:** Recompiled from New Zealand earthquake events and strong-motion recordings: GeoNet earthquake catalogue / waveform archive (quakesearch.geonet.org.nz) plus recomputed hypocentres, moment magnitudes and tectonic classifications. Builds on the 2021 GNS Science Report predecessor that was the central ground-motion event database for the NSHM 2022 ground-motion characterization (GMC) model.
- **Availability:** available (DOI) · DOI [10.1785/0120230184\(2023versionpaper,BSSA\);datasethostedonOSFathhttps://osf.io/q9yrg/](https://doi.org/10.1785/0120230184(2023versionpaper,BSSA);datasethostedonOSFathhttps://osf.io/q9yrg/). Predecessor: GNS Science Report 2021/56, DOI [10.21420/Z20E-5507](https://doi.org/10.21420/Z20E-5507)

The NZGMDB is a unified, expandable New Zealand event-and-ground-motion catalogue purpose-built for ground-motion model development. It compiles all local events above a moment-magnitude threshold together with recomputed earthquake hypocentres, moment magnitudes, event tectonic classifications (crustal, interface, outer-rise, inslab), source-receiver path/site information, and processed ground-motion intensity measures (PGA, 5%-damped pseudo-acceleration response spectra, smoothed Fourier amplitude spectra,

and cumulative/duration metrics). A preceding 2021/2022 version (GNS Science Report 2021/56) served as the central ground-motion database for the 2022 NZ National Seismic Hazard Model GMC modelling; it is maintained as a distinct, separately published dataset from the NSHM Mw-homogenised hazard catalogue.

Key references: [66, 67, 102].

8 Conclusion

This report inventories 107 earthquake catalogues developed for, or covering, Aotearoa New Zealand with coverage spanning 1960–2026, profiled to a common submission schema. The exercise confirms both the strength of the national record and the scale of the discoverability gap below it: temporary deployments and regional catalogues—the most precise local datasets New Zealand seismology produces—are also the hardest to find, compare, and cite. Populating a lightweight, openly versioned registry with descriptors such as these is a low-cost, near-term step that would make the country’s earthquake science measurably more findable, comparable, and reusable. This inventory is offered as a starting population of that registry and as a reproducible reference that can be extended as the long tail is filled in.

A The Catalogue Submission Profile (schema)

Each entry is described by the minimum-attributes submission profile proposed for the national “catalogue of catalogues”. Required fields are kept to a small core to lower the barrier to contribution; controlled drop-downs are preferred over free text.

- **Identification and contact:** catalogue name; contact person(s); data source; brief description; additional notes.
- **Coverage:** start date; end date (or *continuous/ongoing*); bounding box (lat/long); depth range; region of interest; magnitude information (range and/or completeness).
- **Content present** (tick all): hypocentral locations; relative relocations; phase arrival-time picks; amplitude picks; fault-geometry; historical observations; quality information; other.
- **Classification** (controlled): catalogue type (background / aftershock / induced / historical / compilation / large-events-only); focus (geographical / parameter-testing / particular-event-type); detection method (manual / automatic / template-matching / machine-learning / felt-reports / other); review status (non-reviewed / partially reviewed / reviewed).
- **Availability:** available (DOI) / available on request / not available; plus DOI or dataset link where one exists.

This profile complements the richer event-level metadata profile (aligned with QuakeML and FDSN conventions) used for catalogues ingested in full.

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